# SS'16: Optimization for Machine Learning
## (Graphical Models)

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July 13, 2016

# Contents

# 1 Energy Minimization for Graphical Models

**Basic Definitions.** Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph consisting of the finite set of *nodes* $\mathcal{V}$ and the set of node pairs $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. The set $\mathcal{E}$ will be also called *a neighborhood structure* of $\mathcal{V}$.

Let *a finite set of labels* $\mathcal{X}_v$ be associated with each node $v \in \mathcal{V}$. The functions $\theta_v \colon \mathcal{X}_v \to \mathbb{R}$ and $\theta_{uv} \colon \mathcal{X}_u \times \mathcal{X}_v \to \mathbb{R}$ are called *unary* and *pairwise potentials* are associated with nodes $v \in \mathcal{V}$ and edges $uv \in \mathcal{E}$ of the graph $\mathcal{G}$ respectively.

Notation $\mathcal{X}_\mathcal{A}$, $\mathcal{A} \subseteq \mathcal{V}$ will be used for the Cartesian product $\prod_{v \in \mathcal{A}} \mathcal{X}_v$. In particular, $\mathcal{X}_{uv}$ stands for $\mathcal{X}_u \times \mathcal{X}_v$.

The number $\sum_{v \in \mathcal{V}} |\mathcal{X}_v| + \sum_{uv \in \mathcal{E}} |\mathcal{X}_{uv}|$ we will denote as $I$. The vector of labels $x = (x_v \colon v \in \mathcal{V}) \in \mathcal{X}_\mathcal{V}$ is called *labeling*. The vector

$$\mathbb{R}^I \ni \theta := \big(\theta_v(s) \in \mathbb{R} \colon v \in \mathcal{V}, s \in \mathcal{X}_v; \quad \theta_{uv}(s,t) \in \mathbb{R} \colon uv \in \mathcal{E}, (s,t) \in \mathcal{X}_u \times \mathcal{X}_v\big) \in \mathbb{R}^I \quad (1.1)$$

will be called *vector of potentials* or simply *potentials*.

The triple $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$ is called *a graphical model*.

**Definition 1.1** (Energy minimization problem)**.** *The problem*

$$x^* = \operatorname{argmin}_{x \in \mathcal{X}_\mathcal{V}} E(x) = \operatorname{argmin}_{x \in \mathcal{X}_\mathcal{V}} \sum_{v \in \mathcal{V}} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v) \quad (1.2)$$

*of finding a labeling $x^*$ minimizing the total corresponding potential will be called* energy minimization *or* maximum a'posteriori (MAP) inference *problem for the graphical model $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$.*

**Remark 1.1.** *Problem* (1.2) *has also other names depending on community:* maximum likelyhood estimate (MLE) inference *(machine learning, natural language processing community),* weighted or partial constraint satisfaction problem *(constraint satisfaction community).*

**Probabilistic Interpretation.** Names MAP and MLE inference come from the probabilistic interpretation of the problem (1.2). Namely, one associate with the energy $E(x)$ the probability distribution

$$p(x) = \exp\left(-E(\theta, x) - A(\theta)\right) = \frac{1}{Z} \exp\left(-E(\theta, x)\right), \quad (1.3)$$

where the normalising value $A(\theta)$ is called *the log-partition function.* In its turn for its exponent $Z(\theta) = \exp(A(\theta))$ the name *partition function is used.*

Problem (1.2) then is equivalent to finding *the most probable* labeling $x$.

Distributions of the form (1.3) belong to the *exponential family* of distributions. As soon as $E$ is representable in a separable form (1.2) definition (1.3) can be rewritten in a form of a *Gibbs distribution*

$$p(x) = \frac{1}{Z(\theta)} \prod_{v \in \mathcal{V}} \Theta_v(x_v) \prod_{uv \in \mathcal{E}} \Theta_{uv}(x_u, x_v), \quad (1.4)$$

where $\Theta_w = \exp(-\theta_w)$ for $w \in \mathcal{V} \cup \mathcal{E}$.

**Combinatorial Complexity of MAP-Inference** The number of possible labelings $x$ in (1.2) grows exponentially as $\prod_{v\in\mathcal{V}}|\mathcal{X}_v|$ (which results e.g. in $L^{|\mathcal{V}|}$ in the case $|\mathcal{X}_v| = L$, $v \in \mathcal{V}$).

However a combinatorial set of solutions is not sufficient for polynomial non-solvability of the problem. Example: shortest path between two vertices in a directed graph with positive edge weights has a combinatorial set of solutions, but solvable polynomially by Dijkstra's algorithm.

Let us show that the energy minimization (1.2) is *at least* NP-hard. To this end we show that the *Hamiltonial cycle* problem is polynomially reducible to (1.2).

See slides

# 2 Acyclic Graphical Models. Dynamic Programming

## 2.1 MAP-Inference

Let the graph $\mathcal{G}$ of the graphical model $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$ be acyclic. It turns out that in this case the MAP-inference problem (1.2) can be solved efficiently, in a linear time w.r.t. the problem size, with the dynamic programming algorithm.

**Chain-Structured Graphical Models** To simplify our presentation we start with a special case, when $\mathcal{G}$ has a chain structure. In this case $\mathcal{V} = \{1, \dots, n\}$ and $\mathcal{E} = \{(i-1, i) \colon i = 2, \dots, n\}$. The energy minimization problem (1.2) reads:

$$\min_{x_1,\dots,x_n} \underbrace{\theta_1(x_1)}_{q_1(x_1)} + \sum_{i=2}^{n} \underbrace{\Big(\theta_{i-1,i}(x_{i-1}, x_i) + \theta_i(x_i)\Big)}_{q_i(x_{i-1},x_i)} = \min_{x_1,\dots,x_n} q_1(x_1) + \sum_{i=2}^{n} q_i(x_{i-1}, x_i)\,. \tag{2.1}$$

The following transformations lead to a recursive formula for computing (2.1):

$$\begin{aligned}
\min_{x_1,\dots,x_n} q_1(x_1) + \sum_{i=2}^{n} q_i(x_{i-1}, x_i) &= \min_{x_2,\dots,x_n}\left(\underbrace{\min_{x_1\in\mathcal{X}_1}\big[\overbrace{q_1(x_1)}^{F_1(x_1)} + q_2(x_1,x_2)\big]}_{F_2(x_2)} + \sum_{i=3}^{n} q_i(x_{i-1}, x_i)\right) \\
= \min_{x_2,\dots,x_n}\left(F_2(x_2) + \sum_{i=3}^{n} q_i(x_{i-1}, x_i)\right) &= \min_{x_3,\dots,x_n}\left(\underbrace{\min_{x_2\in\mathcal{X}_2}\big[F_2(x_2) + q_3(x_2,x_3)\big]}_{F_3(x_3)} + \sum_{i=4}^{n} q_i(x_{i-1}, x_i)\right) \\
&= \min_{x_3,\dots,x_n}\left(F_3(x_3) + \sum_{i=4}^{n} q_i(x_{i-1}, x_i)\right) = \dots
\end{aligned} \tag{2.2}$$

Complexity of the algorithm is $O(\sum_{i=2}^{n}|\mathcal{X}_{i-1}||\mathcal{X}_t|)$, which in the case of equal number $|\mathcal{X}_0|$ of labels in each node gives $O(|\mathcal{X}_0|^2|\mathcal{V}|)$. The function $F$ is known as a *Bellman* function, in honor of the author of the dynamic programming. Besides the original name, Algorithm 3 is often refereed to as *belief propagation on a chain* or *Viterbi algorithm.*

Add a note about obtaining an optimal labeling

An important theoretical question related to the dynamic programming and acyclic graphs is: Why these problems are easily solvable? We will show that

**Algorithm 1** Dynamic programming algorithm for chain-structured graphical models

Given: $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$ - a chain-structured graphical model
$F_1(s) := \theta_1(s)$, for all $s \in \mathcal{X}_1$
**for** $i = 2$ **to** $n$ **do**
  $F_i(s) := \theta_i(s) + \min_{t\in\mathcal{X}_{i-1}} \left[F_{i-1}(t) + \theta_{i-1,i}(s,t)\right]$
**end for**
**return** $\min_{s\in\mathcal{X}_n} F_n(s)$.

- the local polytope coincides with the marginal polytope in this case and
- the dynamic programming algorithm can be seen as an algorithm, which solves a *dual* relaxed energy minimization problem.

We will address these points after introducing the duality theory.

**Tree-Structured Graphical Models** Two important properties of a tree-structured graphs are:

- There exists always at least one vertex $v \in \mathcal{V}$, which has a single neighbor. Such vertexes we will call the *leaves* of the tree.
- Removing any leave and with its incident edge turns a tree-structured graph into a tree-structured graph.

Note that a chain is a tree and in Algorithm 3 we remove iteratively its leave with the smallest index. This observation leads to the generalization of Algorithm 3 to arbitrary tree-structured graphs.

**Algorithm 2** Dynamic programming algorithm for tree-structured graphical models

Given: $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$ - a tree-structured graphical model
$F_v(s) := \theta_v(s)$, for all $v \in \mathcal{V},\ s \in \mathcal{X}_v$
**while** $|\mathcal{V}| > 1$ **do**
  find $v \in \mathcal{V}\colon \mathcal{N}_b(v) = \{u\}$ for some $u \in \mathcal{V}$
  $F_u(s) := F_u(s) + \min_{t\in\mathcal{X}_v} \left[F_v(t) + \theta_{uv}(s,t)\right]$
  $\mathcal{V} := \mathcal{V}\backslash\{v\}$
**end while**
$v := \mathcal{V}$
**return** $\min_{s\in\mathcal{X}_v} F_v(s)$.

## 2.2 Computation of Min-Marginals

Besides the energy minimization discussed above, with dynamic programming one can compute also so called *min-marginals* for acyclic graphical models. Below in this section we will assume the graphical model to be chain-structured, although all definitions, theoretical results and algorithms generalize straightforwardly to any acyclic structures.

**Definition 2.1.** *Let $m_w$ be mappings of the form $m_w\colon \mathcal{X}_w \to \mathbb{R}$ defined for all $w \in \mathcal{V} \cup \mathcal{E}$. Values*

$$m_w(s) = \min_{x\in\mathcal{X}_\mathcal{V}\colon x_w = s} E(x) \tag{2.3}$$

*are called* min-marginals. *We will distinguish the* label min-marginals $m_v$, $v \in \mathcal{V}$ *and the* edge min-marginals $m_{uv}$, $uv \in \mathcal{E}$.

Note that (2.3) translates to

$$m_w(s) = \min_{x \in \mathcal{X}_\mathcal{V}\colon x_w = s} q_1(x_1) + \sum_{i=2}^{n} q_i(x_{i-1}, x_i) \tag{2.4}$$

for chain-structured models.

To compute min-marginals we will compute another Bellman function $B_i\colon \mathcal{X}_i \to \mathbb{R}$, but computed from the other side of the chain, e.g. $B_n(s) := \theta_n(s)$, $s \in \mathcal{X}_n$ and $B_i(s) := \theta_i(s) + \min_{t \in \mathcal{X}_{i+1}} \left[B_{i+1}(t) + \theta_{i+1,i}(s,t)\right]$ for $i = n-1, \dots, 1$.

It is easy to see that

$$m_i(s) = F_i(s) + B_i(s) - \theta_i(s),\ i \in \mathcal{V},\ s \in \mathcal{X}_i\,. \tag{2.5}$$

$$m_{i,i+1}(s,t) = F_i(s) + B_i(t) + \theta_{i,i+1}(s,t),\ i = 1, \dots, n-1,\ (s,t) \in \mathcal{X}_{i,i+1}\,. \tag{2.6}$$

The algorithm, which computes the functions $F$ and $B$ is often refereed to as *forward-backward algorithm/belief propagation on a chain.*

## 2.3 Cyclic Graphs with Low Tree Width. Junction Tree Algorithm

This topic fits better, than the HMM below. It also provides a motivation: dynamic programming has an exponential complexity as the tree-width grows. We need other computational tools: convex optimization.

## 2.4 Hidden Markov Chains. MAP and Marginalization Inference

In this section we show relation of the chain-structured graphical models to an important probabilistic modeling tool, known as *Hidden Markov Models (Chains)*. We also show that the dynamic programming algorithm can be used to solve exactly several important types of inference problems with these models.

**Definition 2.2.** *Let $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$ be a chain-structured graphical model and $\mathcal{V} = \{1, \dots, n\}$. Let $\mathcal{Y} = \mathcal{Y}^1 \times \mathcal{Y}^2 \dots \mathcal{Y}^n \ni y$ be an observation set, $y$ – an observed sequence. The joint probability distribution of the form*

$$p(x,y) = p_1(x_1) \prod_{i=2}^{n} p_i(y_i, x_i | x_{i-1}) \tag{2.7}$$

*is called a* Hidden Markov Chain *or a* Markov Chain with Latent Variables*. The latent variables are the labels $x$ of the graphical model.*

This model has a lot of applications in various areas of pattern recognition.

Examples

Two usual inference problems are:

- Computing the most probable labeling $x$ given an observation $y$

$$x = \arg\max_{x \in \mathcal{X}_\mathcal{V}} p(x|y) = \arg\max_{x \in \mathcal{X}_\mathcal{V}} \frac{p(x,y)}{p(y)} = \arg\max_{x \in \mathcal{X}_\mathcal{V}} p(x,y) \tag{2.8}$$

  The last equality holds, since $p(y)$ does not depend on $x$. Taking minus-logarithm of (2.7) one obtains

$$x = \arg\min_{x \in \mathcal{X}_\mathcal{V}} E(x;y) = \arg\min_{x \in \mathcal{X}_\mathcal{V}} q_1(x_1) + \sum_{i=2}^{n} q_i(x_{i-1}, x_i; y_1)\,. \tag{2.9}$$

Comparing the last formula to (2.1) we conclude that this problem can be solved with Algorithm 3. At the inference time an observation $y$ is fixed and does not influence computations.

- Computing marginal distributions for each possible state (label) of each variable (graph node) of the graphical model.

$$\hat{p}_i(s) := \sum_{x\in\mathcal{X}\colon x_i=s} p(x|k) = \frac{1}{p(k)} \sum_{x\in\mathcal{X}\colon x_i=s} p(x,k)$$
$$= \frac{1}{p(k)} \sum_{x\in\mathcal{X}\colon x_i=s} p_1(x_1)\prod_{i=2}^{n} p_i(y_i,x_i|x_{i-1}) \quad (2.10)$$

This problem is required if a Bayesian risk minimization problem must be solved, with an additive loss. Write more precisely? The factor $\frac{1}{p(k)}$ can be ignored in computations of (2.10), since it only influences the normalization of $\hat{p}_i(s)$. The latter can be done at the end, by dividing the computed values to their total sum.

Comparing the expression

$$\hat{p}_i(s) \propto \sum_{x\in\mathcal{X}\colon x_i=s} p_1(x_1)\prod_{i=2}^{n} p_i(y_i,x_i|x_{i-1}) \quad (2.11)$$

to (2.4) we can note that they have a similar structure. To translate (2.11) to (2.4) one has to

  - ignore $y$, since it is fixed,
  - rename $p$ to $q$,
  - turn $\prod$ to $\sum$ and
  - $\sum$ to min.

Indeed, both operation pairs $(\prod, \sum)$ and $(\sum, \min)$ are associative, distributive and commutative and exactly these properties are needed to perform computations in the recursive fashion given by (2.2). Therefore, we can apply transformations similar to (2.2) to compute the forward $F$ and backward $B$ Bellman functions. The marginals up to the normalization factor are given by the formulas similar to (2.5) and (2.6):

$$m_i(s) = F_i(s)\cdot B_i(s),\ i\in\mathcal{V},\ s\in\mathcal{X}_i\,. \quad (2.12)$$

$$m_{i,i+1}(s,t) = \frac{F_i(s)\cdot B_i(t)}{p_i(y_i,t|s)},\ i=1,\dots,n-1,\ (s,t)\in\mathcal{X}_{i,i+1}\,. \quad (2.13)$$

## 2.5 Generalized Computational Scheme, $(\oplus,\otimes)$

The triple $(W,\oplus,\otimes)$ of the set $W$ and two operations $\oplus$ and $\otimes$ is called *a commutative semiring with a unit element* if

1. Operations $\oplus$ and $\otimes$ are associative, distributive and commutative.
2. There exist neutral elements (called *zero* and *one*) for both operations.

With such operations the following dynamic algorithm computes

$$\bigoplus_{x_1,\dots,x_n} q_1(x_1)\otimes\big(\otimes_{i=2}^{n} q_i(x_{i-1},x_i)\big) \quad (2.14)$$

**Algorithm 3** Dynamic programming algorithm for chain-structured graphical models and commutative semiring $(W, \oplus, \otimes)$

Given: $(\mathcal{G}, \mathcal{X}_{\mathcal{V}}, \theta)$ - a chain-structured graphical model
$F_1(s) := q_1(s)$, for all $s \in \mathcal{X}_1$
**for** $i = 2$ **to** $n$ **do**
  $F_i(s) := \bigoplus_{t \in \mathcal{X}_{i-1}} \left[F_{i-1}(t) \otimes q_i(s,t)\right]$
**end for**
**return** $\bigoplus_{s \in \mathcal{X}_n} F_n(s)$.

**Important semirings:**

1. $(R, \max, +)$ - energy maximization
2. $(R, \min, +)$ - energy minimization
3. $([0,1], \max, \times)$ - probability maximization
4. $(0, 1, \vee, \wedge)$ - constraint satisfaction
5. $(R, \max, \min)$ - minimax constraint satisfaction problem
6. $(R, \min, \max)$ - minimax constraint satisfaction problem

### 2.6 Bibliography and Further Reading

A classical reference for dynamic programming is the book [3]. Andrew Viterbi proposed a method for finding the most probable sequence of the hidden Markovian chain in his seminal work [50]. Acyclic Hidden Markov models and dynamic programming in general semirings $(\oplus, \otimes, W)$ are presented in great details in the excellent text-book [45], which we recommend for further reading on the topic.

## 3 Basics of Linear Programs and Their Geometry

### 3.1 Polyhedrons and Linear Programs

Let $a$ be a vector in $\mathbb{R}^n$. The set $\{x \in \mathbb{R}^n \colon \langle a, x\rangle = b\}$ is called a *hyperplane* in $\mathbb{R}^n$ and $\{x \in \mathbb{R}^n \mid \langle a, x\rangle \leq b\}$ is a half-space defined by this *hyperplane*.

**Definition 3.1.** *A* polyhedron *$P$ in $\mathbb{R}^n$ is a set of type $P = \{x \in \mathbb{R}^n \colon Ax \leq b\}$ for a matrix $A \in \mathbb{R}^{m \times n}$ and a vector $b \in \mathbb{R}^m$. A bounded polyhedron is called a* polytope.

pictures

**Proposition 3.1.** *The set of the type $\{x \in \mathbb{R}^n \colon Ax = b,\ x_i \geq 0, i \in I \subset \{1, \dots, n\}\}$ is a polyhedron in $\mathbb{R}^n$.*

*Proof.* The constraints

$$Ax = b, \tag{3.1}$$
$$x_i \geq 0, i \in I \tag{3.2}$$

are equivalent to

$$Ax \leq b, \tag{3.3}$$
$$-Ax \leq -b, \tag{3.4}$$
$$-x_i \leq 0, i \in I\,. \tag{3.5}$$

The latter have the form from Def. 3.1. □

The same proof as above can be used to prove that any set of linear constrains defines a polyhedron.

Inverse transformation from $\geq$ to $=$

**Definition 3.2.** *Let $P$ be a polyhedron in $\mathbb{R}^n$ defined by a set of linear constraints. Optimization problems of the form*

$$\min_{x \in P} \langle c, x\rangle \qquad \textit{and} \qquad \max_{x \in P} \langle c, x\rangle \tag{3.6}$$

*are called* linear programs (LP)*.*

Linear programs are polynomially solvable, there exists a number of efficient algorithms. The most universal are interior point and simplex methods, Although the latter one is not polynomial, number of its iterations in practical problems typically grows linearly with the number of variables. The standard optimization packages like CPLEX [14] or Gurobi [1] contain both these types of LP solvers.

Linear programs - picture, discussion about solutions

**Definition 3.3.** *A linear program, equipped with integrality constraints, e.g.*

$$\begin{aligned}
&\min_{x \in \mathbb{R}^N} \langle c, x\rangle \\
&s.t.\ Ax \leq b \\
&x_i \in \{0,1\}, i \in I \subset \{1,\dots,N\}
\end{aligned}$$

*is called an* integer *linear program, if $|I| = N$. Otherwise, if not all coordinates of $x$ must be integer ($|I| < N$), the problem is called a* mixed *integer linear program.*

(Mixed) Integer linear programs are NP-hard to solve. However, there are standard packages like CPLEX [14] or Gurobi [1], which can deal with such problems. Although, because of the MP-hardness, they can be memory extensive and can not guarantee that an exact solution will be obtained in a reasonable time.

**Definition 3.4.** *The vector $x' \in \mathbb{R}^n$ is called a* vertex *of a polyhedron $P$ if there exists a vector $c \in \mathbb{R}^n$ such that $\max_{x\in P}\langle c, x\rangle$ is finite and has a unique solution $x'$.*

Let $\mathbb{R}_+$ denote the set of non-negative real numbers and $\mathbb{R}^n_+$ the non-negative quadrant of the $n$-dimensional vector space.

**Definition 3.5.** *The polytope $\Delta^n \subset \mathbb{R}^n$ defined as $\{p \in \mathbb{R}^n_+ \colon \sum_{i=1}^n p_i = 1\}$ is called $n$-*dimensional (probability) simplex.

picture

**Definition 3.6.** *A set $A \subseteq \mathbb{R}^N$ is called* convex*, if from $\delta^i \in A$, $1 = 1, \dots, n$ and $p \in \mathbb{R}^n$ it follows $(\sum_{i=1}^n p_i \delta^i) \in A$.*

**Definition 3.7.** *Let $\delta^i \in \mathbb{R}^N$, $i = 1, \dots, n$ be arbitrary vectors. The set $\{s \colon s = \sum_{i=1}^n p_i \delta^i,\ p \in \Delta^n\}$ is called a* convex hull *of these vectors and will be denoted as* $\mathrm{conv}\{\delta^i \colon i = 1, \dots, n\}$*.*

From Definition 3.6 it directly follows that convex hull is a convex set.
picture

**Theorem 3.1.** *The set $P \subset \mathbb{R}^n$ is a polytope if and only if it is representable as a convex hull of its vertexes.*

**Theorem 3.2.** *Let $\delta \in \mathbb{R}^N$ be a vertex of* $\mathrm{conv}\{\delta^i \colon i = 1, \dots, n\}$*. Then there exists $i \in \{1, \dots, n\}$ such that $\delta = \delta^i$.*

**Lemma 3.1.** *For any $a \in \mathbb{R}^n$ there holds*

$$\min_{i=1,\dots,n} \{a_i\} = \min_{p \in \Delta^n} \sum_{i=1}^n p_i a_i = \min_{\mu \in \mathrm{conv}\{a_i,\ i=1,\dots,n\}} \mu\,. \tag{3.7}$$

*Proof.* The second equality holds per definition.

To prove the first one let us denote $i^* = \arg\min_{i=1,\dots,n}\{a_i\}$, $a_* := a_{i^*}$ and $p^* = \mathrm{argmin}_{p \in \Delta^n} \sum_{i=1}^n p_i a_i$. It is enough to prove that $p^*_i = 0$ if $a_i > a_*$. Indeed, let $p^*_j > 0$ and $a_j > a_*$. Let

$$p'_i = \begin{cases} p^*_i, & i \notin \{i^*, j\}\,, \\ 0, & i = j\,. \\ p^*_{i^*} + p^*_j, & i = i^*\,. \end{cases} \tag{3.8}$$

Note that $p' \in \Delta^n$ as soon as $p^* \in \Delta^n$. Then

$$\sum_{i=1}^n p'_i a_i = \sum_{i \notin \{i^*, j\}} p^*_i a_i + p^*_{i^*} a_* + p^*_j a_* \overset{a_* < a_j}{<} \sum_{i \notin \{i^*, j\}} p^*_i a_i + p^*_{i^*} a_* + p^*_j a_j = \sum_{i=1}^n p^*_i a_i\,, \tag{3.9}$$

which contradicts definition of $p^*$. □

## 3.2 Marginal Polytope

In what follows we consider the vector space

$$\mathbb{R}^I \ni \mu := \big(\mu_v(s) \in \mathbb{R},\ v \in \mathcal{V},\ s \in \mathcal{X}_v; \mu_{uv}(s,t) \in \mathbb{R},\ uv \in \mathcal{E},\ (s,t) \in \mathcal{X}_u \times \mathcal{X}_v\big)\,. \tag{3.10}$$

Let $[\![\cdot]\!]$ denote *Iverson brackets*, which is for any predicate $A$ there holds $[\![A]\!] = 1$ if and only if $A$ is true.

For each $x \in \mathcal{X}_\mathcal{V}$ we will construct a vector $\delta(x) \in \mathbb{R}^I$ with coordinates $\delta_v(s) = [\![s = x_v]\!], v \in \mathcal{V},\ s \in \mathcal{X}_v$ and $\delta_{uv}(s,t) = [\![(s,t) = (x_u, x_v)]\!],\ uv \in \mathcal{E})) \in \mathbb{R}^I$.

Simple 2-nodes example with $\delta(x)$: see slides.

Using this notation we can briefly write the energy minimization problem (1.2) as

$$\min_{x \in \mathcal{X}_\mathcal{V}} E(x) = \min_{x \in \mathcal{X}_\mathcal{V}} \langle \theta, \delta(x) \rangle\ . \tag{3.11}$$

Let us consider the set $\mathcal{M} := \mathrm{conv}\{\delta(x) \in \mathbb{R}^I \colon x \in \mathcal{X}_\mathcal{V}\}$, which we will call a *marginal polytope*.

Its elements have properties defined by the following lemmas:

**Lemma 3.2.** *Let $\mu \in \mathcal{M}$. Then $\mu_w(s) \in [0,1]$ for all $w \in \mathcal{V} \cup \mathcal{E}$ and $s \in \mathcal{X}_w$.*

*Proof.* According to definition of $\mathcal{M}$ it holds $\mu = \sum_{x\in\mathcal{X}_\mathcal{V}} p_x \delta(x)$, $p \in \Delta^{\mathcal{X}_\mathcal{V}}$, therefore $\mu_w(s) = \sum_{x\in\mathcal{X}_\mathcal{V}} p_x [\delta(x)]_w(s)$. The lemma's claim follows from the fact that $[\delta(x)]_w(s) \in \{0,1\}$. □

**Lemma 3.3.** *Let $\mu \in \mathcal{M}$. Then $\sum_{s\in\mathcal{X}_w} \mu_w(s) = 1$ for all $w \in \mathcal{V} \cup \mathcal{E}$.*

*Proof.* From the proof of Lemma 3.2 it holds $\mu_w(s) = \sum_{x\in\mathcal{X}_\mathcal{V}} p_x [\delta(x)]_w(s) = \sum_{\substack{x\in\mathcal{X}_\mathcal{V}\\ x_w=1}} p_x$. Therefore $\sum_{s\in\mathcal{X}_w} \mu_w(s) = \sum_{s\in\mathcal{X}_w} \sum_{\substack{x\in\mathcal{X}_\mathcal{V}\\ x_w=1}} p_x = \sum_{x\in\mathcal{X}_\mathcal{V}} p_x = 1$. □

Lemmas 3.2 and 3.3 together give

**Corollary 3.1.** *From $\mu \in \mathcal{M}$ it follows $\mu_w \in \Delta^{\mathcal{X}_w}$ for any $w \in \mathcal{V} \cup \mathcal{E}$.*

**Proposition 3.2.**

$$\min_{x\in\mathcal{X}_\mathcal{V}} \langle \theta, \delta(x) \rangle = \min_{\mu\in\mathcal{M}} \langle \theta, \mu \rangle \,. \tag{3.12}$$

*Proof.* Consider Lemma 3.1, where the final set $\mathcal{X}_\mathcal{V} \ni x$ of labelings plays the role of the set $i = 1, \ldots, n$ and $\langle \theta, \delta(x) \rangle$ corresponds to $a_i$. Plugging it to Lemma 3.1 one obtains

$$\min_{x\in\mathcal{X}_\mathcal{V}} \langle \theta, \delta(x) \rangle = \min_{q\in\operatorname{conv}\{\langle\theta,\delta(x)\rangle \colon x\in\mathcal{X}_\mathcal{V}\}} q \tag{3.13}$$

Let us consider constraints in (3.13):

$$\begin{aligned}
\operatorname{conv}\{\langle \theta, \delta(x) \rangle : x \in \mathcal{X}_\mathcal{V}\} &= \left\{ \sum_{x\in\mathcal{X}_\mathcal{V}} p_x \langle \theta, \delta(x) \rangle : p \in \Delta^{\mathcal{X}_\mathcal{V}} \right\} \\
= \left\{ \left\langle \theta, \sum_{x\in\mathcal{X}_\mathcal{V}} p_x \delta(x) \right\rangle : p \in \Delta^{\mathcal{X}_\mathcal{V}} \right\} &= \left\{ \langle \theta, \mu \rangle : \mu \in \operatorname{conv}\{\delta(x) \in \mathbb{R}^I : x \in \mathcal{X}_\mathcal{V}\} \right\} \\
&= \{\langle \theta, \mu \rangle : \mu \in \mathcal{M}\} \,,
\end{aligned} \tag{3.14}$$

which finalizes the proof. □

**Remark 3.1.** *The expression $\min_{\mu\in\mathcal{M}} \langle \theta, \mu \rangle$ defines a linear program. However its explicit representation would require an exponential number of linear constraints, which read*

$$\sum_{x\in\mathcal{X}_\mathcal{V}} p_x \delta(x) = \mu \tag{3.15}$$

$$\sum_{x\in\mathcal{X}_\mathcal{V}} p_x = 1 \tag{3.16}$$

$$p_x \geq 0, \;\; x \in \mathcal{X}_\mathcal{V} \,. \tag{3.17}$$

*Therefore this linear program is NP-hard w.r.t. the problem size $|I|$, just like $\min_{x\in\mathcal{X}_\mathcal{V}} E(x)$ is.*

**Vertices of the Marginal Polytope.** By Theorem 3.2, all vertices of $\mathcal{M}$ belong to the set $\{\delta(x) \colon x \in \mathcal{X}_{\mathcal{V}}\}$.

**Proposition 3.3.** *For any $x \in \mathcal{X}_{\mathcal{V}}$ the vector $\delta(x)$ is a vertex of $\mathcal{M}$.*

*Proof.* The proof is based on Definition 3.4. Consider any $x'$ and the potential vector $\theta_w(s) = \begin{cases} 0, x'_w = s \\ \infty, \text{otherwise} \end{cases}$, $w \in \mathcal{V} \cup \mathcal{E}$, $s \in \mathcal{X}_w$. In other words, $\theta_w(s) = \infty * (\mathbb{1} - \delta(x'))$, where $\infty * 0 = 0$ and $\mathbb{1}$ is the vector of a suitable dimension with all its coordinates equal to 1.

Then $\langle \theta, \delta(x') \rangle = 0$. Since all coordinates of $\theta$ and $\mu$ are non-negative, their scalar product is non-negative and therefore

$$\min_{\mu \in \mathcal{M}} \langle \theta, \mu \rangle \geq 0 = \langle \theta, \delta(x') \rangle \ . \tag{3.18}$$

Therefore, we proved an existence of a vector $\theta$ such that $x'$ is a solution of the linear program (3.18) on $\mathcal{M}$:

$$\min_{\mu \in \mathcal{M}} \langle \theta, \mu \rangle = \langle \theta, \delta(x') \rangle \ . \tag{3.19}$$

It remains to prove uniqueness of this solution. Let $\mu \in \mathcal{M} \backslash \{\delta(x')\}$ and suppose $\langle \theta, \mu \rangle = \langle \theta, \delta(x') \rangle$. The equality can hold only if $\mu_w(s) = 0$ as soon as $[\delta(x')]_w(s) = 0$. However, due to Corollary 3.1 this implies $\mu = \delta(x')$, which is a contradiction that finalizes our proof. $\square$

## 3.3 Local (Marginal) Polytope

As mentioned in Remark 3.1, the marginal polytope can be explicitly represented only with an exponentially large number of constraints and therefore can not be used directly to solve the energy minimization problem as an LP over marginal polytope (3.12). One way to overcome this is to *relax* this LP problem by considering an *outer approximation* of the marginal polytope, i.e. some polytope, with contains $\mathcal{M}$ as its subset. Moreover, for outer approximations usually holds that *the set of vertices of the marginal polytope is a subset of vertices of its outer approximation*. The most common for the energy minimization problem is the outer approximation called a *local (marginal) polytope*. We will consider its precise definition and properties below.

**Definition** To obtain a polynomial number of constraints defining an outer approximation one can not use global constraints, depending on labelings as a whole, and use only local properties. Some of these properties are provided by Lemmas 3.2 and 3.3:

$$\sum_{s \in \mathcal{X}_w} \mu_w(s) = 1, \qquad w \in \mathcal{V} \times \mathcal{E} \tag{3.20}$$

$$\mu_w \geq 0 \,. \tag{3.21}$$

These constraints are insufficient, because they relate only variables within each separate factor and do not provide connections between variables. Due to separability of the objective function (an inner product) the whole problem with those constraints only would spit into separate factors. To avoid this, we must add connections, which have the following meaning (where $\mathcal{N}_b(u)$ is the set of nodes incident to the node $u$):

$$\mu_u(s) = 1 \ \Rightarrow\ \forall v \in \mathcal{N}_b(u)\ \exists t \in \mathcal{X}_v \colon \mu_{uv}(s,t) = 1 \,. \tag{3.22}$$

And symmetrically,

$$\mu_{uv}(s,t) = 1 \ \Rightarrow\ \mu_u(s) = 1 \text{ and } \mu_v(t) = 1 \,. \tag{3.23}$$

These connections should be represented in a form of linear constraints to define a polytope together with the constraints (3.20). One one the ways to do so is defined by the formula below:

$$\sum_{s\in\mathcal{X}_u} \mu_{uv}(s,t) = \mu_v(t),\ \forall\ v\in\mathcal{V},\ t\in\mathcal{X}_v \tag{3.24}$$

$$\sum_{t\in\mathcal{X}_v} \mu_{uv}(s,t) = \mu_u(s),\ \forall\ u\in\mathcal{V},\ s\in\mathcal{X}_u\,. \tag{3.25}$$

It is easy to see that for $\mu\in\{0,1\}^I$ constraints (3.24) are equivalent to those defined by (3.22) and (3.23).

Together with (3.20) these constraints define a *local polytope*

$$\mathcal{L} := \begin{cases} \sum_{s\in\mathcal{X}_u} \mu_{uv}(s,t) = \mu_v(t), & v\in\mathcal{V},\ t\in\mathcal{X}_v & (a)\\ \sum_{t\in\mathcal{X}_v} \mu_{uv}(s,t) = \mu_u(s), & u\in\mathcal{V},\ s\in\mathcal{X}_u & (b)\\ \sum_{t\in\mathcal{X}_v} \mu_v(t) = 1, & v\in\mathcal{V} & (c)\\ \mu\geq 0, & & (d)\\ \sum_{(s,t)\in\mathcal{X}_{uv}} \mu_{uv}(s,t) = 1, & uv\in\mathcal{E},\ (s,t)\in\mathcal{X}_{uv} & (e)\end{cases} \tag{3.26}$$

The constraint (3.26)(e) can be obtained by summing up (3.26)(a) over all $t$ and taking into account (3.26)(c). Therefore we will usually omit it below.

The energy minimization problem on $\mathcal{L}$

$$\min_{\mu\in\mathcal{L}} \langle\theta,\mu\rangle \tag{3.27}$$

is called a *local polytope relaxation* of the energy minimization problem (1.2). We will also called it an *LP relaxation* or just a *relaxed problem*, although the term "LP relaxation" is more general and corresponds to *any* problem of the form $\min_{\mu\in P\supseteq\mathcal{M}}\langle\theta,\mu\rangle$ with $P$ being a polytope. We will use the term *relaxed labelings* for feasible $\mu$ and *relaxed solutions* for solutions of relaxed problems. Linear programs are polynomially solvable. However, their complexity of their iterations grows at least quadratically, which make the standard off-the-shelf solvers practically inapplicable to large-scale ($>10^6$ variables and constraints) instances. Instead approximate solvers are used, with linear complexity of a single iteration. The downside of these solvers is that they require in general an exponential number of iterations to obtain a solution with a numerical accuracy, although practically good solutions are obtained after few dozens/hundreds or thousand iterations. Since we will deal with large scale problems in this lectures, we will consider mainly such solvers here.

Let us first establish show that all integer labelings belong to the local polytope and moreover, they are vertices of this polytope:

**Proposition 3.4.** *For any $x\in\mathcal{X}_\mathcal{V}$ the vector $\delta(x)$ is a vertex of $\mathcal{L}$.*

*Proof.* The proof based on that fact that simplex constraints as formulated in Corollary 3.1 for the marginal polytope, hold also for the local polytope due to (3.26)(c-e). Otherwise is exactly the same as the one for Proposition 3.3, where $\min_{\mu\in\mathcal{M}}$ is exchanged to $\min_{\mu\in\mathcal{L}}$ and reference of Corollary 3.1 is substituted with the reference to (3.26)(c-e). □

Since by construction from $\mu\in\mathcal{L}$ and $\mu\in\{0,1\}^I$ it follows that $\mu=\delta(x)$ for some $x\in\mathcal{X}_\mathcal{V}$, Proposition 3.4 implies

**Corollary 3.2.**

$$\min_{\mu\in\mathcal{L}\cap\{0,1\}^I}\langle\theta,\mu\rangle = \min_{x\in\mathcal{X}_\mathcal{V}}\langle\theta,\delta(x)\rangle \tag{3.28}$$

*and moreover, for each solution $\mu^*$ of the left-hand-side there is a solution $x^*$ to the right-hand-side such that $\mu=\delta(x^*)$.*

The left-hand-side of 3.2 defines an *integer linear program (ILP)*, which is a linear program equipped with the integer constraints. Corollary 3.2 basically claims that the energy minimization problem (1.2) can be represented as an integer liner program. This representation is important, because it defines a vast class of problems, for which efficient off-the-shelf solvers exists, such as e.g. CPLEX [14] or Gurobi [1].

Corollary 3.2 trivially implies that if the solution of the relaxed problem $\min_{\mu\in\mathcal{L}}\langle\theta,\mu\rangle$ is integer (has only $\{0,1\}$ coordinates), then it is also a solution top the non-relaxed problem (1.2).

The following proposition states that in the solution on $\mathcal{L}$ is integer (has only $\{0,1\}$ coordinates), then it is also a solution on $\mathcal{M}$ (solution of the non-relaxed problem, optimal labeling).

The next corollary states that solutions of the local polytope relaxation define a lower bound for solutions of the non-relaxed problem (1.2).

**Corollary 3.3.**

$$\min_{\mu\in\mathcal{L}}\langle\theta,\mu\rangle \le \min_{x\in\mathcal{X}_{\mathcal{V}}}\langle\theta,\delta(x)\rangle \tag{3.29}$$

*Proof.* The proof follows from the observation that $\min_{x\in A} f(x) \le \min_{x\in B} f(x)$ as soon as $A \supseteq B$. Consider $A = \mathcal{L}$ and $B = \mathcal{L}\cap\{0,1\}^{I}$. □

The following proposition formally states that $\mathcal{L}$ is an outer approximation of $\mathcal{M}$:

**Proposition 3.5.** *$\mathcal{M} \subseteq \mathcal{L}$.*

*Proof.* As it follows from the proof of Lemma 3.2, for any $x\in\mathcal{X}_{\mathcal{V}}$ it holds $\delta(x)\in\mathcal{L}$. Since $\mathcal{L}$ is a polytope, it is a convex set. Therefore since $\delta(x)\in\mathcal{L}$ holds for all $x\in\mathcal{X}_{\mathcal{V}}$, it follows that any convex combination of these vectors belongs to $\mathcal{L}$. This implies $\underbrace{\operatorname{conv}\{\delta(x)\colon x\in\mathcal{X}_{\mathcal{V}}\}}_{\mathcal{M}} \subseteq \mathcal{L}$. □

So far we have shown that marginal and local polytopes have the similar properties. The following exercise shows that in general, however, $\mathcal{M} \neq \mathcal{L}$.

**Exercise 3.1** ($\mathcal{M}\subset\mathcal{L}$)**.** *Consider the graphical model with three nodes, as in Fig. 1. Assume that all unary potentials are zero, pairwise potentials corresponding to solid lines are equal to* 1 *and to corresponding dashed lines - to* 100*.*

*Consider the vector $\mu^*$, whose coordinates corresponding to all possible labels in the unary factors and to dashed lines in the pairwise factors are equal to* 0.5 *and zero otherwise.*

- *Show that $\mu^*\in\mathcal{L}$ by checking conditions* (3.26)*.*
- *Show that $\mu^*\notin\mathcal{M}$. To do so, first prove the following lemma:*

  **Lemma 3.4.** *Let $\mu\in\mathcal{M}$. Then there exists $x\in\mathcal{X}_{\mathcal{V}}$ such that $\mu_w(x_w) > 0$ for all $w\in\mathcal{V}\cup\mathcal{E}$.*

- *Additionally show that $\mu^*$ is a vertex of $\mathcal{L}$ using Definition 3.4, similar to the proof of Proposition 3.3.*

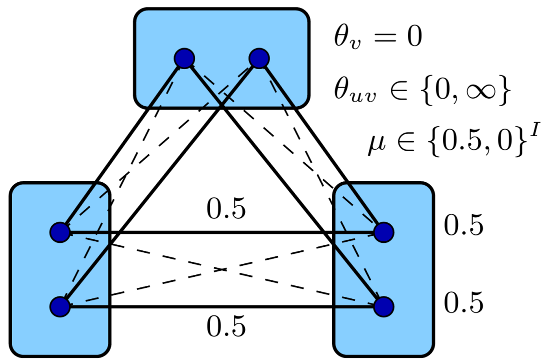


Figure 1: Illustration to Exercise 3.1.

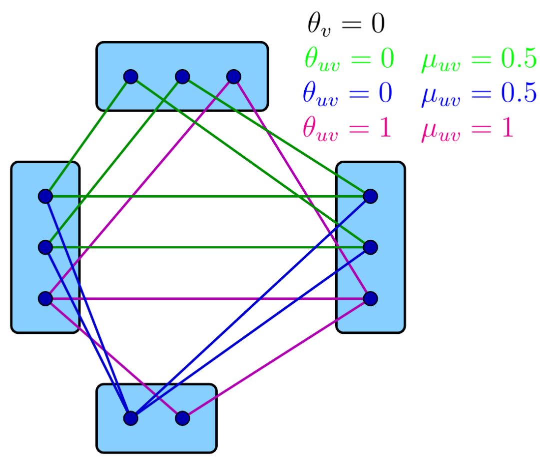


Figure 2: Illustration to Example 3.1.

**Integrality of LP solutions** LP relaxation of an ILP problem is considered to be good, if most of the variables of the relaxed solution have integer values.

A fundamental related theoretical question is whether the integer-valued coordinates are *partially optimal*, which is whether there is an optimal solution to the non-relaxed ILP problem with the same values of these coordinates.

In Section **??** we will show that the local polytope relaxation of binary problems (where only two labels can be assigned to each graph node) indeed possess the partial optimality property. In general it is, however, not the case, as shown by the following example.

**Example 3.1** (Integrality does not imply partial optimality)**.** *Consider the problem in Fig. 2. All unary potentials are assigned zero values, pairwise potentials denoted by green and blue colors are zero as well. Pairwise potentials drawn in red are equal to* 1*, not shown potentials are* $\infty$*. The greed and blue edges correspond to the minimal relaxed labeling, which has an integer coordinates corresponding to the first label of the lowest node. However the best integer labeling is the one consisting of red edges and does not contain the first label in the lowest node, which implies that the latter is not partially optimal.*

Example 3.1 provides a worst-case example. In practice most of integer variables indeed retain their values in the optimal solution. This empirical property allows to obtain good approximate solutions to the non-relaxed energy minimization problem (1.2) with the following simple procedure.

**Rounding of LP solutions** Let $\mu'$ be a relaxed solution of (3.27) and $\bar{\mathcal{V}}(\mu') = \{v \in \mathcal{V} \colon \mu'_v \in \{0,1\}^{|\mathcal{X}_v|}\}$ be the set of nodes assigned an integer value in its solution. LP relaxation is considered practically useful, if $|\mathcal{V}\backslash\bar{\mathcal{V}}(\mu')| \ll |\mathcal{V}|$. An approximate non-relaxed solution to (1.2) can be obtained by rounding of $\mu^*$ in nodes corresponding to fractional values of $\mu^*_v$:

$$x'_v := \arg\max_{s\in\mathcal{X}_v} \mu'_v(s),\ v \in \mathcal{V}. \tag{3.30}$$

Labeling $x'_v$ is usually called *a rounded solution*. Since $x'_v$ is only an approximate solution to (1.2), there holds $E(x') \geq E(x^*)$, where $x^*$ is an exact minimizer of (1.2).

There are other rounding techniques, which we will consider later on.

Insert theorem from Kleinberg-Tardos about theoretically optimal rounding.

Typical size of the LP/ILP problems. Applicability of standard solvers.

**Universality of the Local Polytope** Though the relaxed problem (3.27) has a special format, it turns out that it can be used to encode *any* linear program:

**Theorem 3.3** (Universality of Local Polytope [34])**.** *"Any linear program reduces to the local polytope relaxation of certain graphical model with 3 labels in linear time."*

### 3.4 Bibliography and Further Reading

- Probabilistic view to graphical models, energy minimization and related problems is presented in e.g. [51].
- For the geometry of polyhedra, linear and integer linear programming we recommend the excelled textbooks [25, 11, 31].
- Universality of the local polytope is shown in [34].
- The mentioned optimal probabilistic rounding was introduced in [16].

## 4 Lagrange Duality

### 4.1 Convex Functions

Let $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}$. For a function $f \colon A \to B$ the set $A$ will be called its *domain* and denoted $\mathrm{dom} f$.

Another way to introduce domain is to consider *extended value functions* $f \colon \mathbb{R}^n \to \mathbb{R} \cup \{\infty, -\infty\}$ and define $\mathrm{dom} f := \{x \in \mathbb{R}^n \colon -\infty < f(x) < \infty\}$, which we will do in this and the following lectures.

**Definition 4.1.** *An* epigraph *of a (extended-value) function $f$ is defined as the set* $\mathrm{epi} f := \{(x,t) \colon x \in \mathrm{dom} f,\ t \geq f(x)\}$.

**Example 4.1.** *Epigraph of a linear function $f \colon \mathbb{R}^n \to \mathbb{R}$ is a half-space in $\mathbb{R}^{n+1}$.*

**Definition 4.2.** $f \colon \mathbb{R}^m \to \mathbb{R} \cup \{\infty\}$ *is called* convex *if* $\mathrm{epi} f$ *is a convex set.*

**Definition 4.3.** $f \colon \mathbb{R}^m \to \mathbb{R} \cup \{-\infty\}$ *is called* concave *if $(-f)$ is convex.*

**Definition 4.4.** *Function $f$, which is concave and convex simultaneously is called* linear*.*

**Remark 4.1.** *Compare Definition 4.2 to the classical definition of convexity:*

$$f\left(\sum_{i=1}^{n} p_i x^i\right) \le \sum_{i=1}^{n} p_i f(x^i)\ \forall n,\ p \in \Delta^n\,. \tag{4.1}$$

**Example 4.2.** *Consider a convex function $f\colon D \to \mathbb{R}$ defined on a convex set $D$. The new extended value function $\tilde{f} = \begin{cases} f(x), & x \in D \\ \infty, & x \notin D \end{cases}$ is convex as well.*

**Example 4.3.** *It follows from Definition 4.3 that a concave function $f\colon \mathbb{R}^m \to \mathbb{R}$ and a convex set $D$ define an extended value concave function $\tilde{f} = \begin{cases} f(x), & x \in D \\ -\infty, & x \notin D \end{cases}$*

**Proposition 4.1.** *Let $f\colon \mathbb{R}^m \to \mathbb{R} \cup \{\infty\}$ be convex. Then the set $D = \{x\colon f(x) \le 0\}$ is convex.*

*Proof.* The set $D$ can be represented as an intersection $D = \mathbb{R}^m \cap \mathrm{epi} f$ of two convex sets, which implies $D$ is a convex set as well. □

**Proposition 4.2.** *Let $A$ be a matrix and $b$ be a vector of suitable dimensions. The set $D = \{x\colon Ax = b\}$ is convex.*

*Proof.* Let $p \in \Delta^n$. The proof follows from $A \sum_{i=1}^n p_i x^i = \sum_{i=1}^n p_i A x^i = \underbrace{\sum_{i=1}^{n} p_i}_{=1 \text{ by def.}} b = b$ □

**Proposition 4.3.** *Let $Y \subseteq \mathbb{R}^n$, functions $f\colon \mathbb{R}^n \times Y \to \mathbb{R}$ be linear w.r.t. $x$ for each fixed $y \in Y$. Then the function $g(x) = \sup_{y \in Y} f(x, y)$ is convex.*

*Proof.* Epigraph of each linear function is a half-space, which is a convex set. Epigraph of the function $g$ is an intersection of the epigraphs of the linear functions $f(\cdot, y)\colon \mathbb{R}^n \to \mathbb{R}$ for all $y \in Y$. Therefore $\mathrm{epi}(g)$ is a convex set as an intersection of convex sets. It remains to use Definition 4.2. □

**Corollary 4.1.** *Under assumptions of Proposition 4.3 the function $g(x) = \inf_{y \in D} f(x, y)$ is concave.*

The same holds also for sup/inf of convex/concave functions

## 4.2 Lagrangian Dual

**Definition 4.5.** *Problems of minimizing an extended-value convex function or maximizing an extended-value concave function are called a* convex optimization problems. *Terms* convex minimization *and* concave maximization *respectively are used for such problems as well.*

In what follows we will consider constrained optimization problems, i.e. those having the form

$$\min_{x \in D} f(x)\,, \tag{4.2}$$

for $f\colon \mathbb{R}^n \to \mathbb{R}$. The set $D$ is called a *feasible set* of the optimization problem (4.2) and any $x \in D$ is called a *feasible point*. If $f$ is extended-value, then the feasible set is defined as $D \cap \mathrm{dom} f$.

Let $f_i\colon \mathbb{R}^N \to \mathbb{R} \cup \{\infty, -\infty\}$, $i = 0, \dots, n$ and $h_j\colon \mathbb{R}^N \to \mathbb{R} \cup \{\infty, -\infty\}$, $j = 1, \dots, m$ be arbitrary (potentially non-convex) functions. Let also $D = \cap_{i=0}^n \mathrm{dom} f_i \cap \cap_{j=1}^m \mathrm{dom} h_j \neq \emptyset$. Let us consider the following constraint convex minimization problem:

$$\begin{aligned}
&\min_x f_0(x) \qquad (4.3)\\
&f_i(x) \leq 0,\ i = 1, \dots, n\\
&h_j(x) = 0,\ j = 1, \dots, m
\end{aligned}$$

**Definition 4.6** (Lagrangian)**.** *The function* $L(x, \lambda, \nu) := f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$ *will be called* Lagrangian *or* Lagrange function *and coefficients* $\lambda_i, \nu_j \in \mathbb{R}$ *–* Lagrangian coefficients *or* Lagrangian multipliers*.*

**Definition 4.7.** *The function* $g\colon \mathbb{R}^{n+m} \to \mathbb{R} \cup \{-\infty\}$ *defined as*

$$g(\lambda, \nu) := \inf_x L(x, \lambda, \nu) \tag{4.4}$$

*is called* Lagrangian dual *to* (4.3)*.*

Lagrangian dual is concave as an infinum of linear functions w.r.t. $\lambda$, $\nu$ according to Proposition 4.3. This implies concavity of the dual problem.

**Definition 4.8.** *The* Lagrangian dual problem *of* (4.3) *reads*

$$\begin{aligned}
&\max_{\lambda,\nu} g(\lambda, \nu) && (4.5)\\
&g(\lambda, \nu) > -\infty && (4.6)\\
&\lambda \geq 0\,. && (4.7)
\end{aligned}$$

*Expressions* (4.6) *and* (4.7) *are called the* dual constraints*, whereas* $g(\lambda, \nu)$ *is the* dual objective*.*

**Example 4.4** (Linear program in standard form)**.** *Let us construct a dual problem for*

$$\begin{aligned}
&\min_x \langle c, x\rangle && (4.8)\\
&Ax = b && (4.9)\\
&x \geq 0\,. && (4.10)
\end{aligned}$$

*First, turn the problem into the form* (4.3)*:*

$$\begin{aligned}
&\min_x \langle c, x\rangle && (4.11)\\
&Ax - b = 0 && (4.12)\\
&-x \leq 0\,. && (4.13)
\end{aligned}$$

*The Lagrangian reads*

$$L(x, \lambda, \nu) = \langle c, x\rangle + \langle \nu, (Ax - b)\rangle - \langle \lambda, x\rangle = -\langle \nu, b\rangle + \left\langle c - \lambda + A^\top \nu, x\right\rangle\,. \tag{4.14}$$

*The dual function reads*

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu) = \inf_x \left(-\langle \nu, b\rangle + \left\langle c - \lambda + A^\top \nu, x\right\rangle\right) = -\langle \nu, b\rangle + \inf_x \left(\left\langle c - \lambda + A^\top \nu, x\right\rangle\right)\,. \tag{4.15}$$

*The* $\inf_x$ *above is finite when* $c - \lambda + A^\top \nu = 0$*, which corresponds to the constraint* (4.6) *of the dual problem. Taking this into account the dual problem reads*

$$\max_{\nu,\lambda} - \langle \nu, b\rangle \tag{4.16}$$
$$c - \lambda + A^\top \nu = 0\,, \tag{4.17}$$
$$\lambda \geq 0\,, \tag{4.18}$$

*which can be equivalently rewritten as*

$$\max_{\nu} - \langle \nu, b\rangle \tag{4.19}$$
$$c + A^\top \nu \geq 0\,. \tag{4.20}$$

*Re-denoting* $\nu := -\nu$ *one obtains*

$$\max_{\nu} \langle \nu, b\rangle \tag{4.21}$$
$$A^\top \nu \leq c\,. \tag{4.22}$$

Any linear program can be represented in the standard form:

$$\min_{x} \langle c, x\rangle \tag{4.23}$$
$$Ax = b \tag{4.24}$$
$$x \geq 0\,. \tag{4.25}$$

**Remark 4.2.** *Since any linear program can be represented in the standard form* (4.3)*, Example 4.4 proves that the Lagrange dual of a linear program is a linear program as well.*

## 4.3 Analysis of the Dual Problem

**Proposition 4.4.** *The Lagrangian dual function introduced by Definition 4.7 is concave.*

*Proof.* The property follows from Corollary 4.1 and the fact that $g(\lambda, \nu)$ is an infinum over $x$ of linear on $\lambda, \nu$ functions $L(x, \lambda, \nu)$. □

Note, the fact that the dual function is concave does *not* depend on the cavexity of the primal problem.

Let the feasible set (given by constraints) of (4.3) be non-empty and $p^* < \infty$ be the optimal objective value on this set (which is $f_0$ evaluated in a solution $x^*$ of (4.3)).

**Theorem 4.1** (Weak duality)**.** *For any* $\lambda \geq 0$ *and any* $\nu$ *it holds* $g(\lambda, \nu) \leq f_0(x^*) = p^*$ *.*

*Proof.* For any feasible $x$ there holds:

$$L(x, \lambda, \nu) := f_0(x) + \underbrace{\sum_{i=1}^{n} \lambda_i f_i(x)}_{\leq 0} + \underbrace{\sum_{j=1}^{m} \nu_j h_j(x)}_{=0} \leq f_0(x)\,.$$

Hence $g(\lambda, \nu) \leq f_0(x)$ for any feasible $x$. Since $p^* = \min\limits_{x'-\text{feasible}} f_0(x')$ there holds $g(\lambda, \nu) \leq p^*$. □

Informal explanation of the lower bound.

**Theorem 4.2** (Strong duality)**.** *Let $f_i$ be convex and $h_j$ linear in* (4.3)*. Then the optimal value of* (4.5)-(4.7) *is equal to $p^*$.*

**Remark 4.3.** *Theorem 4.2 basically claims the strong duality holds if the primal problem* (4.3) *is convex.*

**Remark 4.4.** *The Largange dual to maximization problems of the form*

$$\begin{aligned} &\max_x f_0(x) && (4.26)\\ &f_i(x) \le 0,\ \ i = 1, \dots, n\\ &h_j(x) = 0,\ \ j = 1, \dots, m \end{aligned}$$

*is constructed in the same way with the only difference that instead of* $\inf_x$ *one has to consider* $\sup_x$*. In this case there holds $g(\lambda, \nu) \ge p^*$, where $p^*$ is the optimal value of* (4.26)*. The dual problem in this case reads*

$$\min_{\lambda,\nu} g(\lambda, \nu) \tag{4.27}$$

$$g(\lambda, \nu) < \infty \tag{4.28}$$

$$\lambda \le 0\,. \tag{4.29}$$

*An optimal value of* (4.27)-(4.29) *coincides with $p^*$ in particular if $f_0$ is concave, $f_i$, $i = 1 \dots, n$ are convex and $h_j$, $j = 1, \dots, m$ - linear functions.*

**Example 4.5** (Dual of dual)**.** *Let us consider the problem* (4.21)-(4.22) *and find its dual.*

*Lagrange function (with Lagrangian multipliers $x$) reads:*

$$L(\nu, x) = \langle \nu, b\rangle + \left\langle x, A^\top \nu - c\right\rangle = \langle \nu, Ax + b\rangle - \langle c, x\rangle\ . \tag{4.30}$$

*The dual problem reads:*

$$\min_x - \langle c, x\rangle \tag{4.31}$$

$$Ax + b = 0\,, \tag{4.32}$$

$$x \le 0 \tag{4.33}$$

*After re-denoting $x := -x$, it reads*

$$\min_x \langle c, x\rangle \tag{4.34}$$

$$Ax - b = 0\,, \tag{4.35}$$

$$x \ge 0\,, \tag{4.36}$$

*which is exactly the initial primal problem* (4.3)*.*

**Remark 4.5.** *Since any linear program can be represented in the standard form* (4.3)*, Example 4.5 shows that* the dual of the dual linear program is the initial primal linear program. *This property also holds for convex problems in general: the dual to the dual equals to the primal problem.*

**Remark 4.6.** *Primal problem unbounded - dual is infeasible and other way around. Example from exercises.*

## 4.4 Complementary Slackness

Let $x^*$ and $(\lambda^*, \nu^*)$ be any (feasible) solutions of the primal (4.3) and dual (4.5)-(4.7) problems respectively, $p^* = f_0(x^*)$ and $d^* = g(\lambda^*, \nu^*)$ be the corresponding optimal values. Moreover let $-\infty < p^* = d^* < \infty$.

Then there holds

$$f_0(x^*) = g(\lambda^*, \nu^*) = \inf_x L(x, \lambda^*, \nu^*) \leq f_0(x^*) + \sum_{i=1}^{n} \lambda_i^* f_i(x^*) + \sum_{j=1}^{m} \nu_j^* h_j(x^*) \leq f_0(x^*)\,, \quad (4.37)$$

where the second inequality holds due to the feasibility of $x^*$, which also implies $h_j(x^*) = 0$. From latter and (4.37) follows that

$$\sum_{i=1}^{n} \lambda_i^* f_i(x^*) = 0\,, \quad (4.38)$$

which is due to the fact that $f_i(x^*) \leq 0$ and $\lambda^* \geq 0$ implies

$$\lambda_i^* f_i(x^*) = 0,\ i = 1, \ldots, m\,. \quad (4.39)$$

Equalities (4.39) play an important role in the duality theory and are called *complementary slackness conditions*. They imply that if strong duality holds if *at least one* of the two conditions holds:

- primal inequality $f_i(x^*) \leq 0$ holds as equality $f_i(x^*) = 0$;
- the corresponding dual variable $\lambda_i^*$ is equal to zero.

**Karush-Kuhn-Tucker (KKT) conditions** Let $x^*$, and $(\lambda^*, \nu^*)$ be solutions of the primal and dual problems respectively, functions $f_i$, $i = 0, \ldots, n$ and $h_j$, $j = 1, \ldots, m$ be differentiable and the strong duality holds. Then there holds:

$$\left.\begin{array}{rl} f_i(x^*) \leq 0, & i = 1, \ldots, n \\ h_j(x^*) = 0, & j = 1, \ldots, m \end{array}\right\}\text{Primal feasibility}$$
$$\left.\begin{array}{rl} \lambda_i^* f_i(x^*) = 0, & i = 1, \ldots, n \end{array}\right\}\text{Complementary slackness}$$
$$\left.\begin{array}{rl} \lambda_i^* \geq 0, & i = 1, \ldots, n \\ \nabla f_0(x^*) + \sum_{i=1}^n \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^m \nu_j^* \nabla h_j(x^*) = 0 & \end{array}\right\}\text{Dual feasibility and optimality} \quad (4.40)$$

The last condition corresponds to the minimization ($\inf_x$) of the Lagrangian w.r.t. the primal variables to get the dual objective in the differentiable case (in the non-differentiable case there is a sub-gradient in place of the gradient, see its definition in Section **??**).

Importantly, that conditions (4.40) are not only necessary, but also sufficient for optimality:

**Theorem 4.3.** *Let $f_i$, $i = 0, \ldots, n$ be convex differentiable and $h_j$, $j = 1, \ldots, m$ – linear (better to use the term* affine*?) functions. Then if $(x^*, \lambda^*, \nu^*)$ satisfy* (4.40)*, they are optimal.*

*Proof.* From the first two conditions it follows that $x^*$ is feasible. Since $\lambda_i \geq 0$ the Lagrangian $L(x, \lambda, \nu)$ is convex in $x$, hence for its minimum it is necessary and sufficient the last condition to hold. From this we conclude

$$g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \sum_{i=1}^{n} \lambda^* f_i(x^*) + \sum_{j=1}^{m} \nu_j h_j(x^*) = f(x^*)\,, \quad (4.41)$$

where the last equality holds due to the complementary slackness and primal feasibility. This shows that $x^*$ and $(\lambda^*, \nu^*)$ have zero duality gap, which proves their optimality. $\square$

Note that in case the primal problem is a linear problem, the last equation in (4.40) degenerates into the feasibility constraint $g(\lambda, \nu) > -\infty$. This leads to the following

**Corollary 4.2.** *For linear problems any feasible primal/dual pair satisfying complementary slackness constraints is optimal.*

Introduce the reparametrization for general LP's

$$\min_x \langle c, x\rangle \tag{4.42}$$
$$Ax - b = 0\,, \tag{4.43}$$
$$x \geq 0\,. \tag{4.44}$$

Namely, $\langle c, x\rangle = \langle c + c', x\rangle + d$ for any feasible $s$. To compute $c'$ and $d$ let us base on the feasible set. According to it $\langle Ax - b, y\rangle = 0$ for any $y$. Therefore $\langle Ax - b, y\rangle = \left\langle A^\top y, x\right\rangle - \langle b, y\rangle = 0$. Therefore, in general, the reparametrization looks as

$$\langle c, x\rangle = \left\langle c + A^\top y, x\right\rangle - \langle b, y\rangle\,,\ y \in \mathbb{R}^m\,.$$

## 4.5 Further Reading

In this section our presentation mostly follows the excellent text-book on convex optimization [7]. We recommend this book for the first acquaintance with the duality theory and convex optimization in general. More general and deep analysis of the mathematical phenomenon of duality is presented in [35].

# 5 Duality of the LP Relaxation

## 5.1 Lagrange Dual to the LP Relaxation

Let us build a dual to the local polytope relaxation (3.27) of the energy minimization problem:

$$\min_\mu \langle \theta, \mu\rangle \tag{5.1}$$
$$\text{s.t.}\ \begin{array}{l|l}
1 - \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v) = 0,\ v \in \mathcal{V} & \gamma_v \\
1 - \sum_{(x_u,x_v) \in \mathcal{X}_{uv}} \mu_{uv}(x_u, x_v) = 0,\ uv \in \mathcal{E} & \gamma_{uv} \\
\sum_{x_v \in \mathcal{X}_v} \mu_{uv}(x_u, x_v) - \mu_u(x_u) = 0,\ x_u \in \mathcal{X}_u,\ uv \in \mathcal{E} & \phi_{v,u}(x_v) \\
\sum_{x_u \in \mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v) = 0,\ x_v \in \mathcal{X}_v,\ uv \in \mathcal{E} & \phi_{u,v}(x_u) \\
-\mu \leq 0\,. & \beta \in \mathbb{R}^I
\end{array}$$

The Lagrangian reads:

$$L(\mu, \gamma, \phi, \beta) = \langle \theta, \mu\rangle - \langle \beta, \mu\rangle + \sum_{v \in \mathcal{V}} \gamma_v \left[1 - \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v)\right] + \sum_{uv \in \mathcal{E}} \gamma_{uv} \left[1 - \sum_{(x_u,x_v) \in \mathcal{X}_{uv}} \mu_{uv}(x_u, x_v)\right]$$
$$+ \sum_{v \in \mathcal{V}} \sum_{u \in \mathcal{N}_b(v)} \sum_{x_v \in \mathcal{X}_v} \phi_{v,u}(x_v) \left[\sum_{x_u \in \mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v)\right]\,, \tag{5.2}$$

where we regrouped terms in the last item.

Regrouping terms w.r.t. coordinates of $\mu$ gives

$$\begin{aligned}L(\mu,\gamma,\phi,\beta) &= \sum_{v\in\mathcal{V}}\gamma_v + \sum_{uv\in\mathcal{E}}\gamma_{uv}\\ &+ \sum_{uv\in\mathcal{E}}\sum_{(x_u,x_v)\in\mathcal{X}_{uv}}\mu_{uv}(x_u,x_v)\left[\theta_{uv}(x_u,x_v)+\phi_{v,u}(x_v)+\phi_{u,v}(x_u)-\beta_{uv}(x_u,x_v)-\gamma_{uv}\right]\\ &+ \sum_{v\in\mathcal{V}}\sum_{x_v\in\mathcal{X}_v}\mu_v(x_v)\left[\theta_v(x_v)-\sum_{u\in\mathcal{N}_b(v)}\phi_{v,u}(x_v)-\beta_v(x_v)-\gamma_v\right]\end{aligned} \tag{5.3}$$

Taking inf w.r.t. $\mu$ leads to the following dual problem:

$$\max_{\gamma,\phi,\beta}\sum_{v\in\mathcal{V}}\gamma_v + \sum_{uv\in\mathcal{E}}\gamma_{uv} \tag{5.4}$$

$$\gamma_v = \theta_v(x_v) - \sum_{u\in\mathcal{N}_b(v)}\phi_{v,u}(x_v) - \beta_v(x_v),\ v\in\mathcal{V},\ x_v\in\mathcal{X}_v \tag{5.5}$$

$$\gamma_{uv} = \theta_{uv}(x_u,x_v)+\phi_{v,u}(x_v)+\phi_{u,v}(x_u)-\beta_{uv}(x_u,x_v),\ uv\in\mathcal{E},\ (x_u,x_v)\in\mathcal{X}_{uv} \tag{5.6}$$

$$\beta\geq 0 \tag{5.7}$$

Let us introduce two shortcuts, which meaning will become clear soon

$$\theta^{\phi}_v(x_v) := \theta_v(x_v) - \sum_{u\in\mathcal{N}_b(v)}\phi_{v,u}(x_v),\ v\in\mathcal{V},\ x_v\in\mathcal{X}_v \tag{5.8}$$

$$\theta^{\phi}_{uv}(x_u,x_v) := \theta_{uv}(x_u,x_v)+\phi_{v,u}(x_v)+\phi_{u,v}(x_u)\,,\ uv\in\mathcal{E},\ (x_u,x_v)\in\mathcal{X}_{uv}\,.$$

Then (5.4) reads

$$\begin{aligned}&\max_{\gamma,\phi,\beta}\sum_{v\in\mathcal{V}}\gamma_v + \sum_{uv\in\mathcal{E}}\gamma_{uv}\\ &\gamma_v = \theta^{\phi}_v(x_v)-\beta_v(x_v),\ v\in\mathcal{V},\ x_v\in\mathcal{X}_v\\ &\gamma_{uv} = \theta^{\phi}_{uv}(x_u,x_v)-\beta_{uv}(x_u,x_v),\ uv\in\mathcal{E},\ (x_u,x_v)\in\mathcal{X}_{uv}\\ &\beta\geq 0\end{aligned} \tag{5.9}$$

Elimination of $\beta$ gives

$$\begin{aligned}&\max_{\gamma,\phi}\sum_{v\in\mathcal{V}}\gamma_v + \sum_{uv\in\mathcal{E}}\gamma_{uv}\\ &\gamma_v \leq \theta^{\phi}_v(x_v) = \theta_v(x_v) - \sum_{u\in\mathcal{N}_b(v)}\phi_{v,u}(x_v),\ v\in\mathcal{V},\ x_v\in\mathcal{X}_v\\ &\gamma_{uv}\leq\theta^{\phi}_{uv}(x_u,x_v) = \theta_{uv}(x_u,x_v)+\phi_{v,u}(x_v)+\phi_{u,v}(x_u)\,,\ uv\in\mathcal{E},\ (x_u,x_v)\in\mathcal{X}_{uv}\,.\end{aligned} \tag{5.10}$$

**Unconstrained Lagrange Dual.** It is easy to see, that by eliminating $\gamma$ in (5.10), it can be equivalently expressed in the unconstrained form

$$\max_{\phi\in\mathbb{R}^J} D(\phi) := \max_{\phi\in\mathbb{R}^J}\sum_{v\in\mathcal{V}}\min_{s\in\mathcal{X}_v}\theta^{\phi}_v(s) + \sum_{uv\in\mathcal{E}}\min_{(s,t)\in\mathcal{X}_{uv}}\theta^{\phi}_{uv}(s,t)\,. \tag{5.11}$$

Since $\theta^\phi$ depends on $\phi$ linearly, the objective $D(\phi)$ of (5.11) is concave as a sum of point-wise minima of linear functions (Corollary 4.1). Therefore its maximization is a concave optimization problem, exactly as the one corresponding to its constraint formulation (5.10).

Although $D(\phi)$ is concave it is a non-differentiable piece-wise linear function as a point-wise minimum of linear functions, which complicates its optimization. In Section **??** we will discuss possible algorithms for solving (5.11).

In the nex section we will give an alternative interpretation of the dual problem (5.11). This interpretation will also explain the meaning of the shortcuts defined by (5.8).

## 5.2 Interpretation of the Dual LP. Reparametrisation.

It can be seen that the energy minimization representation in the form (1.2) is not unique, i.e. there exist other potentials $\theta'$ such that $\langle \theta, \delta(x)\rangle = \langle \theta', \delta(x)\rangle$ for *all* $x \in \mathcal{X}_\mathcal{V}$.

Pictures from CombiLP presentation.

Let $J := \sum_{uv\in\mathcal{E}} (|\mathcal{X}_u| + |\mathcal{X}_v|)$.

**Theorem 5.1.** *Let $\phi \in \mathbb{R}^J$, $\phi = (\phi_{v,u}(x_v), \phi_{u,v}(x_u) \in \mathbb{R}$, $uv \in \mathcal{E}$, $x_u \in \mathcal{X}_u$, $x_v \in \mathcal{X}_v)$ be* arbitrary *vector, $\theta \in \mathbb{R}^I$ be a vector of potentials and $\theta^\phi \in \mathbb{R}^I$ be defined as in* (5.8)*. Then for any $x \in \mathcal{X}_\mathcal{V}$ it holds $\langle \theta, \delta(x)\rangle = \left\langle \theta^\phi, \delta(x)\right\rangle$.*

*Proof.* The proof follows directly by checking that

$$\sum_{v\in\mathcal{V}} \theta_v(x_v) + \sum_{uv\in\mathcal{E}} \theta_{uv}(x_u, x_v) = \sum_{v\in\mathcal{V}} \theta^\phi_v(x_v) + \sum_{uv\in\mathcal{E}} \theta^\phi_{uv}(x_u, x_v)\,.$$

Alternatively, the theorem is a special case of Theorem 5.2 proved below. □

The vector $\theta^\phi$ is called a *vector of reparametrized potentials*, or shortly, a *reparametrization*.

The claim similar to Theorem 5.1 holds not only for integer labelings $\delta(x)$, $x \in \mathcal{X}_\mathcal{V}$, but also for any relaxed labeling $\mu \in \mathcal{L}$:

**Theorem 5.2.** $\langle \theta, \mu\rangle = \left\langle \theta^\phi, \mu\right\rangle$ *for any $\mu \in \mathcal{L}$ and $\theta^\phi$ defined as in Theorem 5.1.*

*Proof.*

$$\begin{aligned}
\left\langle \theta^\phi, \mu\right\rangle = \langle \theta, \mu\rangle &+ \sum_{v\in\mathcal{V}} \sum_{x_v\in\mathcal{X}_v} \mu_v(x_v) \left( - \sum_{u\in\mathcal{N}_b(v)} \phi_{v,u}(x_v) \right) \\
&+ \sum_{uv\in\mathcal{E}} \sum_{(x_u,x_v)\in\mathcal{X}_{uv}} \mu_{uv}(x_u, x_v) \left(\phi_{v,u}(x_v) + \phi_{u,v}(x_u)\right) \\
= \langle \theta, \mu\rangle - \sum_{v\in\mathcal{V}} \sum_{u\in\mathcal{N}_b(v)} \sum_{x_v\in\mathcal{X}_v} \phi_{v,u}(x_v)\mu_v(x_v) &+ \sum_{v\in\mathcal{V}} \sum_{u\in\mathcal{N}_b(v)} \sum_{x_v\in\mathcal{X}_v} \phi_{v,u}(x_v) \sum_{x_u\in\mathcal{X}_u} \mu_{uv}(x_u, x_v) \\
= \langle \theta, \mu\rangle + \sum_{v\in\mathcal{V}} \sum_{u\in\mathcal{N}_b(v)} \sum_{x_v\in\mathcal{X}_v} \phi_{v,u}(x_v) &\underbrace{\left( \sum_{x_u\in\mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v) \right)}_{=0} = \langle \theta, \mu\rangle
\end{aligned} \tag{5.12}$$

□

Since the statement of Theorem 5.2 holds for the local polytope $\mathcal{L}$, it also holds for any its subset $\mathcal{A} \subseteq \mathcal{L}$. The most important subsets of $\mathcal{L}$ are the set of integer labelings $\delta(\mathcal{X}_\mathcal{V})$ and the marginal polytope $\mathcal{M} = \mathrm{conv}\{\delta(\mathcal{X}_\mathcal{V})\}$. Note that this consideration can be used as an alternative proof of Theorem 5.1, because we did not base on it in the proof of Theorem 5.2.

**Lower Bound** Since the constrained dual problem (5.10) provides a lower bound to the primal one (5.1), so does also its unconstrained equivalent (5.11). The same lower bound property for the non-relaxed problem (1.2) can be straightforwardly proved as follows:

For any potentials $\phi \in \mathbb{R}^J$ it holds

$$\min_{x\in\mathcal{X}_\mathcal{V}} \langle\theta,\delta(x)\rangle = \min_{x\in\mathcal{X}_\mathcal{V}} \left\langle\theta^\phi,\delta(x)\right\rangle = \min_{x\in\mathcal{X}_\mathcal{V}} \sum_{v\in\mathcal{V}} \theta^\phi_v(x_v) + \sum_{uv\in\mathcal{E}} \theta^\phi_{uv}(x_u,x_v)$$
$$\geq \sum_{v\in\mathcal{V}} \min_{s\in\mathcal{X}_v} \theta^\phi_v(s) + \sum_{uv\in\mathcal{E}} \min_{(s,t)\in\mathcal{X}_{uv}} \theta^\phi_{uv}(s,t)\,. \quad (5.13)$$

Maximizing this lower bound w.r.t. $\phi$ constitutes the dual problem (5.11).

**Exercise 5.1.** *When defining the dual problem* (5.10) *we assumed the constraint* $\sum_{(x_u,x_v)\in\mathcal{X}_{uv}} \mu_{uv}(x_u,x_v) = 1$, $uv \in \mathcal{E}$ *to be a part of the local polytope formulation in* (5.1)*. However, this constraint is linearly dependent on others and therefore can be omitted.*

- *Build a dual problem based on the local polytope representation without the mentioned constraints. Show that it reads*

$$\max_{\gamma,\phi} \sum_{v\in\mathcal{V}} \gamma_v \quad (5.14)$$
$$\gamma_v \leq \theta^\phi_v(x_v),\ v\in\mathcal{V},\ x_v\in\mathcal{X}_v$$
$$0 \leq \theta^\phi_{uv}(x_u,x_v)\,,\ uv\in\mathcal{E},\ (x_u,x_v)\in\mathcal{X}_{uv}\,.$$

- *Show that the extended-value objective function $D'$ of the problem* (5.14) *is equivalent to those $D''$ of the problem* (5.10)*, i.e. there is a bijective mapping $\phi' \to \phi''$ such that $D'(\phi') = D''(\phi'')$ for any $\phi', \phi'' \in \mathbb{R}^J$.*

## 5.3 Dual MAP LP. Arc-Consistency

In this section we discuss connection between variable of the primal and dual problems. In other words, we will discuss, how to reconstruct (approiaximately optimal) labelings from reparametrizations. Connection between primal and dual solutions follow from the complementary slackness and KKT conditions.

**Complementary Slackness. Dual Rounding.** Let us now relate the primal constraints $\mu \geq 0$ from (5.1) and the dual variables $\beta$ in (5.9) in the optima of the primal and the dual problems. Each dual variable $\beta_w(s)$, $w \in \mathcal{V}\cup\mathcal{E}$, $s \in \mathcal{X}_w$ corresponds to the inequality $\mu_w(s) \geq 0$. From the complementary slackness condition (4.39) it follows that

$$\beta_w(s)\mu_w(s) = 0\,. \quad (5.15)$$

If $\beta_w(s) = 0$, then $\gamma_w = \theta^\phi_w(s)$, which implies $\theta^\phi_w(s) = \min_{t\in\mathcal{X}_w} \theta^\phi_w(s)$ ($s$ is a minimal element (label or label pair) in the factor $w$). For such $s$ the corresponding primal variable $\mu_w(s)$ can take non-zero values, as follows from (5.15).

An other way around, if $\beta_w(s) = 1$ ($\theta^\phi_w(s) > \min_{t\in\mathcal{X}_w} \theta^\phi_w(s)$, $s$ is **not** a minimal element in the factor $w$), from (5.15) it follows $\mu_w(s) = 0$.

Summing up, the complementary slackness conditions for the primal-dual pair of the relaxed LP reads:

$$\forall w \in \mathcal{V}\cup\mathcal{E},\ s\in\mathcal{X}_w\colon \quad \Big(\mu_w(s) > 0\Big) \Rightarrow \Big(\theta^\phi_w(s) = \min_{s'\in\mathcal{X}_w} \theta^\phi_w(s')\Big) \quad (5.16)$$

This substantiates *the dual rounding*: given a reparametrization $\phi$ one reconstructs a (approximately optimal) labeling $x$ as follows. In each $v \in \mathcal{V}$ assign $x_v$ to any element of the $\arg\min_{s\in\mathcal{X}_v} \theta^{\phi}_v(s)$.

Comparing it to the primal rounding (3.30) it can seem that in the dual rounding is "weaker": according to the complementary slackness from $\mu_v(x_v) > 0$ it follows that $x_v \in \arg\min_{s\in\mathcal{X}_v} \theta^{\phi}_v(s)$. Therefore, an *arbitrary, potentially, but not obligatory, non-zero* primal coordinate is selected instead of the one with the largest value $\mu_v(x_v)$. However, there are no theoretical results, which would claim that such dual rounding is indeed worse than the primal one. Refer to the primal probabilistic rounding, which is theoretically better.

**Arc Consistency.** Let $\phi \in \mathbb{R}^J$ be any (not necessarily optimal) reparametrization. Let also $\gamma_w := \min_{x_w\in\mathcal{X}_w} \theta^{\phi}_w(x_w)$, $w \in \mathcal{V}\cup\mathcal{E}$ and the set $\mathcal{L}(\phi) := \{\mu \in \mathcal{L}\colon \mu_w(x_w) = 0 \text{ if } \theta^{\phi}_w(x_w) > \gamma_w,\ w \in \mathcal{V}\cup\mathcal{E},\ x_w \in \mathcal{X}_w\}$ define possible primal vectors based on the complementary slackness. Then the following holds:

**Theorem 5.3.** *Reparametrization $\phi$ is optimal (is a dual solution) if and only if $\mathcal{L}(\phi) \neq \emptyset$.*

*Proof.* Proof follows directly from Corollary 4.2, however we provide also a specific proof below.

The primal problem is feasible and bounded, hence the dual is feasible and bounded as well. Moreover the strong duality holds.
If $\phi$ is optimal, for any primal solution $\mu \in \mathcal{L}$ the complementary slackness holds, which implies $\mu \in \mathcal{L}(\phi)$ and hence $\mathcal{L} \neq \emptyset$.
If $\mathcal{L} \neq \emptyset$ there is $\mu \in \mathcal{L}(\phi) \subseteq \mathcal{L}$. Due to Theorem 5.2

$$\begin{aligned}\langle\theta,\mu\rangle = \langle\theta^{\phi},\mu\rangle &= \sum_{w\in\mathcal{V}\cup\mathcal{E}}\sum_{x_w\in\mathcal{X}_w}\mu_w(x_w)\gamma_w \\ &= \sum_{w\in\mathcal{V}\cup\mathcal{E}}\gamma_w\underbrace{\sum_{x_w\in\mathcal{X}_w}\mu_w(x_w)}_{=1} = \sum_{w\in\mathcal{V}\cup\mathcal{E}}\min_{x_w\in\mathcal{X}_w}\theta^{\phi}_w(x_w) = D(\phi)\end{aligned} \quad (5.17)$$

Since primal is equal to dual $\mu$ and $\phi$ are optimal solutions of the primal and dual problem respectively. $\square$

**Corollary 5.1.** *Let $\delta(x) \in \mathcal{L}(\phi)$. Then $x$ is the solution of the non-relaxed energy minimization* (1.2)*.*

**Definition 5.1.** *Vector $z \in \{0,1\}^I$ is called* arc-consistent *if*

1. *From $z_{uv}(x_u,x_v) = 1$ it follows that $z_u(x_u) = z_v(x_v) = 1$ for all $uv \in \mathcal{E}$ and*
2. *From $z_u(x_u) = 1$ it follows that for any $v \in \mathcal{N}_b(u)$ there exists $x_v \in \mathcal{X}_v$ such that $z_{uv}(x_u,x_v) = 1$.*

Nice picture.

**Definition 5.2.** *$z \in \{0,1\}^I$ is* strictly *arc-consistent if it is arc-consistent and $\sum_{x_v\in\mathcal{X}_v} z_v(x_v) = 1$ as well as $\sum_{x_{uv}\in\mathcal{X}_{uv}} z_{uv}(x_{uv}) = 1$. This means that for exactly one label in each node and one label pair in each edge the corresponding coordinate of $z$ is equal to $1$.*

**Proposition 5.1.** *For any $x \in \mathcal{X}_{\mathcal{V}}$, $\delta(x)$ is strictly arc-consistent.*

Let us introduce the following notation:

- For $\mu \in \mathbb{R}^I \quad \Rightarrow \mathrm{ce}[\mu]_w(x_w) := \lceil \mu_w(x_w) \rceil$ (**ce**il - closest bigger-equal integer).
- For $\theta \in \mathbb{R}^J \quad \Rightarrow \mathrm{mi}[\theta]_w(x_w) := [\![\theta_w(x_w) = \min_{x_w \in \mathcal{X}_w} \theta_w(x_w)]\!]$ (**mi**n).

**Example 5.1.** $\mathrm{ce}[\delta(x)] = \delta(x)$; $\qquad \mathrm{ce}[(0, 0.2, 0.8, 0)] = (0, 1, 1, 0)$

**Example 5.2.** $\mathrm{mi}[(0, -2, -1, -2, 3)] = (0, 1, 0, 1, 0)$

**Remark 5.1.** *Let $\phi$ be any reparametrization. If $\mathrm{mi}[\theta^\phi]$ is strictly arc-consistent then $\mathrm{mi}[\theta^\phi]$ corresponds to a labeling $x_v = s$ iff $\mathrm{mi}[\theta^\phi]_v(s) = 1$, $v \in \mathcal{V}$, and due to Corollary 5.1 this labeling is an optimal one.*

*This claim can be reformulated as* **a strict arc consistency is sufficient for non-relaxed optimality**.

**Proposition 5.2.** *From $\mu \in \mathcal{L}$ it follows that $\mathrm{ce}[\mu]$ is arc-consistent.*

*Proof.* Follows directly from the constraints of $\mathcal{L}$. □

Therefore, the existence of an arc-consistent subset of coordinates of $\mathrm{ce}[\theta^\phi]$ is *necessary* for optimality of $\phi$. In the following section we give an algorithm, which finds the arc-consistent subset or proves that it does not exists. In the latter case it follows that the corresponding reparametrization is not optimal.

## 5.4 Relaxation Labeling Algorithm

Let us introduce two binary operations: "and", denoted as $\wedge$ and "or", denoted as $\vee$. We assume theri coordinate-wise application to binary vectors.

The necessary optimality condition for the reparametrization $\phi$ follows from Theorem 5.3 and Proposition 5.2: $\phi$ can be optimal only if there is an arc-consistent vector $z \in \{0,1\}^I$, such that $z\,\mathrm{mi}[\theta^\phi] = z$.

**Definition 5.3.** *A vector $z \in \{0,1\}^I$ is called the* kernel *of a vector $z' \in \{0,1\}^I$ and denoted as* $\mathrm{kern}(z')$ *if*

1. *$z \wedge z' = z$ and*
2. *for any other arc-consistent $z'' \in \{0,1\}^I$ it holds $z'' \wedge z = z''$.*

The result $z'$ we will call a *kernel* of $z$ and say that $z$ *has a non-empty kernel* if $z' \neq 0$. The kernel of $z \in \{0,1\}^I$ is denoted by $\mathrm{kern}(z)$.

The second condition assures that $z$ is arc-consistent vector with the largest number of 1's.

**Exercise 5.2** (Uniqueness of kernel)**.** *Show that the kernel is unique. As an intermediate result prove that if $z$ and $z'$ are arc-consistent then $z' \vee z$ is arc-consistent as well. Therefore, one can compute kernel as a logical $\vee$ over all arc-consistent vectors, satisfying the first condition of Definition 5.3.*

We will say that kernel of $z \in \{0,1\}^I$ is *non-empty* if $\mathrm{kern}(z) \neq \overline{0}$, where $\overline{0}$ is a vector with all zero coordinates.

The algorithm, which turns $z$ into $\mathrm{kern}(z)$ iteratively repeats the following two operations until $z$ is not changing anymore:

$$\forall u \in \mathcal{V},\ s \in \mathcal{X}_u,\ v \in \mathcal{N}_b(u): \quad z_u(s) := z_u(s) \wedge \bigvee_{t \in \mathcal{X}_v} z_{uv}(s,t) \tag{5.18}$$

$$\forall uv \in \mathcal{E},\ (s,t) \in \mathcal{X}_{uv}: \quad z_{uv}(s,t) := z_{uv}(s,t) \wedge z_u(s) \wedge z_v(t) \tag{5.19}$$

**$\epsilon$-Arc-Consistency** In practice, however, minimum in $\min_{x_w \in \mathcal{X}_w} \theta_w^\phi(x_w)$ is usually attained uniquely due to finite precision of floating point computations. Therefore for a potential vector $\theta$ the binary vector $\mathrm{mi}[\theta]$ is strictly arc consistent or not arc consistent. Both conditions can be checked directly and without any relaxation labeling algorithm.

Therefore one has to resort to an approximate arc-consistency, so called $\epsilon$-arc-consistency. We denote $\mathrm{mi}_\epsilon[\theta]_w(x_w) = \begin{cases} 1, & \theta_w(x_w) \le \min_{x_w \in \mathcal{X}_w} \theta_w(x_w) + \epsilon \\ 0, & \text{otherwise} \end{cases}$

**Definition 5.4.** *For the potentials $\theta$ the* $\epsilon$-arc-consistency *property holds for $\epsilon \ge 0$ if* $\mathrm{mi}_\epsilon[\theta]$ *is arc-consistent.*

There always exists a large enough $\epsilon$ such that $\mathrm{mi}_\epsilon[\theta]$ has a non-empty kernel. However, one can compute a *minimal* $\epsilon$, which guarantees $\epsilon$-arc-consistency by $\min - \max$ generalization of the relaxation labeling algorithm. This algorithm resembles the relaxation labeling algorithm (5.18)-(5.19) with the only difference that operation $\vee$ is exchanged with min and $\wedge$ is turned to max.

$$\forall u \in \mathcal{V},\ s \in \mathcal{X}_u,\ v \in \mathcal{N}_b(u){:} \quad z_u(s) := \max\left\{ z_u(s), \min_{t \in \mathcal{X}_v} z_{uv}(s,t) \right\} \tag{5.20}$$

$$\forall uv \in \mathcal{E},\ (s,t) \in \mathcal{X}_{uv}{:} \quad z_{uv}(s,t) := \max\left\{ z_{uv}(s,t), z_u(s), z_v(t) \right\}. \tag{5.21}$$

Consider to provide a proof for the or-and and the min-max algorithms

**Example 5.3** (Arc Consistency $\neq$ Optimality)**.** *Arc-consistency is not sufficient for optimality. Picture from slides + explanation.*

Two additional questions relate dual and primal feasible vectors:

- Given $\theta$, can we find $\mu \in \mathcal{L}$ such that from $\mathrm{ce}(\mu_w(x_w)) = 1$ it follows that $\mathrm{mi}(\theta)_w(x_w) = 1$? Yes, it constitutes a linear program $\min_{\mu \in \mathcal{L}'} \langle \theta, \mu \rangle$, where $\mathcal{L}'$ differs from the local polytope $\mathcal{L}$ by the additional constraints:

  $$\mu_w(s) \le \mathrm{mi}[\theta]_w(s),\ \forall w \in \mathcal{V} \cup \mathcal{E},\ s \in \mathcal{X}_w\,. \tag{5.22}$$

  However, for many dual algorithms one can construct a series of the feasible primal vectors $\mu^t$ converging to the optimal one (which satisfies constraints (5.22)) as $\theta^t$ converges to a dual optimum.

- Given $\theta$, can we find $x \in \mathcal{X}_\mathcal{V}$ such that from $x_w = 1$ it follows $\mathrm{mi}[\theta]_w(x_w) = 1$? This problem is NP-hard in general and can be formulated as a constraint satisfaction problem. However, there is a polynomially solvable subclass of such problems. We refer to the corresponding literature at the and of the chapter. Should we give a or-and formulation of the problem here - this could be used for the proof of the relaxation labeling algorithm.

### Arc-Consistency for Acyclic Graphs

**Proposition 5.3.** *Let $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$ be a graphical model and the graph $\mathcal{G}$ be acyclic. Then if* $\mathrm{kern}(\mathrm{mi}(\theta))$ *is not empty, the potentials $\theta$ are dual-optimal. Moreover, there is an integer labeling $x \in \mathcal{X}_\mathcal{V}$ satisfying the complementary slackness conditions* $\delta(x) \wedge \mathrm{kern}(\mathrm{mi}[\theta]) = \delta(x) \wedge \mathrm{mi}[\theta] = \delta(x)$.

*Proof.* We prove it by constructing a labeling $x \in \mathcal{X}_\mathcal{V}$ such that $\delta(x) \wedge \mathrm{mi}[\theta] = \delta(x)$. To do so, we start with arbitrary $v \in \mathcal{V}$ and set $x_v$ to any $s \in \mathcal{X}_v$ such that $\mathrm{mi}[\theta]_v(s) = 1$. Then on each iteration we have the set $A \subset \mathcal{V}$ of labeled nodes. We select any node $v \in \mathcal{V} \backslash A$ and select a label for it, which is connected to ... □

**Proposition 5.4.** *Let $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$ be a graphical model and the graph $\mathcal{G}$ be acyclic, and let $\mu^* = \arg\min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$ be a relaxed solution of the corresponding energy minimization problem. Then from $\mu^*_v(s) > 0$, $v \in \mathcal{V}$, $s \in \mathcal{X}_v$ it follows that there is an optimal integer labeling $x^* = \arg\min_{x \in \mathcal{X}_\mathcal{V}} \langle \theta, \delta(x) \rangle$ such that $x_v = s$.*

*Proof.* Give a nice proof and synchronioze with the proofs of the previous and the following propositions. □

**Proposition 5.5.** *Let $(\mathcal{G}, \mathcal{X}_\mathcal{V}, \theta)$ be a graphical model and the graph $\mathcal{G}$ be acyclic, $\Lambda_\mathcal{G}$ and $\mathcal{M}_\mathcal{G}$ be the corresponding local and marginal polytopes. Then $\mathcal{L}_\mathcal{G} = \mathcal{M}_\mathcal{G}$.*

*Proof.* As follows from Definition 3.4 it is enough to proof that if $\mu \in \mathcal{L}$ is a solution to the relaxed problem, then there is always an integer labeling $x \in \mathcal{X}_\mathcal{V}$, which is a solution as well. This would show that the local polytope for acyclic models has only vertices of the form $\delta(x)$, $x \in \mathcal{X}_\mathcal{V}$ and therefore (due to Theorem 3.1) coincides with the marginal polytope.

Let $\mu \in \mathcal{L}$ be an optimal solution to the relaxed problem (3.27). Then $\mathrm{ce}[\mu]$ is arc-consistent (Proposition 5.2). Therefore, there are optimal dual potentials $\theta$ satisfying the complementary slackness $\mathrm{mi}[\theta] \wedge \mathrm{ce}[\mu] = \mathrm{ce}[\mu]$. It remains to apply Proposition 5.3, since $\mathrm{kern}(\mathrm{mi}(\theta))$ is arc-consistent. □

### 5.5 Further Reading

The local polytope relaxation for graphical models, its dual and its analysis were first provided in [46] and [26]. In computer vision, an algorithm equivalent to the relaxation labeling was proposed [52] and revisited in [36].

The comprehensive overview [53] is a standard reference for the relation of the primal and dual optimal solutions for the energy minimization problem. It includes definitions of the arc-consistency, relaxation labeling algorithm and a lot of related notions and facts.

Getting a primal feasible solution from the dual iterates in a coarse of the dual optimization algorithm is described in [40].

We refer to [37] and references therein for a definition of the constraint satisfaction problems in general. The description of its solvable subclasses was first given in [9], see also [10].

## 6 Basics of Convex Optimization

Our goal now - to optimize the dual LP. We know it is a concave piece-wise linear (which implies - non-smooth) function with a large number $J$ of variables $\phi \in \mathbb{R}^J$. In this section we consider the simplest algorithms for smooth and non-smooth optimization: gradient descent, coordinate descent and the sub-gradient method.

In this section we will assume that $x^* = \mathrm{argmin}_{x \in \mathrm{dom} f} f(x)$, $f^* = f(x^*)$ and $\|x^*\| < \infty$.

### 6.1 Continuity of the Gradient

Let $\|\cdot\|$ denote an $\ell_2$ norm in a vector space $\mathbb{R}^n$.

**Definition 6.1.** *Function $f$ is called* Lipschitz-continuous *on a set $D$ if there is a constant $L \geq 0$ such that for any $x, y \in D$ it holds*

$$|f(x) - f(y)| \leq L\|x - y\|\,.$$

**Proposition 6.1.** *If $f$ is differentiable, then $L \geq \sup_{x\in D} |\nabla f(x)|$.*

We will use the notation $C^{1,1}_L(D)$ for differentiable on $D$ functions with a Lipschitz continuous gradient and the corresponding Lipschitz constant $L$ on $D$.

## 6.2 Gradient Descent

**Theorem 6.1.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be convex and differentiable. Then $\nabla f(x) = 0$ is the necessary and sufficient condition for $x \in \mathbb{R}^n$ to be minimum of $f$.*

Let $x^0$ be a starting point. The iterative process of the form

$$x^{t+1} = x^t - \alpha^t \nabla f(x^t) \tag{6.1}$$

for certain $a^t > 0$ is called *a gradient descent.*

**Theorem 6.2** ((Nesterov,2004), (Polyak,1987))**.** *Let $f \in C^{1,1}_L(\mathbb{R}^n)$ be convex. Then for $\alpha^t = \frac{1}{L}$ for the sequence $x^t$ defined by (6.1) it holds:*

$$\|x^{t+1} - x^*\|^2 \leq \|x^t - x^*\|^2 - \frac{1}{L^2}\|\nabla f(x^t)\|^2 \tag{6.2}$$

$$f(x^{t+1}) \leq f(x^t) - \frac{1}{2L}\|\nabla f(x^t)\|^2 \tag{6.3}$$

$$f(x^t) - f^* \leq \frac{2L\|x^0 - x^*\|}{t+4}\,. \tag{6.4}$$

**Remark 6.1.** *Convergence rate (6.4) is denoted as $O(\frac{L}{t})$, which means that the accuracy $\epsilon = |f(x^t) - f^*|$ improves proportionally to $\frac{L}{t}$. Another often used notation is $O(\frac{L}{\epsilon})$, which means that the number of iterations $t$ needed to attain an accuracy $\epsilon$ grows as $\frac{L}{\epsilon}$.*

**Remark 6.2.** *There are 3 practical ways to define $a^t$ in (6.1):*

- $a^t = \frac{1}{L}$ *- constant step size. Requires knowing the Lipschitz constant for a gradient or its good estimate.*
- $a^t = \operatorname{argmin}_{a\in\mathbb{R}_+} f(x^t - a\nabla f(x^t))$ *- minimization of $f$ in the gradient direction. For efficiency a closed form solution for $a^t$ is required, which is typically not available.*
- $a^t$ *is selected to satisfy*

  $$f(x^{t+1}) \leq f(x^t) - \frac{1}{2a^t}\|\nabla f(x^t)\|^2 \tag{6.5}$$

  *(local estimate of $L$ according to (6.3)), where $a^t$ is searched in a form $a^t = \frac{2a^{t-1}}{2^n}$ for $n = 0, \dots, \infty$ until (6.5) is satisfied.*

*Though the practical behavior can differ for the above three cases, their worst-case analysis is similar and the convergence rate given by Theorem 6.2 is the best you can get for any of these strategies.*

However we can not use gradient descent for our dual MAP-inference problem (5.11) directly, because it is not differentiable.

## 6.3 Sub-gradient method

**Definition 6.2.** *Let $f$ be convex. A vector $g$ is called* subgradient *of $f$ in $x^0 \in \mathrm{dom} f$ if for any $x \in \mathrm{dom} f$ it holds*

$$f(x) \geq f(x^0) + \langle g, x - x^0 \rangle \; . \tag{6.6}$$

*The set $\partial f(x^0)$ of all subgradients of $f$ at $x^0$, is called the* subdifferential *of $f$ at $x^0$.*

Picture!

**Remark 6.3.** *Let $f$ be convex differentiable, then $\partial f(x) = \{\nabla f(x)\}$. In general, however $\partial f$ contains multiple elements.*

**Exercise 6.1** (Nesterov,2004)**.** *Prove, that the set $\partial f(x)$ is non-empty convex closed and bounded set for any $x$.*

**Theorem 6.3.** *Let $f$ be convex. $f(x^*) = \min_{x \in \mathrm{dom} f} f(x)$ iff (if and only if) $0 \in \partial f(x^*)$.*

*Proof.* Indeed, if $0 \in \partial f(x^*)$, then $f(x) \geq f(x^*) + \langle 0, x - x^* \rangle = f(x^*)$ for all $x \in \mathrm{dom} f$. On the other hand, if $f(x) \geq f(x^*)$ for all $x^* \in \mathrm{dom} f$, then $f(x) \geq f(x^*) + \langle 0, x - x^* \rangle = f(x^*)$ and 0 is a subgradient of $f$ in $x^*$. □

Picture!

**Definition 6.3.** *For a convex function $f$ the iterative process*

$$x^{t+1} = x^t - \alpha^t g(x^t) \, , \tag{6.7}$$

*where $g(x^t) \in \partial f(x^t)$, is called* a subgradient method.

**Theorem 6.4.** *Let $f$ be convex and Lipschitz continuous on a ball $\{x \colon \|x^* - x\| \leq R\} \ni x^0$ with constant $M$. Let also $\alpha_t > 0$, $\alpha^t \xrightarrow{t \to \infty} 0$ and $\sum_{t=1}^{\infty} \alpha_t = \infty$. Then it holds*

- $f(x^t) - f(x^*) \xrightarrow{t \to \infty} 0$
- *If ([Bertsekas,1999]) $0 < \alpha^t < 2 \frac{f(x^t) - f(x^*)}{\|g^t\|}$*

$$\|x^{t+1} - x^*\| \leq \|x^t - x^*\| \, . \tag{6.8}$$

- *In particular ([Nesterov,2004]), if $\alpha^t = \frac{R}{\|g^t\| \sqrt{t+1}}$*

$$f^t - f^* \leq \frac{MR}{\sqrt{t+1}} . \tag{6.9}$$

**Remark 6.4.** *Convergence rate (6.9) is denoted as $O(\frac{1}{\sqrt{t}})$, which means that the accuracy $\epsilon = f(x^t) - f^*$ improves proportionally to $\frac{1}{\sqrt{t}}$. Another often used notation is $O(\frac{1}{\epsilon^2})$, which means that the number of iterations $t$ needed to attain accuracy $\epsilon$ grows as $\frac{1}{\epsilon^2}$.*

Picture for (6.8)

**Remark 6.5.** *In general $f(x^{t+1}) \leq f(x^t)$ may not hold for any (arbitrary small) step-size $\alpha^t$, see previous picture.*

**Remark 6.6.** *Since neither $f(x^t) - f(x^*)$ nor $R$ is typically known, the claims of Theorem 11.2 can be simplified as* there exists $\alpha^t$ *such that* (6.8) *holds. And* there exist $\{\alpha^t\}$ *such that* (6.9) *holds.*

Compare to gradient:

- stepsize
- decreasing of the distance to the optimum
- decreasing of the function value
- convergence rate.

**Lemma 6.1.** *Let a convex function $g$ be defined as $g(x) = \max_y f(x, y)$, where $f$ is differentiable in $x$. Then $\partial g(x) = \operatorname{conv}\{\nabla f(x, y^*) \colon y^* \in \operatorname{argmax}_y f(x, y)\}$*

*Proof.* we need it here. □

**Remark 6.7.** *For maximization of concave functions a* super-gradient *term is sometimes used. However, we will stick to use the term* sub-gradient *even for maximization. In order to maximize a concave $f$ one has to minimize the convex $-f$.*

## 6.4 Coordinate Descent

**Definition 6.4.** *The iterative process*

$$x_i^{t+1} = \arg\min_{\xi} f(x_1^{t+1}, \dots, x_{i-1}^{t+1}, \xi, x_{i+1}^t, \dots, x_n^t) \tag{6.10}$$

$$i = i + 1 \mod n \tag{6.11}$$

*is called* cyclic (or Gauss-Seidel) coordinate descent.

**Theorem 6.5** (Proposition 2.7.1, Bertsekas "Nonlinear Programming")**.** *Let $f$ be continuously differentiable on $\mathbb{R}^n$. Furthermore, let for each $i$ and $x \in \mathbb{R}^n$ the minimum in* (6.10) *is uniquely attained. Then for every limit point $x^*$ of $x^k$ it holds $\nabla f(x^*) = 0$.*

If $f$ is convex and differentiable optimality of $x$ follows from $\nabla f(x) = 0$. Theorem 6.5 does not claim existence of the limit points. It only proofs their properties if they exist. In particular, if the set $F(x^0) = \{x \colon f(x) \leq f(x^0)\}$ is bounded, then $x^t \in F(x^0)$ due to $f(x^t) \leq f(x^0)$. Hence $\{x^t\}$ is bounded and has limit points, which correspond to the optimum of $f$ according to Theorem 6.5.

**Example 6.1.** *Picture: Non-differentiable function with a unique optimum in each coordinate*

**Remark 6.8.** *For differentiable function with non-unique optimum in each coordinate where implication of Theorem 6.5 does not hold, see [Powel 1973]*

To prove that certain algorithms perform coordinate descent we will need the following technical lemma:

**Lemma 6.2.** *Let $f$ be convex, $x_j$, $j \neq i$ be fixed and $f_i(\xi) := f(x_1^{t+1}, \dots, x_{i-1}^{t+1}, \xi, x_{i+1}^t, \dots, x_n^t)$. Then $f_i$ is convex and $\xi^* = \arg\min_\xi f_i(\xi)$ iff $\overline{0} \in \partial f_i(\xi)$.*

*Proof.* Epigraph of $f_i$ is an intersection of epigraph of $f$ and the linear subspace given by the linear constraints $x_1 = x_1^{t+1}, \dots, x_{i-1} = x_{i-1}^{t+1}, x_{i+1} = x_{i+1}^t, \dots, x_n = x_n^t$). As intersection of two convex sets the epigraph of $f_i$ is convex. This proves convexity of $f_i$. The claim about optimality of $\xi^*$ follows from Theorem 6.3 and convexity of $f_i$. $\square$

**Remark 6.9.** *If in conditions of Lemma 6.2, $f$ is differentiable and for a given $x$ there holds $0 \in \partial f_i(x_i)$ for all $i$, then also $\nabla f_i(x_i) = \overline{0}$ for all $i = 1, \dots, n$ and therefore $\nabla f(x_i) = \overline{0}$ and $x$ is optimal. Because of the non-uniqueness of sub-gradient this is not the case for non-differentiable $f$.*

### 6.5 Further Reading

In this chapter we have reviewed classical results, which can be found in the excellent textbooks [4, 30, 32]. The counterexample, which shows that coordinate descent may not reach an optimum of a differentiable function belongs to Powell [33].

## 7 Sub-Gradient and Coordinate Ascent for Energy Minimization

### 7.1 Sub-gradient for the Dual LP

Recall the unconstrained dual MAP LP formulation:

$$D(\phi) = \sum_{v\in\mathcal{V}} \min_{x_v\in\mathcal{X}_v} \left[\theta_v(x_v) - \sum_{u\in\mathcal{N}_b(v)} \phi_{v,u}(x_v)\right] + \sum_{uv\in\mathcal{E}} \min_{(s,t)\in\mathcal{X}_{uv}} \left[\theta_{uv}(x_u,x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u)\right] \tag{7.1}$$

Let $x'_u = \operatorname{argmin}_{x_u\in\mathcal{X}_u} \theta^\phi_u(x_u)$ and $(x''_u, x''_v) = \operatorname{argmin}_{x_{uv}\in\mathcal{X}_{uv}} \theta^\phi_{uv}(x_u,x_v)$
The vector defined in the following formula

$$\frac{\partial D}{\partial \phi_{u,v}(x_u)} = \begin{cases} 0, & x_u \neq x'_u \quad \text{and} \quad x_u \neq x''_u \\ 0, & \quad = \quad \text{and} \quad = \\ -1, & \quad = \quad \text{and} \quad \neq \\ 1, & \quad \neq \quad \text{and} \quad =\,, \end{cases} \tag{7.2}$$

is a super-gradient of $D$.

The super-gradient method then reads $\phi^{t+1} = \phi^t + \alpha^t \frac{\partial D}{\partial \phi}$. Picture!

**Exercise 7.1.** *Compute subgradient for the example of arc-consistent potentials, but not optimal for the dual objective. Show that there is no zero subgradient in that case.*

### 7.2 Coordinate Ascent for the Dual LP: Min-Sum Diffusion

Algorithm 4 performs a block-coordinate ascent for the dual LP objective $D$ defined in (5.11) w.r.t. the dual variables $\phi$.

After each step of the algorithm, where operations are performed for a given node $u$, see lines 4-15, the following conditions hold for all $x_u \in \mathcal{X}_u$:

$$\theta^\phi_u(x_u) := 0 \quad \text{and} \quad \min_{s\in\mathcal{X}_v} \theta^\phi_{uv}(x_u, s) = \min_{s\in\mathcal{X}_{v'}} \theta^\phi_{uv'}(x_u, s),\ \forall v, v' \in \mathcal{N}_b(u)\,. \tag{7.3}$$

**Algorithm 4** Min-Sum Diffusion: Block-Coordinate Ascent for $D(\phi)$

1: **Init:** $\phi \in \mathbb{R}^J$
2: **for** $t = 1$ **to** $N$ **do**
3:   **for** $u \in |V|$ **do**
4:     **for** $x_u \in \mathcal{X}_u$ **do**
5:       *Collect weights from edges in the node:* $\theta^\phi_u(x_u) += \sum_{v\in\mathcal{N}_b(v)} \min_{x_v\in\mathcal{X}_v} \theta^\phi(x_u, x_v)$*:*
6:       **for** $v \in \mathcal{N}_b(u)$ **do**
7:         $\phi_{u,v}(x_u) -= \min_{x_v\in\mathcal{X}_v} \theta^\phi(x_u, x_v)$
8:       **end for**
9:       *Distribute the whole weight uniformly to all neighboring edges:*
10:       $\theta^\phi_{uv}(x_u, x_v) += \theta^\phi_u(x_u)/|\mathcal{N}_b(v)|$; $\theta^\phi_u(x_u) := 0$
11:       $w := \theta^\phi_u(x_u)$
12:       **for** $v \in \mathcal{N}_b(u)$ **do**
13:         $\phi_{u,v}(x_u) += \frac{w}{|\mathcal{N}_b(v)|}$
14:       **end for**
15:     **end for**
16:   **end for**
17: **end for**

To show that Algorithm 4 is a block-coordinate ascent we consider its single step for a given node $u \in \mathcal{V}$ in lines 4-15, which optimizes the dual objective w.r.t. the variables $\phi^t_{u,v}(s)$, $v \in \mathcal{N}_b(u)$, $s \in \mathcal{X}_u$. According to Lemma 6.2 it is sufficient to show that there exists a zero super-gradient of $D$ w.r.t. these dual variables.

Indeed, a super-gradient is determined by (7.2). Let us consider any $u \in \mathcal{V}$ and any $v \in \mathcal{N}_b(u)$. Due to (7.3) we can introduce $\alpha(x_u) := \min_{s\in\mathcal{X}_v} \theta^\phi_{uv}(x_u, s)$, which is independent on $v \in \mathcal{N}_b(u)$. Let $x''_u := \arg\min_{x_u\in\mathcal{X}_u} \alpha(x_u)$ and $x'_u = x''_u$ ($x'_u \in \arg\min_{x_u\in\mathcal{X}_u} \theta^\phi_u(x_u)$ due to (7.3)). Note that according to (7.2), the super-gradient is iqual to zero, since $x'_u = x''_u$.

Picture!

**Exercise 7.2.** *Show that the optimum of $D$ w.r.t. the block-coordinate $\{\phi_{u,u}(x_u), v \in \mathcal{N}_b(u), x_u \in \mathcal{X}_u\}$ is not unique. For example, one could apply $\phi_{u,v}(x_u) += \frac{w}{|\mathcal{N}_b(v)|+1}$ instead of the transformation in line 13 of Algortihm 4 and it would correspond to another optimum w.r.t. the same coordinates.*

Since neither $D$ is differentiable nor the $D$ has a unique optimum, there is no guarantee for diffusion to optimize the dual $D(\phi)$. Indeed, the weaker condition holds:

**Theorem 7.1** (Schlesinger, Antoniuk, 2011)**.** *Let $\phi^t$ be produced by Algorithm 4. Then $\epsilon[\theta^{\phi^t}] \xrightarrow{t\to\infty} 0$.*

Loosely speaking, diffusion converges to a non-empty kernel of $\mathrm{mi}[\theta^{\phi^t}]$.

**Exercise 7.3.** *Show that Algorithm ??, known as MPLP, also can be seen as a block-coordinate ascent.*

## 7.3 Coordinate Descent for Energy Minimization: ICM Algorithm

ICM (Iterated Conditional Modes) is one of first and the simplest algorithms for energy minimization. It is indeed an implementation of the coordinate descent (6.10) to the primal non-relaxed energy minimization objective (1.2). Algorithm starts with some labeling $x \in \mathcal{X}_\mathcal{V}$ and iteratively improves it. Each elementary step of the algorithm consists in selection of the optimal label in a current node given a fixed labeling of all other nodes.

The notation $l(x,u,s) := \left(x' \in \mathcal{X}_\mathcal{V} \colon x'_v = \begin{cases} x_v, & v \neq u \\ s, & v = u \end{cases}\right)$ defines a labeling, where all coordinates but $u$ coincide with the labeling $x$ and the coordinate $u$ has the label $s$.

**Algorithm 5** ICM: Block-Coordinate Descent for $E(x)$

```
Init: x ∈ 𝒳_𝒱
repeat
    for u ∈ |V| do
        x_u := arg min_{s∈𝒳_u} E(l(x, u, s))
    end for
until labeling x stops changing
```

Obviously Algorithm 5 guarantees monotonous non-increasing of energy $E$. However, since function $E$ is neither convex nor differentiable (moreover, it is defined on a discrete set only), it does not guarantee attainment of the optimal value of $E$. Also in practice it usually returns significantly worse labellings than most of other existing techniques.

Block-ICM - select a tree, optimize over it. None of the fixed unary factors must be connected to two nodes of the tree

Mean-Field Inference as smoothed coordinate descent in the primal domain

### 7.4 Further Reading

The dual subgradient algorithm in the described form was proposed in [44]. Other works considering another dual problem are [47, 23, 24].

Although the min-sum diffusion algorithm is known at least from 70's, its detailed convergence analysis was given only recently in [43]. The recent work [19] provides a unified analysis and evaluation of different dual coordinate ascent algorithms.

Lazy Flipper.

# 8 Lagrangian (Dual) Decomposition

## 8.1 General Lagrangian Dual Construction Scheme

Let $f \colon \mathbb{R}^n \to \mathbb{R}$ be a function, which has to be minimized, i.e. $\min_{x\in\mathbb{R}^n} f(x)$. Assume, its minimization is difficult, however, it is representable as a sum of two other functions on the same domain, that is $f(x) = f^1(x) + f^2(x)$. Let also $\mathbb{I}_\infty$ denote a function such that $\mathbb{I}_\infty(A) = 0$ if the condition $A$ is fulfilled and $\infty$ otherwise. Then the following equalities and inequalities hold:

$$\min_x f(x) = \min_x (f^1(x)+f^2(x)) = \min_{x^1=x^2}(f^1(x^1)+f^2(x^2)) = \min_{x^1,x^2}(f^1(x^1)+f^2(x^2)+\mathbb{I}_\infty(x^1=x^2))$$
$$\geq \min_{x^1,x^2}(f^1(x^1)+f^2(x^2)+\lambda(x^1-x^2)) = \min_{x^1}(f^1(x^1)+\lambda x^1) + \min_{x^2}(f^2(x^2)-\lambda x^2) \quad (8.1)$$

## 8.2 Decomposition Based Dual. Tree agreement

In the previous lectures we learned two principally different algorithms: dynamic programming and diffusion. Dynamic programming:

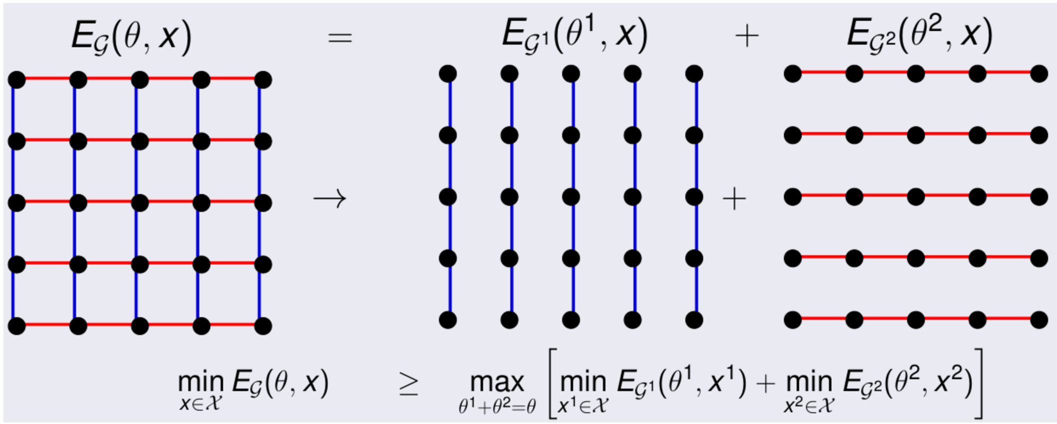


Figure 3: An example of Lagrangean decomposition for grid graphs: sub-graphs formed by columns and rows correspond to separate subproblems

- is a finite step algorithm, linear complexity w.r.t. to the problem size;
- solves the *non-relaxed* energy minimization exactly;
- is limited to acyclic graphs.

Diffusion:

- is an iterative algorithm with unknown complexity (but definitely not polynomially convergent);
- addresses the dual LP and converges to arc-consistency, which is necessary, but not sufficient for optimality;
- is applicable to graphical models with an arbitrary structure.

Moreover, diffusion also converges to the optimum for acyclic graphs, since in this case the arc consistency is sufficient for optimality. However, it requires multiple iterations to attain this condition in general.

**Exercise 8.1** (Difficult)**.** *Let the graph be acyclic and the corresponding energy minimization has a unique minimum. Show that the diffusion algorithm attains the dual optimum in a finite number of iterations.*

Our goal is to construct an algorithm, which would combine advantages of the dynamic programming and diffusion. We will construct such an algorithm, which:

- for acyclic graphs is as efficient as dynamic programming;
- applicable to graphical models with an arbitrary structure;
- addresses the dual LP and converges to arc-consistency.

This algorithm, known as `TRW-S` (**T**ree **Re**weighted message passing **s**equential) (see Kolmogorov, 2006 in the further reading) converges usually significantly faster than diffusion.

We start with a special case of grid graphs to avoid shading the main idea with technical details.

Let $\mathcal{G}$ be a grid graph and $E(\theta, x)$ be the energy of a labeling $x \in \mathcal{X}$. Let us split the graph $\mathcal{G}$ into chain subgraphs, each corresponding to a row or a column of the graph $\mathcal{G}$, see Fig. 3. Energy function of all chains corresponding to columns we denote as $E^c$ and those corresponding to all rows as $E^r$. Let $I^c$ and $I^r$ denote the numbers of primal variables in the corresponding subproblems, similar to $I$ defining it for the *master* problem.

We will split potentials between these two subgraphs so that it guarantees fulfillment of the following equality for all $x \in \mathcal{X}$:

$$E(\theta, x) = E^c(\theta^c, x) + E^r(\theta^r, x)\,, \tag{8.2}$$

where the potentials $\theta^c \in \mathbb{R}^{I^c}$ and $\theta^r \in \mathbb{R}^{I^r}$ are defined as follows:

$$\theta^c_v(x_v) = \frac{\theta_v(x_v)}{2} + \lambda_v(x_v)\,, \qquad \theta^r_v(x_v) = \frac{\theta_v(x_v)}{2} - \lambda_v(x_v) \tag{8.3}$$

$$\theta^c_{uv}(x_u, x_v) = \theta_{uv}(x_u, x_v)\,,\ uv \in \mathcal{E}^c\,, \qquad \theta^r_{uv}(x_u, x_v) = \theta_{uv}(x_u, x_v)\,,\ uv \in \mathcal{E}^r\,. \tag{8.4}$$

Values $\lambda := \{\lambda_v(x_v)\colon v \in \mathcal{V},\ x_v \in \mathcal{X}_v\} \in \mathbb{R}^{\sum_{v\in\mathbb{R}^I} |\mathcal{X}_v|}$ can be arbitrary and in the formulas we will omit the vector space $\mathbb{R}^{\sum_{v\in\mathbb{R}^I} |\mathcal{X}_v|}$ they belong to. We will write $\theta_{[\lambda]}$ to emphasize that potentials $\theta^c$ and $\theta^r$ depend on $\lambda$.

From (8.2) it follows that

$$\min_{x\in\mathcal{X}} E(\theta, x) \geq \min_{x^c\in\mathcal{X}^c} E^c(\theta^c_{[\lambda]}, x^c) + \min_{x^r\in\mathcal{X}^r} E^r(\theta^r_{[\lambda]}, x^r) \tag{8.5}$$

for any $\lambda$. In its turn it implies

$$\begin{aligned}
&\min_{x\in\mathcal{X}} E(\theta, x) \geq \max_{\lambda} \left[\min_{x^c\in\mathcal{X}^c} E^c(\theta^c_{[\lambda]}, x^c) + \min_{x^r\in\mathcal{X}^r} E^r(\theta^r_{[\lambda]}, x^r)\right] \\
&= \max_{\lambda} \left[\min_{x^c\in\mathcal{X}^c} \left\langle \theta^c_{[\lambda]}, \delta(x^c)\right\rangle + \min_{x^r\in\mathcal{X}^r} \left\langle \theta^r_{[\lambda]}, \delta(x^r)\right\rangle\right] = \max_{\lambda} \underbrace{\left[\min_{\mu^c\in\mathcal{L}^c} \left\langle \theta^c_{[\lambda]}, \mu^c\right\rangle + \min_{\mu^r\in\mathcal{L}^r} \left\langle \theta^r_{[\lambda]}, \mu^r\right\rangle\right]}_{U(\lambda)}
\end{aligned} \tag{8.6}$$

Energy minimizations for columns ($E^c$) and rows ($E^r$) are called the *slave (sub)problems*, in contrast to the energy minimization over the whole graph, called a *master problem*. Maximization of $U$ constitutes a dual problem we will called it *tree-reweighted-* or *(tree-)decomposition-based* dual to distinguish it from the Lagrange dual $D(\phi)$ we introduced in Section 5.1.

Since according to (8.3) the potentials $\theta^c_{[\lambda]}$ and $\theta^r_{[\lambda]}$ are linearly depend on $\lambda$, the function $U$ is piecewise linear, concave but **non-smooth**. Note, these are the same properties as those of the LP dual $D(\phi)$ we considered in previous sections.

Note also that the function $U$ can be easily evaluated by dynamic programming due to the fact that both $\mathcal{G}^c$ and $\mathcal{G}^r$ are acyclic. As we will see below its subgradient can be evaluated as easy as its value.

Since our goal is to maximize $U$ we will consider its optimality conditions given by Theorem 6.3. As follows from Lemma 6.1, all subgradients of $U$ have the form:

$$\frac{\partial U}{\partial \lambda_v} = \mu^{c*}_v - \mu^{r*}_v\,, \tag{8.7}$$

where $\mu^{c*} \in \arg\min_{\mu^c\in\mathcal{L}^c} \left\langle \theta^c_{[\lambda]}, \mu^c\right\rangle$ and $\mu^{r*} \in \arg\min_{\mu^r\in\mathcal{L}^r} \left\langle \theta^r_{[\lambda]}, \mu^r\right\rangle$. In particular, vectors with coordinates $\left(\delta(x^{c*})_v - \delta(x^{r*})_v\right)$ are subgradients as well, for $x^{c*}$ and $x^{r*}$ being integer solutions of the slave subproblems.

Since checking existence of zero vector of the form (8.7) is not a trivial task, sufficient conditions were developed. Let $\mathcal{X}_v^c = \left\{x_v^* \colon x^* \in \arg\min_{x\in\mathcal{X}^c} \left\langle \theta^c_{[\lambda]}, \delta(x)\right\rangle\right\}$ and $\mathcal{X}_v^r$ be defined similarly.

**Definition 8.1.** *We will say that there is a* (strong) tree agreement *in the node* $v \in \mathcal{V}$ *if* $\mathcal{X}_v^c = \mathcal{X}_v^r$ *and* $|\mathcal{X}_v^c| = |\mathcal{X}_v^r| = 1$.

**Proposition 8.1.** *Let the strong tree agreement hold in all nodes* $v \in \mathcal{V}$ *for a given* $\lambda$. *Then*

*a)* $\min_{x\in\mathcal{X}} E(\theta, x) = U(\lambda)$*;*

*b) there is a unique* $x^* \in \arg\min_{x\in\mathcal{X}} E(\theta, x)$ *and moreover,*

*c)* $\{x_v^*\} = \mathcal{X}_v^c = \mathcal{X}_v^r$ *for all* $v \in \mathcal{V}$.

*Proof.* Let us denote as $x^*$ the only vector consisting of coordinates fulfilling c) and prove implications a) and b) for it. Item a) follows from (8.2) applied to $x^*$. Let prove b). Let $x' \in \arg\min_{x\in\mathcal{X}} E(\theta, x)$ and $x' \neq x^*$. Then $E^c(\theta^c, x') > E^c(\theta^c, x^*)$ or $E^r(\theta^r, x') > E^r(\theta^r, x^*)$ by definition of $x^*$. Therefore it holds

$$E(\theta, x') \;=\; E^c(\theta^c, x') \;+\; E^r(\theta^r, x') \;>\; E^c(\theta^c, x^*) \;+\; E^r(\theta^r, x^*) \;=\; E(\theta, x^*)\,, \quad (8.8)$$

which is a contradiction. $\square$

In what follows we consider Lagrangian decomposition for arbitrary graphs into arbitrary subgraphs.

## 8.3 General Lagrangian Subgraph Decomposition Scheme for MAP-inference

Let us briefly formulate results of Section 8.2 for general graphs and their general decomposition. Let $\mathcal{T}$ be a collection of subgraphs of the master graph $\mathcal{G}$ such that each edge and each node belong to at least one subgraph of the collection. Let also

For a given subgraph $\mathcal{G}^\tau = (\mathcal{V}^\tau, \mathcal{E}^\tau)$ we define the set

$$I^\tau = \{(w,s) | w \in \mathcal{V} \cup \mathcal{E},\ s \in \mathcal{X}_w\} \quad (8.9)$$

corresponding to the set of indexes associated with vertexes and edges of the subgraph $\mathcal{G}^\tau$.

With each sub-graph $\mathcal{G}^\tau$ we associate a set $\Theta^\tau$ of potentials with coordinates identical to zero outside the set $\mathcal{I}^\tau$:

$$\Theta^\tau = \{\theta^\tau \in \mathbb{R}^I | \theta_w(s)^\tau = 0\ \forall\ (w,s) \in I \backslash I^\tau\}\,. \quad (8.10)$$

We also define the set

$$\Theta^{\mathcal{T}} = \{\theta^{\mathcal{T}} \in \prod_{\tau\in\mathcal{T}} \Theta^\tau | \sum_{\tau\in\mathcal{T}} \theta^\tau = \theta\} \quad (8.11)$$

of all feasible potentials for all subgraphs, whose potentials sum up to the potential of the master graphical model. Let also $\mathcal{M}^\tau := \mathrm{conv}(\delta(\mathcal{X}_{\mathcal{V}^\tau}))$ be the marginal polytope corresponding to the graphical model with the index $\tau$.

Then for any $\theta^{\mathcal{T}} \in \Theta^{\mathcal{T}}$ the following sequence of (in)equalities holds:

$$\min_{x\in\mathcal{X}_\mathcal{V}} E(x) = \min_{x\in\mathcal{X}_\mathcal{V}} \langle\theta, \delta(x)\rangle = \min_{x\in\mathcal{X}_\mathcal{V}} \left\langle \sum_{\tau\in\mathcal{T}} \theta^\tau, \delta(x)\right\rangle$$
$$\geq \sum_{\tau\in\mathcal{T}} \min_{x\in\mathcal{X}_\mathcal{V}} \langle\theta^\tau, \delta(x)\rangle = \sum_{\tau\in\mathcal{T}} \min_{\mu\in\mathcal{M}^\tau} \langle\theta^\tau, \mu\rangle =: U(\theta^{\mathcal{T}}) \quad (8.12)$$

The function $U$ is the Lagrange decomposition-based dual. It is a direct generalization of the Lagrange decomposition-based dual constructed in (8.5) for grid graphs and their decomposition into columns and rows. The feasible set $\Theta^{\mathcal{T}}$ is obviously convex, the dual objective $U$ is concave as a sum of minima of linear functions. Therefore finding its maximal value, which constitutes the best lower bound of the form (8.12) is a concave (non-smooth) maximization problem:

$$\max_{\theta^{\mathcal{T}} \in \Theta^{\mathcal{T}}} U(\theta^{\mathcal{T}})\,. \tag{8.13}$$

In the following, we briefly generalize the results obtained in Section 8.2.

**Proposition 8.2.** *The set of potentials $\theta^{\mathcal{T}} \in \Theta^{\mathcal{T}}$ is a solution to the dual problem (8.13) iff for all $\tau \in \mathcal{T}$ there exist $\mu^\tau \in \arg\min_{\mu \in \mathcal{M}^\tau} \langle \theta^\tau, \mu \rangle$ such that $\mu^\tau_w(s) = \mu^\sigma_w(t)$ for all $\tau, \sigma \in \mathcal{T}$ and $(w,s),(w,t) \in I^\tau \cap I^\sigma$. Moreover, for the vector $\mu^* \in \mathbb{R}^I$ such that $\mu^*_w = \mu^\tau_w$ it holds $\langle \theta, \mu^* \rangle = U(\theta^{\mathcal{T}})$.*

*Proof.* Let us construct the Lagrangian for the constraint minimization problem (8.13). It reads:

$$L(\theta^{\mathcal{T}}, \lambda, \nu) = U(\theta^{\mathcal{T}}) + \left\langle \lambda, \sum_{\tau \in \mathcal{T}} \theta^\tau - \theta \right\rangle + \sum_{\tau \in \mathcal{T}} \sum_{(w,s) \in I \setminus I^\tau} \nu^\tau_w(s) \theta^\tau_w(s) \tag{8.14}$$

for suitably selected dimensions of vectors $\lambda$ and $\nu$. Let us apply the KKT optimality conditions to the problem (8.13). As soon as $\theta^{\mathcal{T}} \in \Theta^{\mathcal{T}}$ the primal and dual feasibility conditions hold ($\lambda$ and $\nu$ are unrestricted), therefore the concavity of the problem (8.13) implies that existence of a zero sub-gradient of the Lagrangian w.r.t. $\theta^{\mathcal{T}}$ is necessary and sufficient for optimality. Indeed, for $(w,s) \in I^\tau$ this subgradient reads:

$$\frac{\partial L}{\partial \theta^\tau_w(s)} = \mu^\tau_w(s) - \lambda^\tau_w(s)\,. \tag{8.15}$$

Therefore $\mu^\tau_w(s) = \lambda^\tau_w(s)$, which implies the conjecture of the proposition and finalizes the proof. ☐

Let $\mathcal{X}^\tau_v$ be the set of labels belonging to some optimal labeling, defined like in Section 8.2:

$$\mathcal{X}^\tau_v = \left\{ x^*_v \mid x^* \in \arg\min_{x \in \mathcal{X}_{\mathcal{V}^\tau}} \langle \theta^\tau, \delta(x) \rangle \right\} \tag{8.16}$$

**Definition 8.2.** *We will say that there is a* (strong) subproblem agreement *in the node $v \in \mathcal{V}$ if $\mathcal{X}^\tau_v = \mathcal{X}^\sigma_v$ for all $\tau, \sigma \in \mathcal{T}$ and $|\mathcal{X}^\tau_v| = 1$ for all $\tau \in \mathcal{T}$ such that $v \in \mathcal{V}^\tau$.*

**Proposition 8.3.** *Let the strong subproblem agreement hold in all nodes $v \in \mathcal{V}$ for a given $\theta^{\mathcal{T}} \in \Theta^{\mathcal{T}}$. Then*

*a) $\min_{x \in \mathcal{X}} E(x) = U(\theta^{\mathcal{T}})$;*

*b) there is a unique $x^* \in \arg\min_{x \in \mathcal{X}} E(x)$ and moreover,*

*c) $\{x^*_v\} = \mathcal{X}^\tau_v$ for all $v \in \mathcal{V}$ and $\tau \in \mathcal{T}$ such that $v \in \mathcal{V}^\tau$.*

The proofs of Proposition 8.3 is similar to those for Proposition 8.1.
We finalize this section with

**Lemma 8.1.** *Let $\mathcal{T}$ and $\mathcal{T}'$ be two decompositions and $\sigma \in \mathcal{T}$. If $\mathcal{G}^{\mathcal{T}'} = \mathcal{G}^{\mathcal{T}} \cup \{\mathcal{G}^\tau\}$ then*

$$\max_{\theta^{\mathcal{T}} \in \Theta^{\mathcal{T}}} U(\theta^{\mathcal{T}}) = \max_{\theta^{\mathcal{T}'} \in \Theta^{\mathcal{T}'}} U(\theta^{\mathcal{T}'}) \tag{8.17}$$

*Proof.* The idea of the proof is based on Proposition 8.2. Let $\tau' \in \mathcal{T}'$ be an index such that $\mathcal{G}^{\tau'} = \mathcal{G}^{\sigma}$. The expressions

$$U(\theta^{\mathcal{T}'}) = \sum_{\tau \in \mathcal{T}'} \min_{\mu \in \mathcal{M}^{\tau}} \langle \theta^{\tau}, \mu \rangle \tag{8.18}$$

and

$$U(\theta^{\mathcal{T}}) = \sum_{\tau \in \mathcal{T}} \min_{\mu \in \mathcal{M}^{\tau}} \langle \theta^{\tau}, \mu \rangle \tag{8.19}$$

differ only by a single term $\min_{\mu \in \mathcal{M}^{\sigma}} \langle \theta^{\sigma}, \mu \rangle$, which appears in (8.18) twice (with indexes $\sigma$ and $\tau'$). Due to Proposition 8.2 there is a minimizer $\mu^* \in \mathcal{M}^{\sigma}$ of this expression, moreover, it is the same for both its copies having potentials $\theta^{\sigma}$ and $\theta^{\tau'}$. Therefore

$$\min_{\mu \in \mathcal{M}^{\tau'}} \left\langle \theta^{\tau'}, \mu \right\rangle + \min_{\mu \in \mathcal{M}^{\sigma}} \langle \theta^{\sigma}, \mu \rangle = \min_{\mu \in \mathcal{M}^{\sigma}} \left\langle (\theta^{\sigma} + \theta^{\tau'}, \mu \right\rangle .$$

Introducing the new potential $\theta^{\sigma} := \theta^{\sigma} + \theta^{\tau'}$ turns expression (8.18) to (8.19), which finalizes the proof. □

**Corollary 8.1.** *Adding subgraphs, which already exist in a decomposition, does not change the optimal lower bound.*

## 8.4 Equivalence of the LP- and Tree-Decomposition-Based Duals

Let us now consider all possible decompositions to acyclic sub-problems. In the following, we will show that all of them are equivalent. This means, the corresponding maximal lower bounds defined by (8.13) coincide and are equal to the bound provided by the local polytope relaxation.

### 8.4.1 A Complete Decomposition

**Definition 8.3.** *The decomposition $\mathcal{G}^{\tau} = (\mathcal{V}^{\tau}, \mathcal{V}^{\tau})$, $\tau \in \mathcal{T}$ is called* complete*, if $\mathcal{T} = \mathcal{V} \cup \mathcal{E}$ and $\mathcal{E}^u = \emptyset$ for all $u \in \mathcal{V}$, $\mathcal{E}^{uv} = \{uv\}$ for $uv \in \mathcal{E}$. That is, each node of the master graph forms a sub-graph in the decomposition as well as each edge with its two incident nodes.*

After introducing the auxiliary variables $\phi_{u,v}(s)$, $s \in \mathcal{X}_u$ for all $uv \in \mathcal{E}$, the constraints $\theta^u_u + \sum_{uv \in \mathcal{E}} \theta^{uv}_u = \theta_u$ defining the feasible set $\Theta^{\mathcal{T}}$ (8.11) can be written explicitly as

$$\theta^{uv}_u(s) := \phi_{u,v}(s), \tag{8.20}$$

$$\theta^{u}_u(s) := \theta_u(s) - \sum_{v \in \mathcal{N}_b(u)} \phi_{u,v}(s),\ s \in \mathcal{X}_u\,. \tag{8.21}$$

Applying a reparametrization in each edge-based subgraph $\mathcal{G}^{uv}$ such that the weight of its unary factors is moved to the weight of its edges one obtains

$$\theta^{uv}_{uv}(s,t) = \theta^{uv}_{uv}(s,t) + \theta^{uv}_u(s) + \theta^{uv}_v(t) = \theta^{uv}_{uv}(s,t) + \phi_{u,v}(s) + \phi_{v,u}(t),\ st \in \mathcal{X}_{uv} \tag{8.22}$$

$$\theta^{u}_u(s) = \theta_u(s) - \sum_{v \in \mathcal{N}_b(u)} \phi_{u,v}(s),\ s \in \mathcal{X}_u\,, \tag{8.23}$$

which is exactly the reparametrization $\theta^{\phi}$ (5.9) based on the Lagrange duality. It is therefore straightforward to see that the Lagrangian-decomposition-based dual $U$ (8.12) as a function of $\phi$ coincides with the unconstrained Lagrange dual $D(\phi)$ (5.11).

This implies the following corollaries:

- The maximum value of the dual $U$ for the complete decomposition is equal to the minimum value of the local polytope relaxation for the master problem.
- The complete decomposition is tight for acyclic graphs.

Indeed, it also implies a simple proof of the following important fact:

**Theorem 8.1.** *All acyclic decompositions are equivalent, which is, the maxima of the corresponding duals are equal to the minumum value of the local polytope relaxation.*

*Proof.* An idea of the proof could be based on Corollary 8.1. First consider a decomposition to arbitrary acyclic subgraphs and then decompose each of them completely. The resulting decomposition may contain multiple copies of the node-wise and edge-wise subproblems, however in total it gives the same optimal lower bound as the complete decomposition of the master problem. Moreover, since the complete decomposition is tight for acyclic graphs, it does not change the bound of the initial acyclic decomposition. On the other side, the complete decomposition has an optimal value equal to the optimal value of the local polytope relaxation of the master problem. This would finalize the proof.

We also give a different proof based on primal arguments. According to (8.12) and since the local and marginal polytopes coincide for acyclic graphs, it holds:

$$U(\theta^{\mathcal{T}}) = \sum_{\tau\in\mathcal{T}} \min_{\mu\in\mathcal{M}^{\tau}} \langle\theta^{\tau},\mu\rangle = \sum_{\tau\in\mathcal{T}} \min_{\mu\in\mathcal{L}^{\tau}} \langle\theta^{\tau},\mu\rangle \tag{8.24}$$

According to Proposition 8.2 in the optimum of $U$ there exist such relaxed labelings $\mu^{\tau} \in \arg\min_{\mu\in\mathcal{L}^{\tau}} \langle\theta^{\tau},\mu\rangle$ that $\mu^{\tau}_w(s) = \mu^{\sigma}_w(t)$ for all $\tau,\sigma$, which denote graphs containing the factor $w \in \mathcal{V}\cup\mathcal{E}$.

Let us construct $\mu^* \in \mathbb{R}^I$ by assigning $\mu^*_w(s) := \mu^{\tau}_w(s)$ for arbitrary $\tau \in \mathcal{T}(w)$. Since all $\mu^{\tau} \in \mathcal{L}^{\tau}$ it follows that $\mu^* \in \mathcal{L}$. Therefore for an arbitrary feasible optimizer $\theta^{\mathcal{T}}$ of $U$ it holds:

$$\max_{\nu^{\mathcal{T}}\in\Theta^{\mathcal{T}}} U(\nu^{\mathcal{T}}) = \sum_{\tau\in\mathcal{T}} \min_{\mu\in\mathcal{L}^{\tau}} \langle\theta^{\tau},\mu\rangle = \min_{\substack{\mu^{\tau}\in\mathcal{L}^{\tau},\tau\in\mathcal{T}\\ \mu^{\tau}=\mu^*|_{\mathcal{G}^{\tau}}}} \sum_{\tau\in\mathcal{T}} \langle\theta^{\tau},\mu^{\tau}\rangle = \min_{\substack{\mu^{\tau}\in\mathcal{L}^{\tau},\tau\in\mathcal{T}\\ \mu^{\tau}=\mu^*|_{\mathcal{G}^{\tau}}}} \left\langle \sum_{\tau\in\mathcal{T}} \theta^{\tau},\mu^{\tau}\right\rangle$$
$$= \min_{\mu^*\in\mathcal{L}} \langle\theta,\mu^*\rangle \tag{8.25}$$

□

Transformation between different dual variables and different dual values.
Any graph decomposition is at least as tight as an acyclic one.

## 8.5 Algorithms for Optimization of $U$.

### 8.5.1 Subgradient Method.

The subgradient method is implemented in Algorithm 6. On each iteration the optimal subgraphs' labelings $x^c$ and $x^r$ are "punished" (their weight is increased, which can eventually make them non-optimal), if they are not coinciding in a given node $v \in \mathcal{V}$.

### 8.5.2 Block-Coordinate Ascent

Let $E^c_v(x_v)$ and $E^r_v(x_v)$ denote energies of the optimal column-wise and row-wise labelings containing the label $x_v$ in the node $v \in \mathcal{V}$.

Let us consider the following update of the variable $\lambda_v$:

$$\lambda_v(x_v) = \lambda_v(x_v) + \frac{E^r_v(x_v) - E^c_v(x_v)}{2} \tag{8.26}$$

**Algorithm 6** Subgradient method for $U(\lambda)$

**Init:** $\lambda^0$
**for** $t = 1$ **to** $N$ **do**
  $x^c = \arg\min_{x\in\mathcal{X}^c} E^c(\theta^c_{\lambda^t}, x)$
  $x^r = \arg\min_{x\in\mathcal{X}^r} E^c(\theta^r_{\lambda^t}, x)$
  $\lambda^{t+1}_v = \lambda^t_v + \alpha^t\,(\delta(x^c)_v - \delta(x^r)_v)$, $v \in \mathcal{V}$
**end for**
**return** $\lambda^{N+1}$

**Lemma 8.2.** *Let $v \in \mathcal{V}$ be some node. Let $E^c_v(x_v)$ and $E^r_v(x_v)$ correspond to the variable $\lambda$ and $E'^c_v(x_v)$ and $E'^r_v(x_v)$ – to $\lambda'$, where $\lambda'_u = \begin{cases}\lambda_u, & u \neq v\\ \lambda_v + \frac{E^r_v - E^c_v}{2}, & u = v\end{cases}$. Then $E'^c_v(x_v) = E'^r_v(x_v)$*

*Proof.* Note that

$$E'^c_v(x_v) = E^c_v(x_v) + \frac{E^r_v(x_v) - E^c_v(x_v)}{2} = \frac{E^r_v(x_v) + E^c_v(x_v)}{2} \tag{8.27}$$

Similarly

$$E'^r_v(x_v) = E^r_v(x_v) - \frac{E^r_v(x_v) - E^c_v(x_v)}{2} = \frac{E^r_v(x_v) + E^c_v(x_v)}{2}\,, \tag{8.28}$$

which finalizes the proof. □

**Proposition 8.4.** *Update* (8.26) *performed for some $v \in \mathcal{V}$ and all $x_v \in \mathcal{X}_v$ is a block-coordinate ascent step for the decomposition dual $U$ w.r.t. the variable(s) $\lambda_v = \{\lambda_v(x_v) \colon x_v \in \mathcal{X}_v\}$.*

*Proof.* Let us assume the update (8.26) is done. Due to Lemma 8.2 the set $A := \mathcal{X}^r_v \cap \mathcal{X}^c_v$ is not empty. Let $s \in A$. By definition of $\mathcal{X}^r_v$ and $\mathcal{X}^c_v$ there are such $x^c \in \arg\min_{x\in\mathcal{X}} \left\langle \theta^c_{[\lambda]}, \delta(x)\right\rangle$ and $x^r \in \arg\min_{x\in\mathcal{X}} \left\langle \theta^r_{[\lambda]}, \delta(x)\right\rangle$ that $x^r_v = x^c_v = s$. Note that $\frac{\partial U}{\partial \lambda_v} = x^c_v - x^r_v = 0$, which proves that $\lambda$ is optimal w.r.t. the coordinate(s) $\lambda_v$. □

Note that $U(\lambda) = U(\lambda')$, where $\lambda'_u(s) = \lambda_u(s) + c_u$ for arbitrary constants $c_u$, $u \in \mathcal{V}$. Therefore the coordinate ascent rule (8.26) can be exchanged with:

$$\lambda_v(x_v) = \lambda_v(x_v) + \frac{\bar{E}^r_v(x_v) - \bar{E}^c_v(x_v)}{2} \tag{8.29}$$

where $\bar{E}^r_v(x_v)$ is the energy of an optimal labeling of the *single* row containing the node $v$ and label $x_v$ and hence $\bar{E}^r_v(x_v) = E^r_v(x_v) - c_v$.

### 8.5.3 Grid-Graphs. TRW-S Algorithm as Block-Coordinate Ascent

Speeding-up the block-coordinate ascent algorithm 7

For the sake of notation we denote $\bar{E}^c_v$ as $E^c_{i,j}$, where $(i, j)$ are coordinates of $v \in \mathcal{V}$. Similar notation is adopted to $\bar{E}^r_v$. Let also $(w, h)$ denote the width and height of the considered grid graph. The values $E^r_{i,j}(s)$, $s \in \mathcal{X}_{i,j}$ can be computed by the dynamic programming (2.5), which recursively computes

$$F_1 \equiv 0, \quad F_i(s) := \min_{t\in\mathcal{X}_{i-1}} \left[F_{i-1}(t) + \theta_{i-1,i}(s,t) + \theta_{i-1}(t)\right] \quad \text{for } i = 1 \to h \tag{8.30}$$

**Algorithm 7** Principal coordinate ascent on $U$

**Init:** $\lambda^0$
**for** $t = 1$ **to** $N$ **do**
  **for** $v \in \mathcal{V}$ **do**
    **for** $x_v \in \mathcal{X}_v$ **do**
      $\lambda_v^t(x_v) = \lambda_v^{t-1}(x_v) + \frac{\bar{E}^r_{[\lambda^{t-1}],v}(x_v) - \bar{E}^c_{[\lambda^{t-1}],v}(x_v)}{2}$
    **end for**
  **end for**
**end for**
**return** $\lambda^N$

and

$$B_h \equiv 0, \quad B_i(s) := \min_{t \in \mathcal{X}_{i-1}} \left[ F_{i-1}(t) + \theta_{i-1,i}(s,t) + \theta_{i-1}(t) \right] \quad \text{for } i = h \to 1\,. \tag{8.31}$$

The resulting *min-marginals* are obtained as

$$m_i(s) = F_i(s) + \theta_i(s) + B_i(s)\,. \tag{8.32}$$

For the sake of notation we omitted the row index $j$ above. Note also that notation in (8.30)-(8.32) slightly differs from the one in (2.5).

Taking (8.32) into account, the computational formula for $E^c_{i,j}$ and $E^r_{i,j}$ reads:

$$E^c_{i,j}(s) = F^c_{i,j}(s) + \theta_{i,j}(s) + B^c_{i,j}(s) \tag{8.33}$$

$$E^r_{i,j}(s) = F^r_{i,j}(s) + \theta_{i,j}(s) + B^r_{i,j}(s). \tag{8.34}$$

Note that $F^r_{i,j}$ can be computed from $F^r_{i-1,j}$ by the number of operations independent on the size of the graph. The same holds for computing of $B^r_{i,j}$ from $B^r_{i+1,j}$, $F^c_{i,j}$ from $F^c_{i,j-1}$ and $B^c_{i,j}$ from $B^c_{i,j+1}$.

Putting all this together, we can rewrite one iteration of Algorithm 7 in the following recursive fashion:

**Algorithm 8** Efficient implementation of a single iteration of Algorithm 7

**Input:** $\lambda^{t-1}$, $B^c$, $B^r$
**for** $j = 1$ **to** $h$ **do**
  **for** $i = 1$ **to** $w$ **do**
    **for** $s \in \mathcal{X}_v$ **do**
      Compute $F^c_{i,j}(s)$ from $F^c_{i,j-1}$
      Compute $F^r_{i,j}(s)$ from $F^r_{i-1,j}$
      Compute $E^c_{i,j}(s)$ and $E^r_{i,j}(s)$ as in (8.33) as functions of $F^c, F^r, B^c, B^r$
      $\lambda^t_{i,j}(s) = \lambda^{t-1}_{i,j}(s) + \frac{\bar{E}^r_{i,j}(s) - \bar{E}^c_{i,j}(s)}{2}$
    **end for**
  **end for**
**end for**
**return** $\lambda^t$

Note that the complexity of the block-coordinate ascent w.r.t. *all* coordinates of the dual vector $\lambda$ by Algorithm 8 only linearly depends on the size of the graph. This is the complexity of the block-coordinate ascent step with respect to the coordinates corresponding to a *single* node $v$ in Algorithm 7.

Note that

- after the run of Algorithm 8 the values of the arrays $F^c$ and $F^r$ are computed;
- inverting the processing order one can compute arrays $B^c$ and $B^r$ based on given arrays $F^c$ and $F^r$.

Therefore the whole cyclic block-coordinate ascent looks as follows:

**Algorithm 9** Cyclic block-coordinate ascent algorithm (TRW-S)

**Input:** $\lambda^0$, $B^c$, $B^r$
**for** $t = 1$ **to** $N$ **do**
  Obtain $\lambda^t$ and $F^c, F^r$ by Algorithm 8.
  Invert order of variables: $i \leftarrow w - i + 1$, $j \leftarrow h - j + 1$, $B^c \leftarrow F^c$, $B^r \leftarrow F^r$
**end for**
**return** $\lambda^t$

Initial values of the arrays $B^c$ and $B^r$ can be either computed by dynamic programming or just have a zero initialization. In the latter case the first iteration of the algorithm does not perform a block-coordinate ascent and may decrease the value of the dual objective.

## 8.6 Further Reading

Key words: dual or Lagrangian decomposition, tree-reweighted message passing.

- ...

# 9 Min-Cut/Max-Flow Based Inference

So far the only special case, when the energy minimization problem was solvable exactly: acyclic models (dynamic programming does the job in this case). We know already that this is due to the fact, that the local polytope is equal to the marginal polytope in this case and therefore, all its vertices correspond to integer solutions. There is, however, another way to force integrality: restrict the type of potentials used and consider only those of them, which guarantee integrality of the solution. Loosely speaking, the vector $(-\theta)$ should point to integer vertices of the local polytope. Such sufficient conditions exist and they are called *submodularity conditions*. Submodular problems is a discrete analog to convex problems and are solvable polynomially in general.

## 9.1 Submodular Set-Functions

Let $\mathcal{V}$ be a discrete set and $2^{\mathcal{V}}$ be its powerset.

**Definition 9.1.** *A set-function* $E\colon 2^{\mathcal{V}} \to \mathbb{R}$ *is called* submodular *if*

$$E(A) + E(B) \geq E(A \cap B) + E(A \cup B)\ \forall A, B \in 2^{\mathcal{V}}, . \tag{9.1}$$

The sub-modular minimization problem consists in computing

$$A^* = \arg\min_{A \in 2^{\mathcal{V}}} E(A)\,, \tag{9.2}$$

where the set-function $E$ is submodular. This problem is known to be polynomially solvable, although the order of the polynom is quite high (6?). Different subclasses of submodular functions may have lower complexity. One of such sub-classes, namely, submodular pairwise energy minimization problems we will consider below.

Note that any set $A \in 2^{\mathcal{V}}$ can be represented as a binary vector with $|\mathcal{V}|$ coordinates indexed by elements of $\mathcal{V}$. Its coordinates equal to 1 define an $A \subset \mathcal{V}$. Therefore, set-functions can be identified with the functions $2^{|\mathcal{V}|} \to \mathbb{R}$ of binary vectors. Binary energy of a graphical model is an example of such functions.

Similarly to convex/concave functions, the function $f$ is called *supermodular*, if $(-f)$ is submodular.

## 9.2 Submodular Energies

In what follows we will assume that the sets $\mathcal{X}_v$, $v \in \mathcal{V}$, of labels are completely ordered. This implies that for any $s, t \in \mathcal{X}_v$ their maximum and minimum, denoted as $s \vee t$ and $s \wedge t$ respectively, are well-defined. Moreover, we will assume operations $+1$ and $-1$ defined on the label set, as soon as they results remain inside the set. These operations return the next and previous elements in the defined order.

Energy $E$ is called *submodular* if for any two labelings $x^1, x^2 \in \mathcal{X}_{uv}$ it holds:

$$E(x^1) + E(x^2) \geq E(x^1 \vee x^2) + E(x^1 \wedge x^2)\,. \tag{9.3}$$

Note that in the case of binary label sets, which is $\mathcal{X}_v = \{0, 1\}$ for $v \in \mathcal{V}$, the definition (9.3) is equivalent to Definition 9.2. The multilable case corresponds to considering submodular functions on a distributive lattice (a partially ordered set with operations $\vee$ and $\wedge$ satisfying the distributive law). It is known that any distributive lattice is isomorphic to the lattice of sets with operations $\cap$ and $\cup$ [12].

**Definition 9.2.** *Function $E\colon \mathcal{X}_{\mathcal{V}} \to \mathbb{R}$ is called* modular*, if*

$$E(x^1) + E(x^2) \geq E(x^1 \vee x^2) + E(x^1 \wedge x^2) \tag{9.4}$$

**Proposition 9.1.** *Function $E\colon \mathcal{X}_{\mathcal{V}} \to \mathbb{R}$ is modular, if it is representable as a sum of unary factors*

$$E(x) = \sum_{v \in \mathcal{V}} \theta_v(x_v) \tag{9.5}$$

The sufficiency of the implication of Proposition 9.1 follows from the two following lemmas:

**Lemma 9.1.** *Function of a single discrete variable, e.g. $\theta_v\colon \mathcal{X}_{\mathcal{V}} \to \mathbb{R}$, is modular, which can straightforwardly follows from Definition 9.2 considered with $|\mathcal{V}| = 1$.*

**Lemma 9.2.** *If $E^1$ and $E^2$ are (sub-)modular, their sum $E^1 + E^2$ is (sub-)modular.*

We leave the proof of the necessity as an exercise to a reader.

Let us consider the case $|\mathcal{V}| = 2$. Then the potential $\theta_{uv}$ is submodular if for any $x^1, x^2 \in \mathcal{X}_{uv}$ it holds

$$\theta_{uv}(x^1) + \theta_{uv}(x^2) \geq \theta_{uv}(x^1 \vee x^2) + \theta_{uv}(x^1 \wedge x^2)\,. \tag{9.6}$$

Picture!

In the binary case, when $\mathcal{X}_v = \{0, 1\}$, $v \in \mathcal{V}$, the submodularity condition (9.6) reads:

$$\theta_{uv}(0,1) + \theta_{uv}(1,0) \geq \theta_{uv}(0,0) + \theta_{uv}(1,1)\,. \tag{9.7}$$

Easy to see that in this case each factor is either submodular or supermodular.

**Example 9.1** (Ising model)**.** *Let the energy function be binary and $\theta_{uv}(s,t) = \lambda_{uv}[\![s \neq t]\!]$ for some constants $\lambda_{uv}$. If $\lambda_{uv} \geq 0$, the corresponding potential is submodular, otherwise – supermodular. In the following, we will show that using reparametrization any pairwise binary energy can be transformed into the form of the Ising model. Moreover, submodular pairwise factors are transformed into submodular with $\lambda_{uv} \geq 0$ and supermodular –into supermodular with $\lambda_{uv} \leq 0$.*

The formula (9.6) can be also written explicitly: For all $x_u^1, x_u^2 \in \mathcal{X}_u$ and all $x_v^1, x_v^2 \in \mathcal{X}_v$ such that $x_u^1 \leq x_u^2$ and $x_v^1 \leq x_v^2$, it holds

$$\theta_{uv}(x_u^1, x_v^1) + \theta_{uv}(x_u^2, x_v^2) \leq \theta_{uv}(x_u^2, x_v^1) + \theta_{uv}(x_u^1, x_v^2)\,. \tag{9.8}$$

**Example 9.2** (Potts model)**.** *The Potts model is the generalization of the Ising model into the multilabel case. That is, the pairwise factors have the same format as in the Ising model $\theta_{uv}(s,t) = \lambda_{uv}[\![s \neq t]\!]$, however the label sets $\mathcal{X}_u$ and $\mathcal{X}_v$ may have more than two elements. The Potts model is not submodular even for $\lambda > 0$, because already for 3 labels it holds*

$$\theta(1,2) + \theta(2,3) = 2\lambda \geq \lambda = \theta(1,3) + \theta(2,2)\,. \tag{9.9}$$

**Exercise 9.1** (Convex functions)**.** *Let the labels sets $\mathcal{X}_v$ and $\mathcal{X}_u$ be equal to $\{1, 2 \ldots, n\}$, i.e. be subsets of integer numbers. Let the subtraction operation "$-$" be defined in a natural way. Let $f\colon \mathbb{R} \to \mathbb{R}$ be a convex function and $\theta_{uv}(s,t) = f(|s-t|)$. Prove that $\theta_{uv}$ is submodular. In particular, the following functions are submodular:*

$$\theta_{uv}(s,t) = |s-t| \tag{9.10}$$

$$\theta_{uv}(s,t) = (s-t)^2\,. \tag{9.11}$$

**Exercise 9.2** (Truncated convex)**.** *Show that* **neither**

$$\theta_{uv}(s,t) = \min\{a, |s-t|\},\ a > 0$$

**nor**

$$\theta_{uv}(s,t) = \min\{a, (s-t)^2\},\ a > 0$$

*are submodular in multilabel case.*

Due to Lemmas 9.1 and 9.2, the submodularity of energy follows from the submodularity of its non-unary potentials, since the unary potentials a always modular. The inverse implication is less trivial:

**Proposition 9.2** (Reference or proof)**.** *Let the energy $E$ be defined as in* (1.2)*, i.e. contains only unary and pairwise potentials. Then, if $E$ is submodular, all its pairwise potentials $\theta_{uv}$, $uv \in \mathcal{E}$ are submodular as well.*

**Remark 9.1.** *Proposition 9.2 does not hold for higher order models. In most cases, however, energies with submodular potentials can be minimized more efficiently than general submodular energies, since the latter class coincides with the class of all submodular set-functions.*

Written as in (9.8) the submodularity definition requires $O(|\mathcal{X}_{uv}|^2)$ operations to check, which can be computationally expensive for big label sets. Indeed, an equivalent definition provided by the following theorem allows to check submodularity in linear time $O(|\mathcal{X}_{uv}|)$:

**Proposition 9.3.** *The potential $\theta_{uv}$ is submodular iff it holds*

$$\theta_{uv}(x_u, x_v) + \theta_{uv}(x_u+1, x_v+1) \leq \theta_{uv}(x_u+1, x_v) + \theta_{uv}(x_u, x_v+1)\,. \tag{9.12}$$

*for $x_u \in \{1, \ldots, |\mathcal{X}_u| - 1\}$ and $x_v \in \{1, \ldots, |\mathcal{X}_v| - 1\}$.*

The technique, used in the following proof, plays an important role for this section and will also be used several times later on. The following lemma constitutes the core of the technique:

**Lemma 9.3.** *Let $\theta\colon \mathbb{Z}^2 \to \mathbb{R}$ be a function of two discrete variables. Then*

$$\theta(x^2, z^2) + \theta(x^1, z^1) - \theta(x^1, z^2) - \theta(x^2, z^1) = \sum_{s=x^1}^{x^2-1} \sum_{t=z^1}^{z^2-1} \frac{\partial^2 \theta(s,t)}{\partial s \partial t}\,, \tag{9.13}$$

*where $x^2 > x^1$, $z^2 > z^1$ and $\frac{\partial^2 \theta(s,t)}{\partial s \partial t} := \theta(s+1, t+1) + \theta(s,t) - \theta(s, t+1) - \theta(s+1, t)$.*

*Proof.* Essentially, the proof is based on the following fact: for a function $f\colon \mathbb{Z} \to \mathbb{R}$ and $a > b$ it holds

$$f(a) - f(b) = \sum_{i=b}^{a-1} f(i+1) - f(i)\,, \tag{9.14}$$

which in continuous case reads

$$f(a) - f(b) = \int_b^a df = \int_b^a f'(t) dt\,. \tag{9.15}$$

Here, the derivative $f'$ is represented by its discrete analog $\frac{f(t+dt)-f(t)}{dt}$ with $dt = 1$.

Similar holds, indeed, also for a function of two variables:

$$f(a,b) + f(a',b') - f(a',b) - f(a,b') = \int_a^{a'} \int_b^{b'} \frac{\partial^2 f(s,t)}{\partial s \partial t} ds dt\,. \tag{9.16}$$

Nevertheless, we provide the rigorous proof below.

Let us consider the differences $\theta(x^2, z^1) - \theta(x^1, z^1)$ and $\theta(x^2, z^2) - \theta_{uv}(x^1, z^2)$. Comparing them to (9.14) one obtains

$$\theta(x^2, z^1) - \theta(x^1, z^1) = \sum_{s=x^1}^{x^2-1} \theta(s+1, z^1) - \theta(s, z^1)\,, \tag{9.17}$$

$$\theta(x^2, z^2) - \theta(x^1, z^2) = \sum_{s=x^1}^{x^2-1} \theta(s+1, z^2) - \theta(s, z^2)\,. \tag{9.18}$$

Subtracting the second equality from the first one, we obtain

$$\theta(x^2, z^2) + \theta(x^1, z^1) - \theta(x^1, z^2) - \theta(x^2, z^1) = \sum_{s=x^1}^{x^2-1} \theta(s+1, z^2) + \theta(s, z^1) - \theta(s, z^2) - \theta(s+1, z^1)\,. \tag{9.19}$$

The similar procedure w.r.t. each summand of the right-hand-side of (9.19) gives:

$$\theta(s+1, z^2) + \theta(s, z^1) - \theta(s, z^2) - \theta(s+1, z^1) = \sum_{t=z^1}^{z^2-1} \theta(s+1, t+1) + \theta(s,t) - \theta(s, t+1) - \theta(s+1, t)\,. \tag{9.20}$$

Plugging (9.20) into (9.19) one obtains the implication of the lemma. □

To prove Proposition 9.3, first note that (9.12) follows directly from (9.8), since the former defines a subset of the latter one. The inverse implication follows from Lemma 9.3: the non-negativity of the left-hand-side of (9.13) follows from the non-negativity of each summand in the right-hand-side.

Note that the submodularity condition (9.12) can be expressed as $\frac{\partial^2 \theta_{uv}(s,t)}{\partial s \partial t} \leq 0$.

It can be shown that the same necessary and sufficient submodularity condition holds also for a submodular (potentially higher-order) energy $E$:

**Theorem 9.1** ([27])**.** *$E$ is submodular iff*

$$\frac{\partial^2 E(x)}{\partial x_i \partial x_j} := E(s_1, \dots, s_i, \dots, s_j, \dots, s_n) + E(s_1, \dots, s_i + 1, \dots, s_j + 1, \dots, s_n) \\ - E(s_1, \dots, s_i + 1, \dots, s_j, \dots, s_n) - E(s_1, \dots, s_i, \dots, s_j + 1, \dots, s_n) \leq 0 \quad (9.21)$$

*for any two neighboring nodes $i$ and $j$.*

**Properties of Submodular Potentials**

**Proposition 9.4** (straightforward proof)**.** *Let $\theta^\phi$ be a reparametrization of $\theta$. Then $\theta$ is submodular iff $\theta^\phi$ is submodular.*

**Theorem 9.2** (proof)**.** *Let $E$ be submodular. Then arc-consistency of* $\mathrm{mi}[\theta^\phi]$ *implies existence of the integer labeling $x \in \mathcal{X}$ such that $E(\theta, x) = D(\phi)$, i.e. optimality of $\phi$ and tightness of the LP relaxation.*

Theorem 9.2 implies that submodular problems are solvable *exactly* with such algorithms as diffusion and TRW-S. In particular, it implies that the local polytope relaxation is tight for such problems. In what follows we will show even more powerful tools for these problems.

**Permuted Submodularity** Submodularity is the property dependent on the order of labels. The order, however, does not play any role for energy itself.

**Definition 9.3.** *Problem defined by potentials $\theta$ is called* permuted submodular *if there exists an ordering on each label set $\mathcal{X}_v$, $v \in \mathcal{V}$ such that the corresponding problem is submodular.*

**Example 9.3.** *Binary problem on a tree is always permuted submodular. One should traverse the tree and change the order of labels if needed to switch from supermodular potential to submodular.*

**Example 9.4.** *Binary supermodular problems on bipartite graphs (e.g. 2D grid) are permuted submodular.*

**Remark 9.2** (D.Schlesinger-2007)**.** *For pairwise models there an efficient algorithm, which either finds the permutation turning the problem into submodular or proves, that it is impossible.*

## 9.3 Binary Submodular Problems as Min-Cut

### 9.3.1 st-Min-Cut Problem

**Definition 9.4.** *Let $\mathcal{G}' = \{\mathcal{V}', \mathcal{E}'\}$ be a directed graph, $c\colon \mathcal{E} \to \mathbb{R}$ be the* cost function of the edges *and $s, t \in \mathcal{V}$ be two different nodes, which will be called* source *and* `sink` *respectively.*

*A* cut $C = (S, T)$ *be called such a partitioning of* $\mathcal{V}'$ *into two parts* $S$ *and* $T$*, which satisfy:*

$$S \cap T = \emptyset \tag{9.22}$$

$$S \cup T = \mathcal{V} \tag{9.23}$$

$$s \in S,\ t \in T\,. \tag{9.24}$$

The cost of the cut $C$ is a total cost of edges that go from $S$ to $T$:

$$c(S,T) := \sum_{\substack{u\in S, v\in T\\ uv\in\mathcal{E}'}} c_{uv}\,. \tag{9.25}$$

Picture!

The st-min-cut (or shortly *min-cut*) problem is NP-hard in general. However, if all costs $c(u,v)$ are non-negative, its linear programming relaxation is tight and therefore the problem is polynomially solvable. Moreover, the corresponding linear programming relaxation has a special structure, which allows to find its minimum with polynomial finite-step algorithms with the worst-case complexity ???. Typically, the Lagrange dual of the problem is solved instead of the primal one. The former constitutes a *max-flow* problem and has a number of efficient polynomial finite-step solvers, those best-known worst-case complexity is ???.

In what follows we will show equivalence of pairwise energy minimization and min-cut problems. Moreover, we will show that pairwise submodular energy minimization is equivalent to the min-cut problem with non-negative costs and the other way around. Moreover, we show that the local polytope relaxation of binary energy minimization also reduces to the min-cut problem with non-negative costs.

### 9.3.2 Oriented Quadratic Pseudo-Boolean Functions (O-QPBF)

Let $\mathbb{B} = \{0,1\}$ be the set of binary variables and $\mathbb{B}^n$ - the set of all binary vectors of the length $n$.

**Definition 9.5.** *Functions* $q\colon B^n \to \mathbb{R}$ *are called* pseudo-Boolean*. Pseudo-Boolean function of the form* $q(x) = q_0 + \sum_{i=1}^n q_i x_i + \sum_{1\le i<j\le n} q_{ij} z_i z_j$*, where* $z_i$ *denotes either* $x_i$ *or* $\bar{x}_i$ *are called* quadratic *and its minimization (maximization) problem* $\min_{x\in\mathbb{B}^n} q(x)$ *is called* quadratic pseudo-Boolean optimization (QPBO)*.*

In the following, we will use the shortcut QPBF for quadratic pseudo-boolean functions.

**Remark 9.3.** *Since* $q_{ii}x_ix_i = q_{ii}x_i$*, such terms are excluded from the definition above. Term* $q_0$ *is technical and does not influence optimization.*

**Definition 9.6.** *QBPF of the form* $q(x) = q_0 + \sum_{i=1}^n q_i x_i + \sum_{1\le i<j\le n} (q_{ij} x_i \bar{x}_j + \bar{q}_{ij} \bar{x}_i x_j$ *are called* oriented *QPBF (shortcut O-QPBF).*

**Example 9.5.** *The following QPBF are O-QPBF:*

$-5 + 7x_3\bar{x}_1 + 12x_3 + \bar{x}_1$

$2\bar{x}_1x_2 + 3\bar{x}_2x_1 - 6x_3\bar{x}_4$

**Example 9.6.** *The following QPBF are not O-QPBF::*

$-5 + 7x_3x_1 + 12x_3 + \bar{x}_1$

$2x_1x_2 + 3\bar{x}_2x_1 - 6x_3\bar{x}_4$

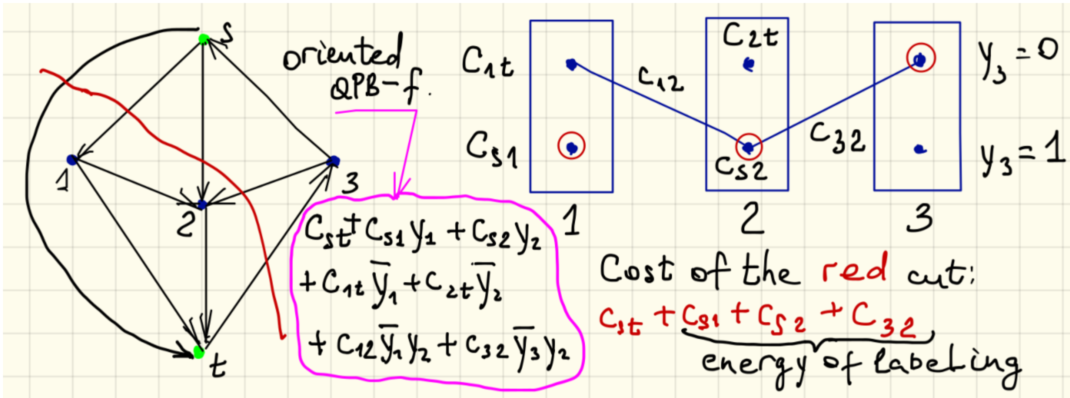


Figure 4: $st$-min-cut graph, corresponding QPB-function in oriented representation and energy minimization problem. Labeling corresponding to the cut is shown with red circles around labels

#### 9.3.3 st-Min-Cut As O-QPBF

Let $\mathcal{G}'(\mathcal{V}', \mathcal{E}')$, $s, t \in \mathcal{V}'$, be an oriented graph with costs $c_{uv}$, $uv \in \mathcal{E}'$ of its edges. A cut in $\mathcal{G}'$ can be seen as a mapping $x\colon \mathcal{V} \to \{0, 1\}$ such that $y_s = 0$ and $y_t = 1$. Abusing notation we will write $c(x)$ for $c(S, T)$ if $x$ corresponds to the cut $(S, T)$. Let us map edges of $\mathcal{G}'$ to elementary O-QPBFs as follows (see Fig. 4):

1. Edges $(u, v) \in \mathcal{E}'$, $u, v \notin \{s, t\}$ are mapped to $c_{uv}\bar{x}_u x_v$.
2. Edges $(s, v) \in \mathcal{E}'$, $v \neq t$ are mapped to $c_{sv}x_v$.
3. Edges $(v, t) \in \mathcal{E}'$, $v \neq t$ are mapped to $c_{vt}\bar{x}_v$.
4. If the edge $(s, t)$ belongs to $\mathcal{E}'$ it is mapped to the constant factor $c_{st}$.
5. In case the graph has edges $(t, v)$ or $(v, s)$, their representation is 0.

The last two type of edges do not influence finding a minimal cut, hence can be ignored w.l.o.g. in what follows.

The Fig. 4 illustrates the mapping with an example.

It is easy to see that any cut $x$ has the cost equal to the sum of the O-QPBFs corresponding to all edges of a graph. And the other way around: any O-QPBF $q$ can be interpreted as an $st$-cut on a specially constructed graph $\mathcal{G}'$. Non-negativity of the graph edges is equivalent to non-negativity of coefficients defining the O-QPBF. The min-cut problem is equivalent to the problem of O-QPBF minimization over binary vectors constituting its argument.

#### 9.3.4 Binary Energy As O-QPBF

Let $\theta$ be potentials of a binary energy minimization problem, i.e. $\mathcal{X}_v = \{0, 1\}$. Each labeling is a binary vector from $\{0, 1\}^{|\mathcal{V}|}$.

**Unary Potentials** By adding a constant if needed one can always transform unary potentials $\theta_u$, $u \in \mathcal{V}$ to one of two forms below and represent them as oriented monomials with non-negative coefficients:

1. $\theta_u(0) = 0$, $\theta_u(1) = c_u \geq 0$, the corresponding oriented monomial reads $c_u x_u$.

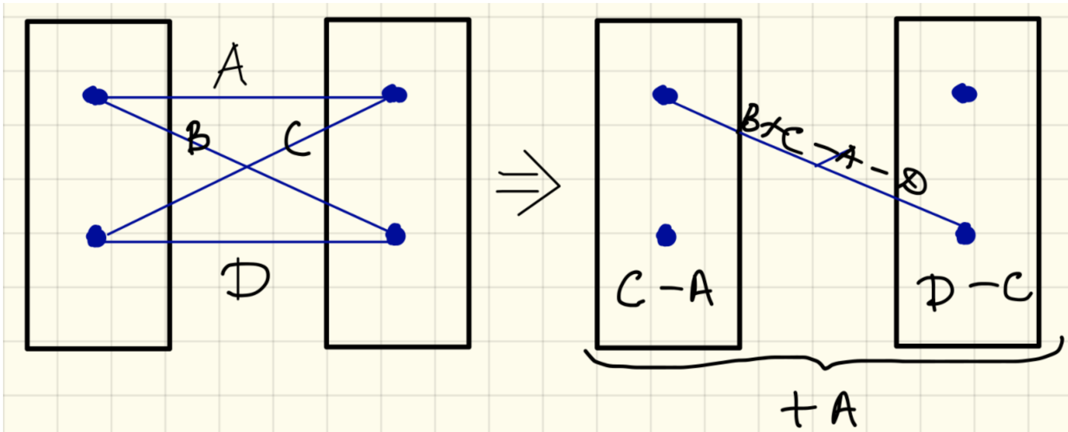


Figure 5: Transformation of a pairwise factor into the "diagonal" form. Only non-zero edges are shown.

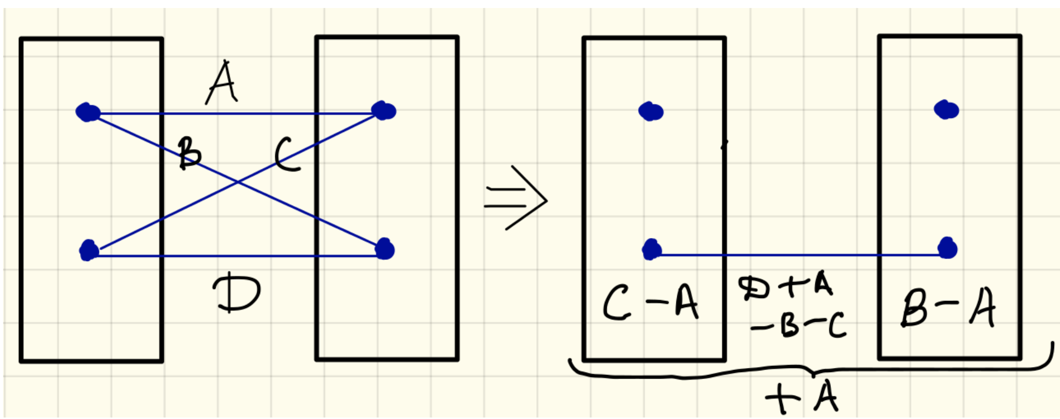


Figure 6: Transformation of a pairwise factor into the "parallel" form. Only non-zero edges are shown.

2. $\theta_u(1) = 0$, $\theta_u(0) = c_u \geq 0$, the corresponding oriented monomial reads $c_u \bar{x}_u$.

**Pairwise Potentials** Using reparametrisation (and adding a constant to a factor) one can transform all pairwise factors $\theta_{uv}$, $uv \in \mathcal{E}$ to any of the two forms:

- "diagonal" form, when $\theta_{uv}(0,0) = \theta_{uv}(1,1) = \theta_{uv}(1,0) = 0$, see Fig. 5 and
- "parallel" form, when $\theta_{uv}(0,0) = \theta_{uv}(0,1) = \theta_{uv}(1,0) = 0$, see Fig. 6.

In these forms the pairwise factors can be further represented as QPBF:

- The "diagonal" form is equal to O-QPBF $c_{uv}\bar{x}_u x_v$ and
- The "parallel" form is equal to $c'_{uv} x_u x_v$ .

for suitably selected $c_{uv}$ and $c'_{uv}$, see Fig. 5 and 6.

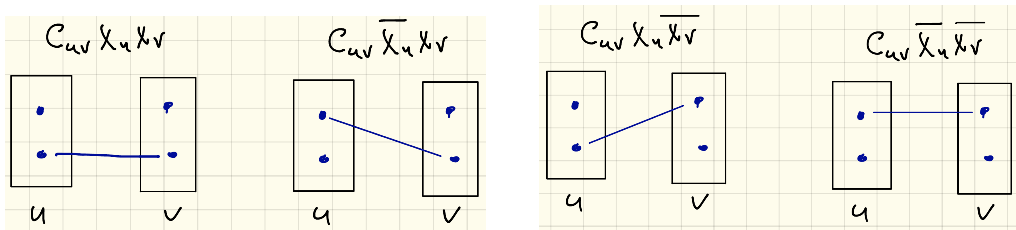


Figure 7: Quadratic monomials as pairwise factors. Potentials with zero values are not shown

Till the end of Section 9 we will stick to the "diagonal" form as it represents a O-QPBF. The "parallel" will be used later on in the Section 10. Note that since the reparametrisation does not influence the submodularity property, submodularity of a pairwise factor is equivalent to non-negativity of the coefficient $c_{uv}$ in its "diagonal" representation.

Summing up all monomials corresponding to pairwise and unary factors (in the "diagonal" representation) a binary energy can be represented as a O-QPBF. Submodular energies correspond to O-QPBF with non-negative coefficients.

This reduction can be applied also in the opposite direction: any quadratic monomial can be considered as a binary pairwise factor, as show in the Fig. 7.

#### 9.3.5 Equivalence: Binary Energy Minimization = Min Cut

The reductions described in Section 9.3.3 and 9.3.4 prove the following theorem:

**Theorem 9.3.** *Binary energy minimization and st-min-cut problem are linearly reducible to each other. Moreover, submodular binary energy minimization is linearly reducible to a st-min-cut with non-negative edge costs and back.*

### 9.4 Transforming Multilabel Problems Into Binary Ones

Let the graphical model $\{\mathcal{G} = (\mathcal{V}, \mathcal{E}), \mathcal{X} = \times_{v\in\mathcal{V}}\mathcal{X}_v, \theta\}$ be given. We will construct a binary graphical model $\{\bar{\mathcal{G}} = (\bar{\mathcal{V}}, \bar{\mathcal{E}}), \bar{\mathcal{X}} = \times_{v\in\mathcal{V}}\bar{\mathcal{X}}_v, \bar{\theta}\}$ with $\bar{\mathcal{X}}_v = \{0,1\}$ from it and build a mapping $x \to \bar{x}$ such that $E_{\mathcal{G}}(\theta, x) = E_{\bar{\mathcal{G}}}(\bar{\theta}, \bar{x})$. Let us construct the binary problem according to the following rules (see illustration in Fig. 8):

- Node set: $\bar{\mathcal{V}} = \{(v,l)\colon\ : v \in \mathcal{V}, l \in \mathcal{X}_v\}$.
- Edge set:
  $\bar{\mathcal{E}} = \{[(v,l),(v,l+1)]\colon v \in \mathcal{V}, 1 \le l < |\mathcal{X}_v|\} \cup \{[(u,s),(v,t)]\colon uv \in \mathcal{E}, (s,t) \in \mathcal{X}_{uv}\}$
- Labels set: $\bar{\mathcal{X}}_{\bar{v}} = \{0,1\}$, $\bar{v} \in \bar{\mathcal{V}}$.
- The mapping $x \to \bar{x}$ is constructed as: $x_v \to \forall l \in \mathcal{X}_v\ \bar{x}_{(v,l)} = [\![ l \ge x_v ]\!]$.
- Potentials:

$$\bar{\theta}_{(v,l)}(0) = \begin{cases} 0, & 1 \le l < |\mathcal{X}_v| \\ \infty, & l = |\mathcal{X}_v| \end{cases} \tag{9.26}$$

$$\bar{\theta}_{(v,l)}(1) = \begin{cases} 0, & 1 < l \le |\mathcal{X}_v| \\ \theta_v(0), & l = 0 \end{cases} \tag{9.27}$$

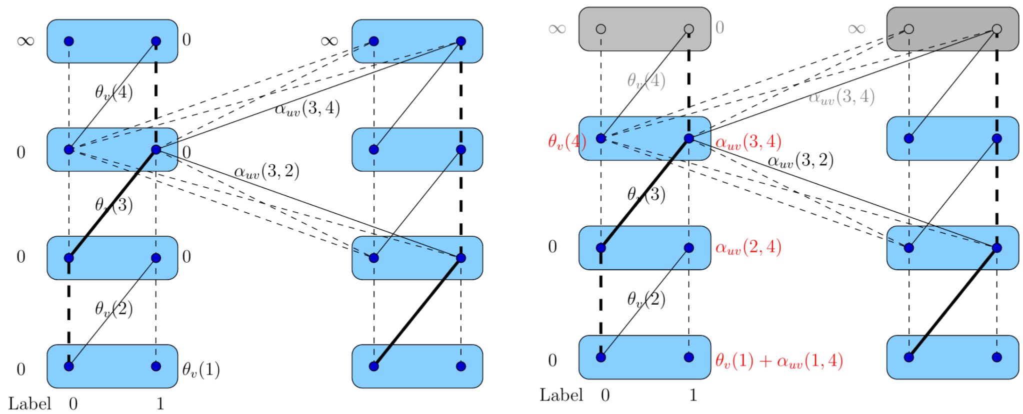


Figure 8: (Left) Transformation of a multilabel problem into a binary one (Right)Eliminating the top node

For $v \in \mathcal{V}$ and $1 \le s < |\mathcal{X}_v|$: $\theta_{(v,s)(v,s+1)} =$

| | **0** | **1** |
|---|---|---|
| **0** | 0 | $\theta_v(s+1)$ |
| **1** | $\infty$ | 0 |

For $uv \in \mathcal{V}$ and $s \in \mathcal{X}_u$, $t \in \mathcal{X}_v$: $\theta_{(u,s)(v,t)} =$

| | **0** | **1** |
|---|---|---|
| **0** | 0 | 0 |
| **1** | 0 | $\alpha_{uv}(s,t)$ |

Here

$$\alpha_{uv}(s,t) = \begin{cases} \theta_{uv}(s,t) + \theta_{uv}(s+1,t+1) - \theta_{uv}(s,t+1) - \theta_{uv}(s+1,t), & s < |\mathcal{X}_u|, t < |\mathcal{X}_v| \\ \theta_{uv}(s,|\mathcal{X}_v|) - \theta_{uv}(s+1,|\mathcal{X}_v|), & s < |\mathcal{X}_u|, t = |\mathcal{X}_v| \\ \theta_{uv}(|\mathcal{X}_u|,t) - \theta_{uv}(|\mathcal{X}_u|,t+1), & s = |\mathcal{X}_u|, t < |\mathcal{X}_v| \\ \theta_{uv}(|\mathcal{X}_u|,|\mathcal{X}_v|), & s = |\mathcal{X}_u|, t = |\mathcal{X}_v| \end{cases} \tag{9.28}$$

As soon as we show that

$$\theta_v(x_v) = \sum_{l \in \mathcal{X}_v} \bar{\theta}_{(v,l)}(\bar{x}_{(v,l)}) + \theta_{(v,l),(v,l+1)}(\bar{x}_{(v,l)}, \bar{x}_{(v,l+1)}) \tag{9.29}$$

and

$$\theta_{uv}(s,t) = \sum_{l=s}^{|\mathcal{X}_v|} \sum l' = t^{|\mathcal{X}_u|} \alpha_{uv}(l,l'). \tag{9.30}$$

we will prove that $E_{\mathcal{G}}(\theta, x) = E_{\bar{\mathcal{G}}}(\bar{\theta}, \bar{x})$.

The proof of (9.29) is straightforward. The proof of (9.30) follows from Lemma 9.3, which implies for $s < |\mathcal{X}_u|, t < |\mathcal{X}_v|$ (for the sake of notation we omit the indexes $uv$ in the next formula):

$$
\begin{aligned}
\sum_{l=s}^{|\mathcal{X}_v|}\sum_{l'=t}^{|\mathcal{X}_u|}\alpha(l,l') &\overset{(9.28)}{=} \sum_{l=s}^{|\mathcal{X}_v|-1}\sum_{l'=t}^{|\mathcal{X}_u|-1}\theta(s,t)+\theta(s+1,t+1)-\theta(s,t+1)-\theta(s+1,t)\\
&+\sum_{l=s}^{|\mathcal{X}_v|-1}a(l,|\mathcal{X}_v|)+\sum_{l'=t}^{|\mathcal{X}_u|-1}a(|\mathcal{X}_u|,l')+a(|\mathcal{X}_u|,|\mathcal{X}_v|)\\
&\overset{(9.13)}{=}\theta(s,t)+\theta(|\mathcal{X}_v|,|\mathcal{X}_u|)-\theta(s,|\mathcal{X}_u|)-\theta(|\mathcal{X}_v|,t)\\
\overset{(9.28)}{+}\sum_{l=s}^{|\mathcal{X}_v|-1}\theta(l,|\mathcal{X}_v|)-\theta(l+1,|\mathcal{X}_v|) &\overset{(9.28)}{+}\sum_{l'=t}^{|\mathcal{X}_u|-1}\theta(|\mathcal{X}_u|,l')-\theta(|\mathcal{X}_u|,l'+1)+a(|\mathcal{X}_u|,|\mathcal{X}_v|)\\
&\overset{(9.14)}{=}\theta(s,t)+\theta(|\mathcal{X}_v|,|\mathcal{X}_u|)-\theta(s,|\mathcal{X}_u|)-\theta(|\mathcal{X}_v|,t)\\
+\theta(s,|\mathcal{X}_v|)-\theta(|\mathcal{X}_u|,|\mathcal{X}_v|)&+\theta(|\mathcal{X}_u|,t)-\theta(|\mathcal{X}_u|,|\mathcal{X}_v|)\overset{(9.28)}{+}\theta(|\mathcal{X}_u|,|\mathcal{X}_v|)\\
&=\theta(s,t)\,. 
\end{aligned}
\tag{9.31}
$$

The described transformation is valid in general, for any input pairwise graphical model. Note, however, that if the pairwise potentials of the multilabel problem are submodular, so are the pairwise potentials of the corresponding binary problem. This follows directly from the definition of the pairwise potentials of the binary problem.

This shows equivalence of the min-cut and submodular pairwise energy minimization in general, including multilable energies as well.

**Computational Complexity** Let a binary graphical model be the reduction of an $L$-label one. Then the number of pairwise factors (and therefore, the size of the problem) grows as $O(|\mathcal{E}|\times|L^2|)$, where $\mathcal{E}$ is the number of edges in the multilabel model. Up to a constant, this coincides with the size of the multilable model. Therefore, the considered reduction has a *linear* complexity. In a special case, however, when the pairwise potentials constitute an $\ell_1$-norm (9.10), an equivalent binary problem can be constructed, which size grows as $O(|\mathcal{E}|\times|L^2|)$, only linearly with the number of labels $L$. We refer to [15? ] for details.

**Technical Note.** The top-most nodes in Fig. 8 can be eliminated, since they contain a single allowed label (with a finite weight). The elimination changes only unary potentials, as it is shown in Fig. 8, and therefore it does not destroy a possible submodularity of the problem.

## 9.5 Move-Making Algorithms

Although the class of multilabel submodular energies is rich, it does not include a number of important pairwise potentials, such as Potts (Example 9.2), truncated linear and quadratic potentials (Example 9.2) with the non-negative constant $\lambda$. Indeed, all these three types are equivalent to the Ising model (see Example 9.1) in the binary case and therefore submodular in that case. The algorithms described below use this fact to employ efficient min-cut solvers and obtain approximate solutions to the mentioned non-submodular multilabel energies.

### 9.5.1 $\alpha$-expansion

Let $\mathcal{X}_v=\mathcal{X}_0$ for all $v\in\mathcal{V}$.

**Algorithm 10** $\alpha$-expansion

1: **Init:** $x^0$, $t = 0$
2: **repeat**
3:     **for** $\alpha \in \mathcal{X}_0$ **do**
4:
$$x^{t+1} = \arg \min_{x\colon x_v \in \{x_v^t, \alpha\}} E(x) \tag{9.32}$$
5:         $t = t + 1$
6:     **end for**
7: **until** $E(x^t) = E(x^{t-|\mathcal{X}_0|})$
8: **return** $x^t$

Let us assume that the minimization (9.32) at the step 4 of Algorithm 10 can be done efficiently. Then $E(x^t) \geq E(t+1)$, which is, Algorithm 10 monotonously decreases the energy $E$. On each iteration it builds a binary subproblem, where one label corresponds to the current labeling $x^t$ and the second label is $\alpha$. In this way algorithm tries to substitute labels in the current labeling with the label $\alpha$ and finds the best substitution of this type, which is the one minimizing the energy. This procedure is repeated iteratively in for each $\alpha$ until none of them leads to energy improvement.

Let us now figure out, when the minimization (9.32) can be solved efficiently. In particular, when this problem is submodular.

**Exercise 9.3.** *Show that if the energy $E$ is submodular, the minimization problem* (9.32) *is submodular. Show it also for higher order energies.*

However, the $\alpha$-expansion algorithm can be used also with non-submodular energies. It is sufficient that pairwise factors define a metrics:

**Definition 9.7.** *For any set $L$ the function $f\colon L \times L \to \mathbb{R}$ is called* a semi-metrics *if for all $x, x' \in L$ it holds: (i) $f(x, x') \geq 0$; (ii) $f(x, x') = 0$ iff $x = x'$; (iii) $f(x, x') = f(x', x)$.*

**Definition 9.8.** *Function $f\colon L \times L \to \mathbb{R}$ is called* a metrics *if it is a semi-metrics and additionally it holds: $f(x, x') + f(x', x'') \geq f(x, x'')$, $\forall x, x', x'' \in L$.*

**Proposition 9.5.** *Let $\theta_{uv}(s, t)$ be a metrics, then the binary optimization problem* (9.32) *is submodular.*

*Proof.* The submodularity follows from

$$\underbrace{\theta_{uv}(\alpha, \alpha)}_{=0} + \underbrace{\theta_{uv}(x_u, x_v) - \theta_{uv}(x_u, \alpha) - \theta_{uv}(\alpha, x_v)}_{\leq 0} \leq 0\,. \tag{9.33}$$

□

**Example 9.7.** *The following pairwise potentials are metrics:*

- *Potts potentials $\theta_{uv}(s, t) = [\![ s \neq t ]\!]$.*
- *Truncated $\ell_1$-norm: $\theta_{uv}(s, t) = \min\{|s - t|, M\}$*
- *Truncated $\ell_2$-norm: $\theta_{uv}(s, t) = \min\{|s - t|^2, M\}$*

From the last example it follows that a metrics need not be submodular. Conversely, submodularity does not imply metric properties, e.g. if $\theta_{uv}$ is metrics, then for any constant $a$ potentials $\theta_{uv} + a$ are not metrics. However if $\theta_{uv}$ is submodular, $\theta_{uv} + a$ is submodular as well.

The $\alpha$-expansion algorithm does not solve energy minimization optimally (even submodular), it does in a certain sence "local search" in a quite big vicinity. From this point of view it is similar to the ICM method, although its results are usually notably better. Each iteration of $\alpha$-expansion requires only linear in $|\mathcal{X}_0|$ number of graph cuts and typically a small number of iterations is required. This is in contrast to the construction described in 9.4, which has a quadratically growing number of pairwise potentials and therefor is significantly slower in practice.

The following theorem provides the worst-case optimality bound for Algorithm 10. In practice, the algorithm attains significantly better energy values than the guaranteed by the bound.

**Theorem 9.4** (Multiplicative optimality bound [8]). *Let $\theta_{uv}$ be metrics for all $uv \in \mathcal{E}$, $x^*$ be an optimal solution and $x'$ be the labeling obtained by $\alpha$-expansion. Then for*

$$c := \max_{uv \in \mathcal{E}} \frac{\max_{x_v \neq x_u} \theta_{uv}(x_u, x_v)}{\min_{x_v \neq x_u} \theta_{uv}(x_u, x_v)} \tag{9.34}$$

*it holds:*

$$\sum_{v \in \mathcal{V}} \theta_v(x_v^*) + 2c \sum_{uv \in \mathcal{V}} \theta_{uv}(x_u^*, x_v^*) \geq \sum_{v \in \mathcal{V}} \theta_v(x'_v) + \sum_{uv \in \mathcal{V}} \theta_{uv}(x'_u, x'_v) \tag{9.35}$$

**Theorem 9.5** (Multiplicative optimality bound of LP relaxation [16]). *Let $\theta_{uv}$ be a metrics. There is a randomized (primal) rounding procedure for the solution of the LP relaxation, which provides a solution within the factor $c = 1$ for Potts models and $c = O(\log|\mathcal{X}_0| \log\log|\mathcal{X}_0|)$ for arbitrary multilable problem.*

### 9.5.2 $\alpha\beta$-swap

The triangle inequality in Definition 9.8 can still be quite restrictive for certain applications, such as e.g. the panorama stitching problem, see Example 9.9 below. Algorithm 11, known as $\alpha\beta$-swap, runs also for semi-metric pairwise potentials. The set $\mathcal{X}^{\alpha,\beta}(s)$, $s, \alpha, \beta \in \mathcal{X}_0$ is defined as

$$\mathcal{X}^{\alpha,\beta}(s) = \begin{cases} \{s\}, & s \in \mathcal{X}_0 \backslash \{\alpha, \beta\} \\ \{\alpha, \beta\}, & s \in \{\alpha, \beta\}\,. \end{cases} \tag{9.36}$$

**Algorithm 11** $\alpha\beta$-swap

**Init:** $x^0$, $t = 0$
**repeat**
  // The line below defines $\binom{\mathcal{X}_0}{2}$ iterations
  **for** $(\alpha, \beta) \in \mathcal{X}_0^2$ **do**
    $x^{t+1} = \arg\min_{x\colon x_v \in \mathcal{X}^{\alpha,\beta}(x_v^t)} E(x)$
    $t = t + 1$
  **end for**
**until** $E(x^t) = E(x^{t - \binom{\mathcal{X}_0}{2}})$
**return** $x^t$

The algorithm considers a mutual substitution of labels $\alpha$ and $\beta$ in the current labeling, which decreases the energy.

Similar to $\alpha$-expansion, the $\alpha\beta$-swap can be applied to submodular functions, however it is applicable to semi-metrics, as we mentioned above. This follows from the fact, that on each iteration only labels $\alpha$ and $\beta$ are active. The following submodularity condition for this restricted problem follows directly from the semi-metrics conditions:

$$\underbrace{\theta_{uv}(\alpha,\beta)}_{>0}+\underbrace{\theta_{uv}(\beta,\alpha)}_{>0}>\underbrace{\theta_{uv}(\alpha,\alpha)}_{=0}+\underbrace{\theta_{uv}(\beta,\beta)}_{=0} \tag{9.37}$$

Each outer iteration of $\alpha\beta$-swap has a quadratic $(\binom{\mathcal{X}_0}{2})$ complexity with respect to the number of labels, which is more than the linear complexity of $\alpha$-expansion. However since only those nodes participate in optimization, which have labels $\alpha$ or $\beta$, each min-cut computation is cheaper than those of $\alpha$-expansion.

Contrary to $\alpha$-expansion, $\alpha\beta$-swap does not provide any a optimality bound. Moreover, the following example shows that the algorithm can return arbitrary bad results. However, in practice it typically performs only somewhat worse than $\alpha$-expansion.

**Example 9.8** (Arbitrary bad optimality bound (Boykov-Veksler-Zabih-2001))**.** *Consider an graphical model consisting of* 3 *nodes:* $\mathcal{V}=\{1,2,3\}$. *The edges connect the nodes* 1,2 *and* 2,3*:* $\mathcal{E}=\{\{1,2\},\{2,3\}\}$. *Each node is associated with* 3 *labels:* $\mathcal{X}_0=\{a,b,c\}$.

*The values of unary potentials are defined by the following table:*

| | **1** | **2** | **3** |
|---|---|---|---|
| **a** | 0 | $K$ | $K$ |
| **b** | $K$ | 0 | $K$ |
| **c** | 2 | 2 | 0 |

*Pairwise potentials are homogeneous* ($\theta_{12}=\theta_{23}=\theta$) *and equal to* $\theta(a,b)=\theta(b,c)=\frac{K}{2}$ *and* $\theta(a,c)=K$. *It is easy to see that configuration* $(a,b,c)$ *is a fix-point for Algorithm 11. Its energy is* $K$*, while the optimal configuration is* $(c,c,c)$ *and has energy* 4. *Since* $K$ *is not bounded from above, the energy of a fix-point of the* $\alpha\beta$*-swap algorithm can be arbitrary worse than the energy of the optimal labeling.*

**Exercise 9.4.** *Show that* $\alpha$*-expansion Algorithm 10 obtains the optimal solution when started from the fix-point of* $\alpha\beta$*-swap in Example 9.8.*

**Example 9.9** (Panorama stitching, the image credint is to [49])**.** *The "thread-free" panorama stitching problem is an example of semi-metric pairwise potentials. The set of nodes forms a two-dimensional grid. Each node corresponds to a pixel of the resulting panoramic image. Labels index images which contain the corresponding pixel of the panorama. The task is to find a lableing, which minimizes color jumps between neighboring pixels. Therefore, the unary potentials are all zero and pairwise potentials are equal to the difference of colors in the points of the corresponding input images:* $\theta_v(x_v)=0$*;* $\theta_{uv}(x_u,x_v)=\begin{cases}0, & x_u=x_v\\ |Color_u(x_u)-Color_v(x_v)|, & x_u\neq x_v\end{cases}.$

*Panorama stitching, the image credit is to [49].*

## 9.6 Further Reading

The classical paper, which describes reduction of the binary energy minimization problem to the min-cut problem is the seminal work [22]. The transformation of multilabel problems into the min-cut problem was proposed independently in [2] and [15] for convex pairwise potentials (like in Exercise 9.1). Later on this result was generalized in [42] to the submodularity-preserving transformation of general multilabel problems into binary ones. An algorithm

Figure 9:

determining whether a given problem is permuted submodular or not is due to the publication [41]. Energies with metric potentials are analyzed in details in [16]. The $\alpha$-expansion and $\alpha\beta$-swap algorithms were proposed and analyzed in [8].

# 10 Relaxed Binary Energy as $st$-Min-Cut

## 10.1 Half-Integrality of the Local Polytope

**Arc consistency ⇒ Optimality** Recall the definition of $\mathcal{L}(\phi)$ from Section 5.3 as the set of all relaxed labelings satisfying the complementary slackness condition w.r.t. the reparametrized potentials $\theta^{\phi}$.

**Theorem 10.1** (e.g. [53, 21])**.** *Let $\mathcal{X}_v = \{0, 1\}$, $v \in \mathcal{V}$ and let $\mathrm{mi}[\theta^{\phi}]$ have a non-empty kernel (arc-consistent subset). Then $\mathcal{L}(\phi) \cap \{0, \frac{1}{2}, 1\}^I \neq 0$.*

**Remark 10.1.** *The theorem claims, that for binary problems the arc-consistency is sufficient for optimality. And moreover, that there is always a solution with half-integer coordinates. In particular this implies that the corresponding local polytope has only vertices having half-integer coordinates.*

*Proof.* Consider the kernel of $\mathrm{mi}[\theta^{\phi}]$ and $n_v$ be the number of locally minimal lables in a node $v \in \mathcal{V}$: $n_v = \mathrm{mi}[\theta^{\phi}]_v(0) + \mathrm{mi}[\theta^{\phi}]_v(1)$. Up to the label flipping it has only the following 5 possible configurations:

Let us now assign 0's to the edges crossed with red strokes and all other black labels and label pairs get the non-zero weights $\frac{1}{n_v}$ and $\frac{1}{\max\{n_v, n_u\}}$ respectively. Such weights form a feasible solution from $\mathcal{L}(\phi) \cap \{0, \frac{1}{2}, 1\}$. □

**Full Characterisation of Relaxed Solutions** Although we have proved above that half-integral values are sufficient to describe all vertices of the local polytope in binary case, not all feasible combinations of such vertices define some vertex of the local polytope. Let $\mu \in \mathcal{L}$

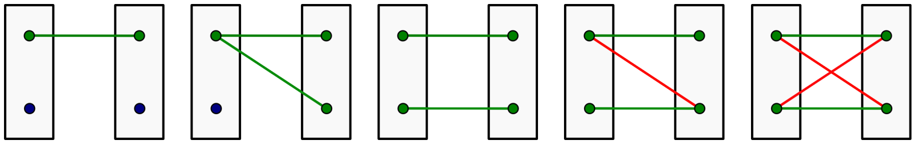

Figure 10: Possible configurations of locally optimal labels end label pairs belongin to the kern(mi$[\theta^\phi]$). Both green and red colors mean the elements from this set. To construct the primal value assign 0 to red edges and $\frac{1}{n_v}$ and $\frac{1}{\max\{n_v,n_u\}}$ to greed lables nad edges respectively.

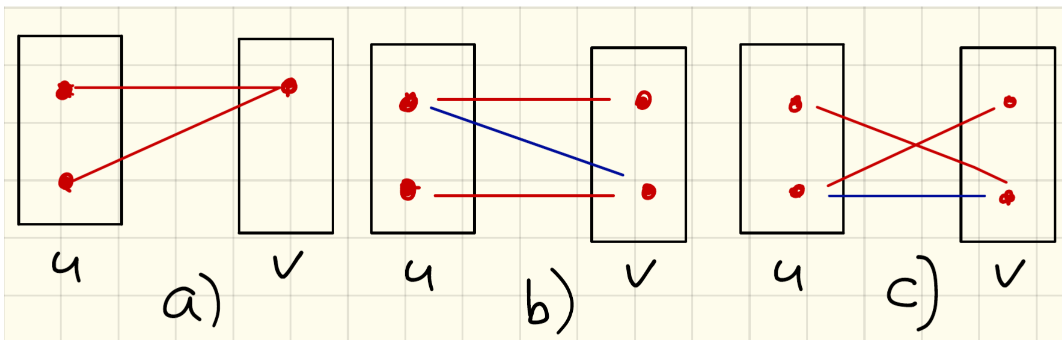


Figure 11: Basis primal solutions for pairwise factors. Red lines correspond to the value $\frac{1}{2}$ of the corresponding primal variable, red labels - either to 1 (case a), node $v$) or to $\frac{1}{2}$ (all other cases). Blue lines on b) and c) correspond to "diagonal" and "parallel" forms

be a primal vector. What concerns its unary coordinates $\mu_v$, there is a clear situation: only $(0,1)$, $(1,0)$ and $(\frac{1}{2},\frac{1}{2})$ configurations of $(\mu_v(0),\mu_v(1))$ are allowed. All of them can be used to construct a vertex of the local polytope (see e.g. Exercise 3.1 for the case $(\frac{1}{2},\frac{1}{2})$) as soon as $\mathcal{L}\neq\mathcal{M}$ (like e.g. for acyclic graphs). Let us analyse the pairwise variables $\mu_{uv}$.

To do so note that the local polytope (for arbitrary, not only binary problems) can be written in the following separable form:

$$\mu\in\mathcal{L}\Leftrightarrow\{\mu\in\mathbb{R}^I;\ \mu_v\in\Delta_{|\mathcal{X}_v|},\ v\in\mathcal{V};\ \mu_{uv}\in\mathcal{L}_{uv}(\mu_u,\mu_v),\ uv\in\mathcal{E}\}\,, \tag{10.1}$$

where $\mathcal{L}_{uv}(\mu_u,\mu_v)=\left\{\mu_{uv}\in\mathbb{R}^{|\mathcal{X}_{uv}|}\colon \begin{cases}\sum_{x_u\in\mathcal{X}_u}\mu_{uv}(x_u,x_v)=\mu_v(x_v), & x_v\in\mathcal{X}_v\\ \sum_{x_v\in\mathcal{X}_v}\mu_{uv}(x_u,x_v)=\mu_u(x_u), & x_u\in\mathcal{X}_v\\ \mu_{uv}(x_{uv}\geq 0, & x_{uv}\in\mathcal{X}_{uv}\end{cases}\right\}$

This separable structure implies the following equality, which shows that given unary terms of a solution one can find pairwise terms by optimizing over to each pairwise factor independently:

$$\min_{\mu\in\mathcal{L}}\langle\theta,\mu\rangle=\min_{\substack{\mu_v\in\Delta_{|\mathcal{X}_v|}\\ v\in\mathcal{V}}}\sum_{uv\in\mathcal{E}}\min_{\mu_{uv}\in\mathcal{L}_{uv}(\mu_u,\mu_v)}\langle\theta_{uv},\mu_{uv}\rangle\ . \tag{10.2}$$

W.l.o.g. we will assume that all submodular pairwise factors of the energy will be represented in the "diagonal" form and supermodular – in the "parallel" form. Then given optimal values of the unary factors amounts minimization of each factor independently w.r.t. the pairwise one. The existing possibilities are shown on Fig. 11, where configuration b) appears if $\theta_{uv}$ is submodular, and c) – if it is supermodular.

## 10.2 LP Relaxation as Oriented Posiform

As we learned from Section 9.3.4, a binary energy can be represented as a QPBF with non-negative coefficients, however it is O-QPBF with non-negative coefficients (and hence can be reduced to a min-cut problem with non-negative weights) only if all pairwise factors are submodular. In this section we show that the local polytope relaxation of a binary energy minimization (with arbitrary, submodular or/and supermodular factors) can be represented as minimization of a O-QPBF with non-negative coefficients and therefore, it is reducible to min-cut. To do so we will introduce an auxiliary variable $y = \bar{x}$ and write all potentials symmetrically w.r.t. $x$ and $y$:

- Unary terms:

$$c_v x_v \to \frac{1}{2} c_v x_v + \frac{1}{2} c_v \bar{y}_v \tag{10.3}$$

$$c_v \bar{x}_v \to \frac{1}{2} c_v \bar{x}_v + \frac{1}{2} c_v y_v \tag{10.4}$$

- Submodular pairwise terms:

$$c_{uv} \bar{x}_u x_v \to \frac{1}{2} c_{uv} \bar{x}_u x_v + \frac{1}{2} c_{uv} y_u \bar{y}_v \tag{10.5}$$

- Supermodular pairwise terms:

$$c_{uv} x_u x_v \to \frac{1}{2} c_{uv} x_u \bar{y}_v + \frac{1}{2} c_{uv} \bar{y}_u x_v \tag{10.6}$$

Let $E$ be an energy and $g(x, y)$ – the O-QPBF with positive coefficients, constructed according to (10.3)-(10.6). By construction $E(x) = g(x, \bar{x})$, therefore it holds:

$$\min_{x,y \in \mathcal{B}^{|\mathcal{V}|}} g(x, y) \le \min_{\substack{x,y \in \mathcal{B}^{|\mathcal{V}|} \\ y = \bar{x}}} g(x, y) = \min_{x \in \mathcal{B}^{|\mathcal{V}|}} E(x)\,. \tag{10.7}$$

The following theorem relates the first term in (10.7) to the LP relaxation of the energy minimization:

**Theorem 10.2.** *Let $\theta$ be potentials of a binary energy minimization problem and its submodular and supermodular pairwise terms are in the "diagonal" and "parallel" forms respectively. Let $g$ be constructed according to* (10.3)-(10.6). *Then it holds:*

$$\min_{x,y \in \mathcal{B}^{|\mathcal{V}|}} g(x, y) = \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle \tag{10.8}$$

*and moreover, a minimizer $\mu'$ of the right-hand-side can be constructed from a minimizer $(x', y')$ of the left-hand-side as follows:*

$$\mu'_v(1) = \begin{cases} x_v, & x_v = \bar{y}_v \\ \frac{1}{2}, & x_v \ne \bar{y}_v\,, \end{cases} \tag{10.9}$$

$$\mu'_v(0) = 1 - \mu'_v(1) \tag{10.10}$$

$$\mu'_{uv} = \arg \min_{\mu_{uv} \in \mathcal{L}_{uv}(\mu'_u, \mu'_v) \cap \{0, 1, \frac{1}{2}\}^{|\mathcal{X}_{uv}|}} \langle \theta_{uv}, \mu_{uv} \rangle \tag{10.11}$$

*Proof.* The proof of the theorem is quite straightforward: except for a special case noted separately, for each factor $w \in \mathcal{V} \cup \mathcal{E}$ we will show that if $\mu'_w$ is constructed from *arbitrary* $(x'_w, y'_w)$ the relaxed energy $\langle \theta_w, \mu'_w \rangle$ is equal to the corresponding term of $g(x', y')$.

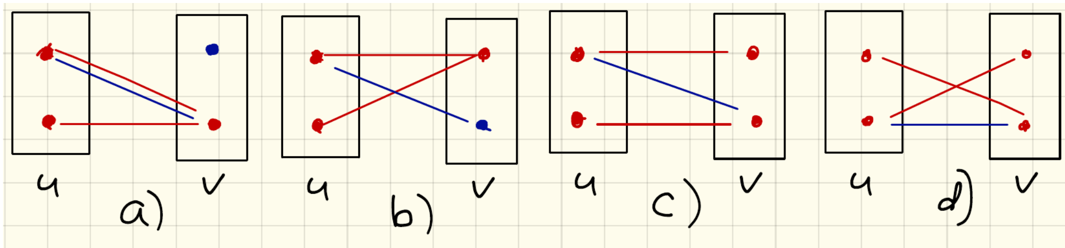


Figure 12: Optimal (basis) primal solutions for pairwise factors. Red lines correspond to the value $\frac{1}{2}$ of the corresponding primal variable, red labels - either to 1 (if case of a single red lable per node) or to $\frac{1}{2}$ (two red labels per node). Blue lines on b) and c) correspond to "diagonal" and "parallel" forms

1. If $x_w = \bar{y}_w$ the condition above holds by construction.
2. Unary terms (10.3)-(10.4). If $x_v \neq \bar{y}_v$ then the corresponding term of $g$ is equal to $\frac{1}{2}c_v$, which is equal to $\langle\theta_v, \mu'_v\rangle$.
3. Submodular terms (10.5). Let $x_v = \bar{y}_v$ and $x_u \neq \bar{y}_u$.
   (a) if $x_v = \bar{y}_v = 1$ the corresponding term (10.5) of $g$ turns into $\frac{1}{2}c_{uv}(x_u + \bar{y}_u)$, which is $\frac{1}{2}c_{uv}$ (independently on the values of $x_u \neq \bar{y}_u$) and is equal to $\langle\theta_{uv}, \mu'_{uv}\rangle$ according to Fig. 12,a).
   (b) if $x_v = \bar{y}_v = 0$ the corresponding term (10.5) of $g$ is equal to 0 (independently on the values of $x_u \neq \bar{y}_u$), which is equal to $\langle\theta_{uv}, \mu'_{uv}\rangle$ according to Fig. 12,b).
4. Supermodular terms (10.6) with $x_v = \bar{y}_v$ and $x_u \neq \bar{y}_u$ turns into 3 by changing $x_v$ to $\bar{x}_v$.
5. Submodular (10.5) and supermodular (10.6) terms. Let $x_v \neq \bar{y}_v$ and $x_u \neq \bar{y}_u$. This is the mentioned above exceptional case, there holds $\langle\theta_{uv}, \mu'_{uv}\rangle = 0$ (see Fig. 12,c) and d)). The same (0) value the terms (10.5) and (10.6) have when $x_v = y_v = x_u = y_u = 0$. There are combinations of values of $x_v$ and $x_u$ (e.g. $x_v = y_v = 1$ and $x_u = y_u = 0$ for (10.5)), corresponding to bigger values of the considered terms, however they are never optimal. This is due to the fact that as soon as $x_u \neq \bar{y}_u$ the change from $x_u = y_u = 1$ to $x_u = y_u = 0$ can either decrease the value of the corresponding components of $g(x, y)$ (as in the current case) or leave it unchanged (as in the situations described above). This precisely means that considering $x_u = y_u = 0$ a soon as $x_u \neq \bar{y}_u$ gives smaller or equal value of $g$.

□

## 10.3 Persistency Definition and Criterion

In this subsection we will consider general (non-binary) energy minimization.

**Motivation.** Let $\mu' = \arg\min_{\mu\in\mathcal{L}} \langle\theta, \mu\rangle$ be an optimal relaxed labeling. Its coordinate $\mu'_v(s)$ is called *(weakly) persistent* or *(weakly) partially optimal*, if (i) $\mu'_v(s) \in \{0, 1\}$ and (i) there exists $\mu^* \in \arg\min_{\mu\in\mathcal{L}\cap\{0,1\}} \langle\theta, \mu\rangle$ such that $\mu^*_v(s) = \mu'_v(s)$. It is *strongly persistent* if this holds for any $\mu^* \in \arg\min_{\mu\in\mathcal{L}\cap\{0,1\}} \langle\theta, \mu\rangle$.

**Definition 10.1** (Persistency)**.** *A labeling $x' \in \mathcal{X}_\mathcal{A}$ on a subset $\mathcal{A} \subseteq \mathcal{V}$ is* (weakly) partially optimal *or* (weakly) persistent *if $x' = x^*|_\mathcal{A}$, where $x^* \in \operatorname{argmin}_{x\in\mathcal{X}} \langle\theta, \delta(x)\rangle$.*

The following proposition formulates a sufficient persistency condition:

**Proposition 10.1** ([48])**.** *A partial labeling $x' \in \mathcal{X}_\mathcal{A}$ is persistent, if for all $\tilde{x} \in \mathcal{X}_{\mathcal{V}\setminus\mathcal{A}}$ it holds*

$$x' \in \operatorname{argmin}_{x\in\mathcal{X}_\mathcal{A}} E((x,\tilde{x}))\,, \tag{10.12}$$

*where $(\cdot,\cdot)$ stands for concatenation.*

*Proof.* Consider the equation

$$\min_{x\in\mathcal{X}_\mathcal{V}} E(x) = \min_{\tilde{x}\in\mathcal{X}_{\mathcal{V}\setminus\mathcal{A}}} \min_{x\in\mathcal{X}_\mathcal{A}} E((x,\tilde{x}))\,. \tag{10.13}$$

Let $\tilde{x} \in \mathcal{X}_{\mathcal{V}\setminus\mathcal{A}}$ be such that it leads to a minimal value on the right hand side of (10.13). Then $\tilde{x}$ is part of an optimal solution. By the assumption (10.12), $x'$ is an optimal solution to the inner minimization problem of (10.13), hence $(x',\tilde{x})$ minimizes $E$. □

## 10.4 Persistency of the Binary LP Relaxation

Let $\mathcal{A}(\mu) = \{v \in \mathcal{V}\colon \mu_v \in \{0,1\}^2\}$ be the set of nodes corresponding to integer coordinates of $\mu$.

**Theorem 10.3** (Similar to [21])**.** *Let us consider a binary energy with potentials $\theta$ and $\mu' \in \arg\min_{\mu\in\mathcal{L}} \langle\theta,\mu\rangle$. Then $\mu'|_{\mathcal{A}(\mu')}$ is persistent.*

*Proof.* The proof is illustrated in Fig. 13. Let us denote $\mathcal{A} := \mathcal{A}(\mu')$. Let also $x' \in \mathcal{X}_\mathcal{A}$ be selected such that $\delta(x') = \mu'|_\mathcal{A}$. The criterion of Proposition 10.1 does not depend on reparametrization of $\theta$. W.l.o.g. we will assume it is (dual) optimal and for all $w \in \mathcal{V}\cup\mathcal{E}$ it holds

$$\min_{s\in\mathcal{X}_w} \theta_w(s) = 0 \tag{10.14}$$

Let us consider (10.12). For a fixed $\tilde{x} \in \mathcal{X}_{\mathcal{V}\setminus\mathcal{A}}$ it holds

$$\arg\min_{x\in\mathcal{X}_\mathcal{A}} E((x,\tilde{x})) = \arg\min_{x\in\mathcal{X}_\mathcal{A}} \left( E_\mathcal{A}(x) + \sum_{\substack{uv\in\mathcal{E}:\\ u\in\mathcal{A}, v\in\mathcal{V}\setminus\mathcal{A}}} \theta_{uv}(x_u,\tilde{x}_v) \right) \geq 0\,, \tag{10.15}$$

where the energy $E_\mathcal{A}$ is restricted to the induced subgraph of $\mathcal{A}$ and the inequality holds due to (10.14).

Since $\mu'$ is primal (relaxed) optimal and $\theta$ is a dual optimal, it holds $E_\mathcal{A}(x') = 0$. Moreover, since the solution $\mu'$ outside $\mathcal{A}$ is non-integer, it holds $\mu'_v(s) > 0$, $s \in \mathcal{X}_v$, $v \in \mathcal{V}\backslash\mathcal{A}$. Therefore, due to the local polytope constraints, $\mu_{uv}(x'_u, x_v) > 0$, $u \in \mathcal{A}$, $x_v \in \mathcal{X}_v$. Due to the complementary slackness it holds $\theta_{uv}(x'_u, x_v) = 0$, in particular $\theta_{uv}(x'_u,\tilde{x}_v) = 0$. Hence, we obtain

$$\left( E_\mathcal{A}(x') + \sum_{\substack{uv\in\mathcal{E}_{\partial\mathcal{A}}:\\ u\in\mathcal{A}}} \theta_{uv}(x'_u,\tilde{x}_v) \right) = 0\,. \tag{10.16}$$

Comparing to (10.15) we obtain $x' \in \arg\min_{x\in\mathcal{X}_\mathcal{A}} E((x',\tilde{x}))$, which finalizes the proof. □

**Remark 10.2.** *An important property of $x'$ follows from the fact that it minimizes* (10.15)*. It reads:*

$$E((x,\tilde{x})) \geq E((x',\tilde{x})), \quad \forall x \in \mathcal{X}_\mathcal{A} \text{ and } \tilde{x} \in \mathcal{X}_{\mathcal{V}\setminus\mathcal{A}}\,. \tag{10.17}$$

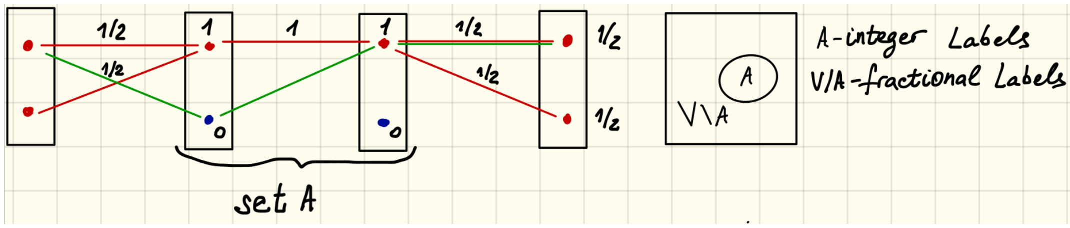


Figure 13: Illustration to the proof of Theorem 10.3. Red labels and edges correspond to the non-zero primal variables of a given relaxed solution $\mu'$. By complementary slackness they also correspond to locally optimal edges and labels in an optimal reparametrization $\theta$. We assume these locally optimal labels and edges to have weight 0. Green edges correspond to an arbitrary selected alternative labeling.

**Fusion Moves** The property (10.17) allows algorithms like $\alpha$-expansion (or $\alpha\beta$-swap) to work with arbitrary pairwise potentials. Instead of solving the corresponding binary problems exactly, one solves its relaxation and uses (10.17) to improve the current energy. In this scenario the current labeling $(x, \tilde{x})$ is replaced with the new one $(x', \tilde{x})$, if the inequality (10.17) holds strictly.

### 10.5 Further Reading

Although there is quite some literature of quadratic pseudo-Boolean optimization and its relation to binary labeling problems and min-cut, the presentation of the lecture is quite new. Typically the proof of Theorem 10.2 is done for dual objectives. This is more practical for applications, however much more technical and less illustrative. Moreover, it is less straightforwardly related to the persistency. The typical proof for the persistency property is done by reducing the problem to the vertex cover problem and operating by known results in that domain. It turned out that this result can be easily proved directly, see Theorem 10.3. Related works on quadratic pseudo-boolean optimization (QPBO) are [5, 6, 13, 18]. QPBO-related papers from the computer vision community are [38, 28, 17, 20, 29].

## 11 Discriminative Learning of Graphical Models

### 11.1 Problem Formulation

**Discussion.** Let $y \in \mathcal{Y}$ be an observation, e.g. an element of some *observation set* $\mathcal{Y}$. Potentials, previously considered as a vector $\theta \in \mathbb{R}^{\mathcal{I}}$ become a function of observation, e.g. $\theta\colon \mathcal{Y} \to \mathbb{R}^{\mathcal{I}}$. Here, $\theta(y)$ is the vector of potentials corresponding to the the observation $y$. *Training set* $D = \{y^i, x^i\}_{i=1}^m$, $y^i \in \mathcal{Y}$ - observations and $x^i \in \mathcal{X}_{\mathcal{V}}$ – the corresponding *ground truth* labelings. Loosely speaking, we want to find such parameters $\theta(y)$ that

$$x^i \in \arg\min_{x \in \mathcal{X}_{\mathcal{V}}} \left\langle \theta(y^i), \delta(x) \right\rangle . \tag{11.1}$$

Let us write this condition in a different form:

$$\left\langle \theta(y^i), \delta(x^i) \right\rangle \le \left\langle \theta(y^i), \delta(x) \right\rangle,\ x \in \mathcal{X},\ i = \overline{1, m}\,. \tag{11.2}$$

or, which is the same,

$$\left\langle \theta(y^i), \delta(x^i) - \delta(x) \right\rangle \le 0,\ x \in \mathcal{X},\ i = \overline{1, m}\,. \tag{11.3}$$

It suggests that learning is solving a linear system of inequalities...
However, before starting using this linear system, several questions must be answered:

- Does the system (11.3) always have a solution?
- What if it has multiple solutions, which of them must be preferred?
- How the resulting vector $w$ generalizes from the training set $D$ to the data, which is not included to the training set?

**Loss function.** Function of the form $l\colon \mathcal{X}_\mathcal{V} \times \mathcal{X}_\mathcal{V} \to \mathbb{R}$ is called *a loss function*. Its value $l(d(y^i), x^i)$ defines a loss or a penalty, which must be paid for a *decision* (labeling) $d(y^i) := \arg\min_{x\in\mathcal{X}_\mathcal{V}} \langle\theta(y^i), \delta(x)\rangle$ given that a correct *object state* (labeling) is $x^i$. It is natural to assume that (i) $l(x', x) \geq 0$ for all possible pairs $x', x$ and (ii) $l(x, x) = 0$.

The learning problem consists in minimizing the mathematical expectation of the risk $\mathbb{E}_{p(y)} l(d(y^i), x^i)$. Assuming that samples $y^i$ in the training set are i.i.d. and taken from the same distribution as those appearing in the testing (working) stage, we can substitute the mathematical expectation with its statistical estimate – the mean $\frac{1}{m}\sum_{i=1}^m l(d(y^i), x^i)$. This substitution is called *an empirical risk*.

Additionally a *regularizer* $R(w)$ is used to avoid biasing of the resulting parameter set towards the training set. As a result, the following

$$R(w) + \frac{C}{m}\sum_{i=1}^m l(d(y^i), x^i) \tag{11.4}$$

is considered as the *structured learning* problem.

**Example 11.1.** *As an example explaining the role of regularizer let us consider the following (indeed, very popular and important in practice) setting: to compute potentials $\theta(y)$ one computes $k$* feature vectors *or simply* features $\psi^j(y) \in \mathbb{R}^m$, $j = \overline{1,k}$. *One assumes that $\theta(y) = \sum_{j=1}^k w_j\psi^j(y) = \Psi(y)^\top w$ for a suitably selected* feature *matrix $\Psi(y)$ consisting of the feature vectors as its rows. The number of potentially useful features can be very large and one wants to keep only a small number of them with a non-zero weight. This implies to use $\ell_0$-norm $\sum_{i=1}^k [\![w \neq 0]\!]$ as a regularized. However, such a regularizer in non-convex, which does not allow to construct efficient algorithms for the learning problem* (11.4). *A convex relaxation (the largest convex function constituting a lower bound) of the $\ell_0$-norm is the $\ell_1$-norm. Indeed, under certain technical conditions [**?** ] its optimization provides exactly the same results as those of the $\ell_0$-norm. Therefore, one assumes the $\ell_1$-norm of $w$ to be a regularizer $R(w) = \|w\|_1$.*

**Remark 11.1.** *Regularizer - log of prior probability of the parameter distribution...*

**Example 11.2** (0/1 loss)**.** *The simples penalization strategy one can imagine is to penalize all but a correct result equally:*

$$l(x', x) = [\![x \neq x']\!]\,. \tag{11.5}$$

*Note that this loss does not take into account a possible internal structure of $x$.*

*Draw a plot $l$ vs. $x - x'$*

**Example 11.3** (Hamming loss)**.** *Assume the problem of semantic segmentation and one penalizes the total number of wrongly classified pixels. If pixels correspond to nodes of a graphical model, one obtains the* Hamming loss

$$l(x', x) = \sum_{v\in\mathcal{V}} [\![x_v \neq x'_v]\!]\,. \tag{11.6}$$

*Note that the Hamming loss is just a sum of* $0/1$ *losses for nodes of the model.*

**Example 11.4** (Intersection-Over-Union)**.** *Indeed, for semantic segmentation another loss is a standard de facto: intersection-over-union. Its simplified version for a single data instance and a binary (foreground/background) segmentation* ($x_v \in \{0,1\}$) *reads:*

$$iou(x', x) = \frac{\sum_{v\in\mathcal{V}} [\![x_v = 1 \ \textit{and}\ x'_v = 1]\!]}{\sum_{v\in\mathcal{V}} [\![x_v = 1 \ \textit{or}\ x'_v = 1]\!]} \,. \tag{11.7}$$

*The loss growing with the difference between* $x$ *and* $x'$ *and equal to zero when* $x = x'$ *reads*

$$l(x', x) = 1 - iou(x', x)\,. \tag{11.8}$$

**Example 11.5** (Sum of absolute or squared distances)**.** *Assume the problem of stereo-reconstruction. Then the sum of absolute or squared distances between two surfaces is a reasonable measure:*

$$l(x', x) = \sum_{v\in\mathcal{V}} |x_v - x'_v| \tag{11.9}$$

*or*

$$l(x', x) = \sum_{v\in\mathcal{V}} (x_v - x'_v)^2\,. \tag{11.10}$$

**Hinge Loss.** The most natural loss functions $l$ are non-convex with respect to their first parameter, which in its turn leads to non-convex optimization problems. Consider the "0/1"-loss (11.5) It is not only just non-convex, it is not continuous. Optimizing this loss without regularizer would correspond to finding the largest solvable subsystem (w.r.t. $i$) of the system of linear inequalities (11.3), which is NP-hard. Therefore, one considers an upper bound on a loss, known as *hinge loss*:

$$\hat{l}_\theta(x) = \max_{x'\in\mathcal{X}_\mathcal{V}} l(x', x) + \langle \theta(y), \delta(x) - \delta(x')\rangle\ . \tag{11.11}$$

**Example 11.6.** *Let us consider the* $0/1$*-loss* $l(x', x) = [\![x \neq x']\!]$*. In this case the the hinge-loss* (11.11) *reads*

$$\hat{l}_\theta(x) = \max_{x'\in\mathcal{X}_\mathcal{V}} [\![x \neq x']\!] + \langle \theta(y), \delta(x) - \delta(x')\rangle = \tag{11.12}$$

$$\max\{0,\ 1 + \max_{x'\in\mathcal{X}_\mathcal{V}\setminus\{x\}} \langle \theta(y), \delta(x) - \delta(x')\rangle\}\,. \tag{11.13}$$

*It implies that the loss is equal to* $0$*, if* $x$ *is the minimizer of* $\langle\theta(y), \delta(x)\rangle$ *and moreover, the difference in the energy* $E(x) = \langle\theta(y), \delta(x)\rangle$ *between* $x$ *and a second-best labeling is at least* $1$*. Otherwise, if* $x$ *is not an optimum, the loss grows linearly with the difference in energies between* $x$ *and an optimum.*

*Note that to evaluate the hinge-loss in this case we need only to compute an optimal labeling, i.e. solve the following energy minimization problem* $\max_{x'\in\mathcal{X}_\mathcal{V}} \langle\theta(y), -\delta(x')\rangle = -\min_{x'\in\mathcal{X}_\mathcal{V}} \langle\theta(y), \delta(x')\rangle$*.*

Properties of the hinge-loss:

1. $\hat{l}_\theta(x)$ – convex w.r.t. $\theta(y)$;
2. $\hat{l}_\theta(x) \geq l(d(y), x)$.

*Proof.*

$$\hat{l}_\theta(x^i) = \max_{x\in\mathcal{X}_\mathcal{V}} l(x,x^i)+\big\langle\theta(y^i),\delta(x^i)-\delta(x)\big\rangle \ge l(x^*,x^i)+\big\langle\theta(y^i),\delta(x^i)-\delta(x^*)\big\rangle \ge l(x^*,x^i)\,, \tag{11.14}$$

where $x^* = \arg\min_{x\in\mathcal{X}_\mathcal{V}} \big\langle\theta(y^i),\delta(x)\big\rangle$. The last inequality holds, because $\big\langle\theta(y^i),\delta(x^*)\big\rangle \le \big\langle\theta(y^i),\delta(x^i)\big\rangle$ by definition of $x^*$. □

After all, the *structured SVM* problem reads:

$$\min_w R(\theta) + \frac{C}{m}\sum_{i=1}^m \max_{x\in\mathcal{X}_\mathcal{V}} l(x,x^i) + \big\langle\theta(y^i),\delta(x^i)-\delta(x)\big\rangle\,. \tag{11.15}$$

## 11.2 Algorithms for Structured SVM

### 11.2.1 Loss-Augmented Inference

Consider the structured SVM problem (11.15). To evaluate the objective one has to be able to compute the hinge-loss

$$\max_{x\in\mathcal{X}_\mathcal{V}} l(x,x^i) + \big\langle\theta(y^i),\delta(x^i)-\delta(x)\big\rangle = \big\langle\theta(y^i),\delta(x^i)\big\rangle - \min_{x\in\mathcal{X}_\mathcal{V}} \big\langle\theta(y^i),\delta(x)\big\rangle - l(x,x^i)\,. \tag{11.16}$$

The minimization problem in the right-hand-side is called *the loss-augmented inference problem*:

$$\min_{x\in\mathcal{X}_\mathcal{V}} \big\langle\theta(y^i),\delta(x)\big\rangle - l(x,x^i),\ \overline{i=1,m} \tag{11.17}$$

Let us consider several important cases, when this problem is as difficult as the non-augmented inference $\min_{x\in\mathcal{X}_\mathcal{V}} \big\langle\theta(y^i),\delta(x)\big\rangle$.

**Fully Separable Loss** Let the loss $l$ be representable as a sum of unary and pairwise factors (the losses from Examples 11.3 and 11.5 are special cases of this situation):

$$l(x',x) = \sum_{v\in\mathcal{V}} l_v(x'_v,x_v) + \sum_{uv\in\mathcal{E}} l_{uv}(x'_{uv},x_{uv})\,. \tag{11.18}$$

Then the loss-augmented inference (11.17) reads (with the index $i$ changed to $'$)

$$\min_{x\in\mathcal{X}_\mathcal{V}} \big\langle\theta(y^i),\delta(x)\big\rangle - l(x,x^i) \tag{11.19}$$

$$= \min_{x\in\mathcal{X}_\mathcal{V}} \sum_{v\in\mathcal{V}} \theta_v(x_v;y') + \sum_{uv\in\mathcal{E}} \theta_{uv}(x_{uv};y') - \sum_{v\in\mathcal{V}} l_v(x'_v,x_v) - \sum_{uv\in\mathcal{E}} l_{uv}(x'_{uv},x_{uv}) \tag{11.20}$$

$$= \min_{x\in\mathcal{X}_\mathcal{V}} \sum_{v\in\mathcal{V}} \Big(\theta_v(x_v;y') - l_v(x'_v,x_v)\Big) + \sum_{uv\in\mathcal{E}} \Big(\theta_{uv}(x_{uv};y') - l_{uv}(x'_{uv},x_{uv})\Big) \tag{11.21}$$

Since $y'$ and $x'$ are fixed, the loss-augmentation means just subtracting certain values from each unary and pairwise factor independently, according to the formula (11.21).

Therefore, solvability of the loss-augmented problem (11.17) is determined by solvability of (11.21). The latter is solvable for (a) acyclic models and (b) for the case when augmentation with the loss-components $l_{uv}$ does not destroy submodularity properties of the initial problem, given $\theta_{uv}$ are submodular. The case (b) holds, in particular, when $l_{uv} = 0$ for all $uv \in \mathcal{E}$ as e.g. in Examples 11.3 and 11.5.

Solving problems with non-separable losses, like the intersection-over-union considered in Example 11.4 is a subject of research for each particular combination of a loss and a graphical model.

**Linear SVM** In the following, we will assume that the potentials are linear functions of $w$, e.g. $\theta(y) = \Psi(y)^\top w$ as in Example 11.1. Then the Structured SVM problem reads

$$\min_w R(w) + \frac{C}{m}\sum_{i=1}^m \max_{x\in\mathcal{X}_\mathcal{V}} l(x,x^i) + \left\langle w, \Psi(y)(\delta(x^i)-\delta(x))\right\rangle \tag{11.22}$$

$$=: \underbrace{\min_w R(w) + \frac{C}{m}\sum_{i=1}^m \max_{x\in\mathcal{X}_\mathcal{V}} l(x,x^i) + \left\langle w, \Psi(y,x^i)-\Psi(y,x)\right\rangle}_{\Lambda(w)}, \tag{11.23}$$

where we reused the notation $\Psi(y,x)$ for $\Psi(y)\delta(x)$.

We will also assume that the regularizer $R$ is convex and continuous as a function of the parameter vector $w$.

### 11.2.2 Subgradient Method

The objective (11.15) is convex, but non-smooth, therefore it can be optimized with the subgradient method. A subgradient of the objective in (11.23) reads (see Lemma 6.1):

$$\frac{\partial\Lambda}{\partial w} = \frac{\partial R(w)}{\partial w} + \frac{C}{m}\sum_{i=1}^m \left\langle w, \Psi(y,x^i)-\Psi(y,x^{i*})\right\rangle, \tag{11.24}$$

where $x^{i*} \in \arg\max_{x\in\mathcal{X}_\mathcal{V}} l(x,x^i) + \left\langle w, \Psi(y,x^i)-\Psi(y,x)\right\rangle = \arg\min_{x\in\mathcal{X}_\mathcal{V}} \langle w, \Psi(y,x)\rangle - l(x,x^i)$ is a solution of the loss-augmented inference problem.

Therefore, for a subgradient update step

$$w^{t+1} = w^t - \alpha^t\frac{\partial\Lambda(w^t)}{\partial w} \tag{11.25}$$

one must solve *all* (for $i=\overline{1,m}$) loss-augmented inference problems.

**Example 11.7** (Acyclic Graphical Models)**.** *The subragient method can be used, when the loss-augmented inference is tractable. For example, it is so in the case, when the graphical model is acyclic and the loss function is fully separable.*

Let the loss be fully separable with pairwise components being supermodular (e.g. equal to 0). The loss-augmented inference (11.17) is tractable if the potentials $\theta$ are submodular. To keep the potentials submoodular we should add the additional, submodularity constraints to the SVM-rpoblem formulation. It turns it into constrained minimization and the subgradient method does not apply. The following section addresses this issue.

### 11.2.3 Projected Subgradient Method

Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex (but possibly non-smooth) function and $D$ be a closed convex set. We are interested in solving the following *constrained* optimization problem:

$$\min_{x\in D} f(x)\,. \tag{11.26}$$

**Theorem 11.1** (Necessary and sufficient optimality conditions)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex (but possibly non-smooth) function and $D$ be a closed convex set. Then $x^*$ is a solution to* (11.26) *iff there is a subgradient $g \in \partial f(x^*)$ such that*

$$\langle g, x-x^*\rangle \geq 0 \text{ for all } x\in D\,. \tag{11.27}$$

*Proof.* See the proof in [32]. □

Note that in the unconstrained case, when $D = \mathbb{R}^n$, the condition (11.27) is equivalent to $0 \in \partial f(x^*)$, which we learned before.

Picture: if $x^*$ on the border.

The simplest method addressing the constrained minimization problem (11.26) is the *projected subgradient method*, which consists in the iterative updates of the form:

$$x^{t+1} = \mathcal{P}_D(x^t - \alpha_t g^t)\,, \tag{11.28}$$

where $g^t \in \partial f(x^t)$ and $\mathcal{P}_D(x) := \arg\min_{x' \in D} \|x - x'\|_2^2$ denotes an Euclidean projection to the set $D$.

Picture with the projection

**Theorem 11.2.** *Let $\alpha_t > 0$, $\alpha_t \xrightarrow{t\to\infty} 0$ and $\sum_{t=1}^{\infty} \alpha_t = \infty$. Then it holds*

- $f(x^t) - f(x^*) \xrightarrow{t\to\infty} 0$
- *If* $0 < \alpha_t < 2\frac{f(x^t)-f(x^*)}{\|g^t\|}$

$$\|x^{t+1} - x^*\| \leq \|x^t - x^*\|\,. \tag{11.29}$$

*Proof.* See the proof in [4]. □

Moreover, the worst-case convergence estimate of the projected suggradient on the set of all convex funcitons is the same as for the ordinary subgradeint method: $O(\frac{1}{\epsilon^2})$.

**Example 11.8** (Training binary submodular energies)**.** *Let us consider the linear structured SVM problem* (11.23) *with the additional submodularity constraints:*

$$\min_w R(w) + \frac{C}{m}\sum_{i=1}^{m} \max_{x \in \mathcal{X}_\mathcal{V}} l(x, x^i) + \langle w, \Psi(y, x^i) - \Psi(y, x)\rangle \tag{11.30}$$

$$s.t.\ \theta_{uv}(0,0) + \theta_{uv}(1,1) \leq \theta_{uv}(0,1) + \theta_{uv}(1,0),\ uv \in \mathcal{E}\,, \tag{11.31}$$

*where $\theta_{uv} = [\Psi^\top]_{uv} w$ is a linear function of $w$ defined by the matrix $\Psi$.*

*Application of the projected subgradient method* (11.28) *requires to project to the set of the submodularity constraints* (11.31)*. In case when the parameter vector $w$ is the vector of potentials $\theta$ itself this projection decomposes into independent projections to inequalities corresponding to each pairwise potential and can be done efficiently. Moreover, in this case, there is an explicit formula for the projection. Let*

$$\theta_{uv}(0,0) + \theta_{uv}(1,1) - \theta_{uv}(0,1) - \theta_{uv}(1,0) = a > 0\,, \tag{11.32}$$

*the Euclidean projection to the constraint* (11.31) *corresponds to the following update*

$$\theta_{uv}(0,0) -= \frac{a}{4},\ \theta_{uv}(1,1) -= \frac{a}{4},\ \theta_{uv}(1,0) += \frac{a}{4},\ \theta_{uv}(0,1) += \frac{a}{4}\,. \tag{11.33}$$

*An alternative way is to use the parallel form of potentials and therefore, train a non-zero number $\theta_{uv}(1,1)$. As discussed in Section* **??** *the submodularity constraint* (11.31) *reduces in this case to $\theta_{uv}(1,1) \leq 0$. Projection to such a set is particularly easy: $\mathcal{P}_{\{w<0\}}(w') = w'[\![w' < 0]\!]$.*

**Example 11.9** (Training multilabel submodular potentials)**.** *Let us consider the linear structured SVM problem* (11.23) *with the additional submodularity constraints:*

$$\min_{w} R(w) + \frac{C}{m}\sum_{i=1}^{m} \max_{x\in\mathcal{X}_{\mathcal{V}}} l(x,x^i) + \left\langle w, \Psi(y,x^i) - \Psi(y,x)\right\rangle \tag{11.34}$$

$$s.t.\ \theta_{uv}(s,t) + \theta_{uv}(s+1,t+1) \le \theta_{uv}(s+1,t) + \theta_{uv}(s,t+1),\ uv\in\mathcal{E},\ (s,t)\in\mathcal{X}_{uv}\,, \tag{11.35}$$

*where $\theta_{uv} = [\Psi^\top]_{uv} w$ is a linear function of $w$ defined by the matrix $\Psi$.*

*Application of the projected subgradient method* (11.28) *requires to project to the set of the submodularity constraints* (11.35)*. Although this projection may decompose into independent projections to subsets corresponding to each pairwise potential, it may still be a time-consuming operation, when the number of labels is big. Instead, one can use a binary representation (see Fig. 8) of multilabel problems, where the projection is particularly simple, as described in Example 11.8.*

**Example 11.10** (Learning with LP Relaxation - decomposition-based)**.** *In cyclic non-submodular case the loss-augmented inference problem* (11.17) *is NP-hard, even for fully decomposable losses. Let us however fix to this case and consider $\hat\theta(x|y^i) := \theta(x) - l(x,x^i)$ .*

*A work-around the NP-hardness of the loss-augmented iunference, is to consider an LP relaxation of the loss-augmented inference problem, which reads:*

$$\min_{\mu\in\mathcal{L}} \left\langle \hat\theta(y^i), \mu\right\rangle \tag{11.36}$$

*Picture with the local polytope and the vector $\theta$!*

*However, the most of efficient dedicated LP solvers, such as TRW-S work in the dual domain and do not provide access to the primal variables $\mu$. Therefore, one usually refomulates the relaxed loss-augmented problem* (11.36) *to contain the dual objective:*

$$\min_{\phi\in\mathbb{R}^{\mathcal{J}}} D(\phi;\hat\theta(y^i)) \tag{11.37}$$

*Here $D$ is the dual LP objective*

$$D(\phi;\hat\theta(y^i)) = \sum_{\xi\in\mathcal{V}\cup\mathcal{E}} \min_{s\in\mathcal{X}_\xi} \hat\theta_\xi(s;y^i) \tag{11.38}$$

*The corresponding sub-gradient reads:*

$$\frac{\partial\Lambda(\phi,w)}{\partial w} = \frac{\partial R(w)}{\partial w} + \frac{C}{m}\sum_{i=1}^{m} \frac{\partial D(\phi;\hat\theta(y^i))}{\partial\hat\theta(y^i)}\frac{\partial\hat\theta(y^i)}{\partial w}; \tag{11.39}$$

$$\frac{\partial\Lambda(\phi,w)}{\partial \phi} = \frac{C}{m}\sum_{i=1}^{m} \frac{\partial D(\phi;\hat\theta(y^i))}{\partial\phi} \tag{11.40}$$

*Here:*

- $\frac{\partial\hat\theta(y^i)}{\partial w} = \Psi^\top$ *is the matrix of the corresponing derivatives;*
- $\frac{\partial D(\phi;\hat\theta(y^i))}{\partial\hat\theta(y^i)}$ *is the vector with coordinates* $\mu_\xi(s) = [\![\hat\theta^\phi_\xi(s;y^i) = \min_{s\in\mathcal{X}_\xi}\hat\theta_\xi(s;y^i)]\!]$, $\xi \in \mathcal{V}\cup\mathcal{E}$, $s\in\mathcal{X}_\xi$;

- $\frac{\partial D(\phi;\hat{\theta}(y^i))}{\partial \phi}$ *is defined by* (7.2). *See also Section 7 for details.*

*Note that the subgradient method updates both the parameter vector $w$ and the reparametrization $\phi$. This means, one solves the inference and learning problems simultaneously. Besides the slow convergence of the subgradient method, the drawback is that one has to keep in memory current reparametrizations for all datasets $i = \overline{1, m}$.*

**Example 11.11** (Learning with the decomposition-based (LP) relaxation)**.** *TBD*

### 11.2.4 Stochastic Subgradient Method

Although the subgradient method can be usually well-paralellized, if the number $m$ of training samples is large, such updates become computationally quite expensive. Therefore, one may resort to the stochastic subgradient method, described below.

**Definition 11.1.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex function. A random vector $\tilde{g} \in \mathbb{R}^n$ is called* a noisy (unbiased) subgradient *of $f$ at $x^0 \in \mathrm{dom} f$, if $g = \mathbb{E}\tilde{g} \in \partial f(x^0)$, which is, we have*

$$f(x) \geq f(x^0) + \left\langle \mathbb{E}\tilde{g}, x - x^0 \right\rangle \tag{11.41}$$

*for all $x$. In other words, $\tilde{g}$ is a noised unbiased subgradient, if $\tilde{g} = g + v$ and $v$ has a zero mean.*

Let $f$ be a function of the form

$$f(x) = \sum_{i=1}^{m} f_i(x) \tag{11.42}$$

Then its subdifferential reads $\partial f(x) = \sum_{i=1}^{m} \partial f_i(x)$. Let us consider a random vector $\tilde{g}$, which is equal to $\frac{\partial f_i(x)}{\partial x}$ for $i$ randomly generated from the uniform distribution over $\{1, \dots, m\}$. Then the expectation of this vector is equal to $\mathbb{E}\tilde{g} = \frac{1}{m}\sum_{i=1}^{m} \frac{\partial f_i(x)}{\partial x} = \frac{1}{m}\frac{\partial f(x)}{\partial x}$ and therefore $m\tilde{g}$ is a noisy unbiased subgradient of $f$.

The stochastic subgradient method uses the same update rule as the subgradient method, which is

$$x^{t+1} = x^t - \alpha_t \tilde{g}^t\,, \tag{11.43}$$

where $x^t$ is the $t$-th iterate, $\alpha_t > 0$ is the $t$-th step-size and $\tilde{g}^t$ is the noisy subgradient of $f$ in $x^t$, which is

$$\mathbb{E}(\tilde{g}^t | x^t) = g^t \in \partial f(x^t)\,. \tag{11.44}$$

Even more so than with an ordinary subgradient method, the value $f^t$ may increase during the algorithm. Therefore we keep track on $f^t_{best} = \min\{f(x^1), \dots, f(x^k)\}$. The following theorem determines the convergence conditions of the stochastic sub-gradient method:

**Theorem 11.3** (Nedic,Bertsekas, Boyd,Mutapcic)**.** *Let the following conditions hold:*

- *There is $x^*$, which minimizes $f$;*
- *the mean of the squared stochastic subgradient norm is bounded, i.e. $\mathbb{E}\|\tilde{g}^t\|_2^2 \leq C < \infty$ for all $t$.*
- *$\alpha_t$ being square-summable but not summable, i.e.*

$$\alpha_t > 0, \quad \sum_{t=1}^{\infty} \alpha_t^2 < \infty, \quad \sum_{t=1}^{\infty} \alpha_t = \infty \tag{11.45}$$

*Then* $\mathbb{E}f^t_{best} \overset{t\to\infty}{\to} f^*$ *and for any* $\epsilon > 0$ *it holds* $\mathbf{Prob}(f^t_{best} - f^* \geq \epsilon) \overset{t\to\infty}{\to} 0$.

Compare the step-size rule (11.45) and convergence guarantees to those provided by Theorem 11.2 for the ordinary subgradient method.

Although the convergence results for the stochastic sub-gradient method are somewhat weaker than those for the ordinary subgradient method, the stochastic method typically performs better on large training sets, because it does not require to go through the whole dataset to perform a single update step. Moreover, assuming that the training set grows to an infinite size, one can use the stochastic subgradient update for an *on-line* learning, when new datasets are added to the training dataset at any time and the parameter vector must be updated accordingly.

To apply the stochastic sub-gradient to the SVM-problem (11.23), we represent its objective in the form (11.42) as

$$\Lambda(w) = \sum_{i=1}^{m} \frac{1}{m} R(w) + \frac{C}{m} \max_{x \in \mathcal{X}_\mathcal{V}} l(x, x^i) + \left\langle w, \Psi(y, x^i) - \Psi(y, x) \right\rangle . \tag{11.46}$$

Therefore,

$$\tilde{g} = R(w) + C(l(x^*, x^i) + \left\langle w, \Psi(y, x^i) - \Psi(y, x^*) \right\rangle) \tag{11.47}$$

is a noisy unbiased subgradient of $\Lambda$, for any solution $x^*$ of the loss-augmented inference problem.

The projected stochastic subgradient method

$$x^{t+1} = \mathcal{P}_D(x^t - \alpha_t \tilde{g}^t) , \tag{11.48}$$

generalizes the stochastic subgradient method to the constrained optimization. It theoretical guarantees remain basically unchanged.

## 11.3 A Note About Non-Linear Parameter Dependence

Let us consider the regularized hinge-loss minimization problem:

$$\min_w \Lambda(w) = \min_w R(w) + \frac{C}{m} \sum_{i=1}^{m} \max_{x \in \mathcal{X}_\mathcal{V}} l(x, x^i) + \left\langle \theta(w, y^i), (\delta(x^i) - \delta(x)) \right\rangle . \tag{11.49}$$

The case when $\theta$ is a linear function of $w$ was considered above. This linearity is required to guarantee that the whole objective remains convex (recall, the convexity of the hinge-loss w.r.t. $\theta$ is guaranteed by construction of the hinge-loss). Strictly speaking, the notion of subgradient is defined for convex functions only. However, in all but the non-smoothness points a gradient is defined. In most practical cases one never attains the point of the non-smoothness exactly, since they have the measure 0. Loosely speaking, it suggests that the gradient of a non-smooth and potentially non-convex function is well-defined in all points of an iterative algorithm. Therefore, one can use the subgradient descent method to minimize most of the practically important non-smooth and non-convex functions, although the convergence to an optimum is not guaranteed anymore. Let us write such a general sugbradient method for the learning problem (11.49). The gradient of the function (in all, but the non-smoothness points) reads:

$$\frac{\partial \Lambda(w)}{\partial w} = \frac{\partial R(w)}{\partial w} + \frac{C}{m} \sum_{i=1}^{m} \left[ \frac{\partial \theta(w, y^i)}{\partial w} \right]^\top \cdot (\delta(x^i) - \delta(x^{*i})) . \tag{11.50}$$

Here the derivative $\frac{\partial\theta(w,y^i)}{\partial w}$ is a matrix with the components $\frac{\partial[\theta(w,y^i)]_k}{\partial w_l}$. In the considered above linear case $\theta(y) = \Psi(y)^\top w$ this matrix is the matrix $\Psi(y)^\top$. The subgradient update rule reads as before:

$$w^{t+1} = w^t - \alpha_t \frac{\partial\Lambda(w^t)}{\partial w}\,. \tag{11.51}$$

In practice it is implemented by the so called "back-propagation" algorithm [39].

## 11.4 Constrained Representation and Cutting Plane Method

Let us rewrite the SVM-problem (11.23) in the following *constrained* form:

$$\min_{w,\xi} R(w) + \frac{C}{m}\sum_{i=1}^{m}\xi_i \tag{11.52}$$

$$\text{s.t. } \left\langle w, \Psi(y^i,x) - \Psi(y^i,x^i)\right\rangle \geq l(x,x^i) - \xi_i,\ x \in \mathcal{X}_{\mathcal{V}},\ i = \overline{1,m} \tag{11.53}$$

Note that when $x = x^i$ the corresponding constraint reduces to $\xi_i \geq 0$ and if the inequalities $\left\langle w, \Psi(y,x) - \Psi(y,x^i)\right\rangle \geq l(x,x^i)$ hold for all $x \in \mathcal{X}_{\mathcal{V}}$, the corresponding optimal value of the variable $\xi_i$ is equal to 0. Therefore, these variables determine a slack, which lacks to satisfy all inequalities. This explains why the variables $x_i$ are called *slack variables*.

Let us consider different regularizers.

- For $R(w) = \|w\|_2^2$ the problem (11.52) becomes a *convex quadratic problem*, which is, the objective is a convex quadratic function and the constraints are linear.

- For $R(w) = \|w\|_1$ the problem (11.52) can be transformed into a linear program by introducing two parameter vectors $w_- \leq 0$ and $w_+ \geq 0$ such that the non-zero coordinates of $w_-$ ($w_+$) correspond to the negative (positive) coordinates of $w$. Therefore, as $w = w_- + w_+$, $\|w\|_1 = w_+ - w_-$, constraints $w_- \leq 0$ and $w_+ \geq 0$ added to the constraint set, the problem (11.52) becomes a linear program.

Both, quadratic and linear formulations of (11.52) are convex and would be polynomially solvable, if the number of constraints would be polynomial. However, it contains $m$ constraints for *each* labeling $x \in \mathcal{X}_{\mathcal{V}}$ and the latter set grows exponentially with the size of the graphical model.

To cope with this difficulty a *cutting plane* Algorithm 12 can be used.

**Algorithm 12** Cutting Plane Algorithm

1: **Given** $f_0$ - convex and $\min_x f_0(x) < \infty$, $f_i$, $i = \overline{1,m}$ - linear, $x^0$ - starting point, $\epsilon$-solution accuracy
2: **Returns** an approximate solution of the optimization problem: $\min_x\{f_0(x) \mid f_k(x) \leq 0,\ k = \overline{1,M}\}$
3: $K^1 = \emptyset$; $t = 0$
4: **repeat**
5: $\quad t = t + 1$
6: $\quad$ Find $x^t = \arg\min_x\{f_0(x) \mid \ f_k(x) \leq 0,\ k \in K^t\}$
7: $\quad$ Find $k_t = \arg\max_{i=\overline{1,M}} f_k(x^t)$
8: $\quad I^{t+1} = I^t \cup \{k_t\}$
9: **until** $f_{k_t}(x^t) \leq \epsilon$
10: **return** $x^t$

The critical part of Algorithm 12 is to find the most violated constraint $i_k = \arg\max_{i=\overline{1,M}} f_k(x^t)$. In many important cases it need not even be the most violated, but rather violated enough, which is $f_k(x^t) \geq \epsilon$ for some $\epsilon$. Finding such a constraint, when $M$ is very large (like e.g. in (11.52)) can be quite a non-trivial task. Let us consider it for the problem (11.52).

Let $K^{t+1}$ be the set of selected constraints for the problem (11.52), $w^t$ and $\xi^t$ – the corresponding optimal parameters.

Then finding the most violated constraint amounts in computing

$$\begin{aligned}
&\arg\max_{i=1,m}\max_{x\in\mathcal{X}_\mathcal{V}} \left\langle w, \Psi(y^i,x^i) - \Psi(y^i,x)\right\rangle + l(x,x^i) - \xi_i^t \\
&= \arg\max_{i=1,m}\left(\left\langle w, \Psi(y^i,x^i)\right\rangle - \underbrace{\min_{x\in\mathcal{X}_\mathcal{V}}\left[\left\langle w, \Psi(y^i,x)\right\rangle - l(x,x^i)\right]}_{\text{loss-augmented inference}} - \xi_i^t\right). \qquad (11.54)
\end{aligned}$$

In other words, it requires to perform the loss-augmented inference for all training samples and compare the actual slack $\hat{\xi}_i^t := \left\langle w, \Psi(y^i,x^i)\right\rangle - \min_{x\in\mathcal{X}_\mathcal{V}}\left[\left\langle w, \Psi(y^i,x)\right\rangle - l(x,x^i)\right]$ with the value of $\xi_i^t$. The largest difference defines the most violated constraint, which must be then added to the active set $K^t$ to upgrade it to the set $K^{t+1}$.

As in the case of the subgradient optimization, it is usually very costly to perform loss-augmented inference for all training sample to find a single cutting plane. Therefore the cutting plane is added for each dataset, as soon as $\hat{\xi}_i^t - \xi_i^t > \epsilon$.

**Stopping condition** Let $(w^*, \xi^*)$ be a solution to (11.52). Let us consider the following inequality:

$$\Lambda(w^t, \hat{\xi}^t) \leq \Lambda(w^*, \hat{\xi}^*) \leq \Lambda(w^t, \hat{\xi}^t)\,. \qquad (11.55)$$

The first inequality holds, because $(w^t, \hat{\xi}^t)$ is a solution to the problem with a subset of constraints only. The second one - because $(w^t, \hat{\xi}^t)$ is a *feasible* point in the fully constrained problem (11.52) and $(w^*, \hat{\xi}^*)$ is its minimum.

The inequalities (11.55) imply

$$\Lambda(w^t, \hat{\xi}^t) - \Lambda(w^*, \hat{\xi}^*) \leq \Lambda(w^t, \hat{\xi}^t) - \Lambda(w^t, \xi^t)\,. \qquad (11.56)$$

The right-hand-side is in its turn is equal to:

$$\Lambda(w^t, \hat{\xi}^t) - \Lambda(w^t, \xi^t) = \frac{C}{m}\sum_{i=1}^m (\hat{\xi}^t - \xi^t)\,. \qquad (11.57)$$

If $(\hat{\xi}^t - \xi^t) \leq \epsilon$ it implies

$$\Lambda(w^t, \hat{\xi}^t) - \Lambda(w^*, \hat{\xi}^*) \leq \Lambda(w^t, \hat{\xi}^t) - \Lambda(w^t, \xi^t) \leq \frac{C}{m}\sum_{i=1}^m \epsilon = C\epsilon\,, \qquad (11.58)$$

which establishes a sound stopping condition for the algorithm.

**Computational complexity** It is known that the computation complexity of the algorithm is similar to the one of the subgradient method. Namely, to attain precision $\epsilon$ it has to perform $O(\frac{1}{\epsilon^2})$ steps. However, empirically it performs much better and, in addition, has a sound stopping condition. Typically, the number of required operations grows linearly with the dimensionality of the parameter vector $w$.

**Additional constraints** can be easily incorporated into the cutting plane method. For instance, to learn submodular potentials, the submodularity constraints can be added to the initial set of constraints at the very beginning. This would guarantee submodularity of the corresponding loss-augmented inference problem.

**Advantages and Disadvantages** of the cutting plane method: Compared to the (projected stochastic) subgradient method the cutting plane approach has two major advantages:

- usually, it converges much faster (typically the number of required iterations grows linearly with the dimensionality of the parameter vector $w$);
- it has a sound and easy-to-check stopping condition.

On the down-side:

- The set of selected constraints ($K^t$) in Algorithm 12 grows fast with the dimensionality of the parameter vector $w$ and the size of the training set. Solving the auxiliary problem at the step 6 of Algorithm 12 becomes more and more computationally expensive.
- It is hard to incorporate a good starting point. A close-to-the-optimal starting value of $w$ may influence only the first iteration of the algorithm and right after this first iteration its value may jump away. For a warm start the whole set of constraints must be kept in memory. In general, the selected cutting planes are very much dependent on the starting point of the algorithm (not an optimal value, this is a convex problem!).

There are also methods, which do not have these disadvantages, based on bundle methods and (stochastic, block-coordinate) Frank-Wolfe decomposition.

### 11.5 Further Reading

TBD

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